AutoISF Fork —Operations Manual

AIM of the program.

The aim is to make the requirement for intervention. very minimal., given still the expectation of. at least intermittent. bolus calculation. This represents a lesser goal than the “boost.” Variant aAPS programme, which I think really aims at no bolus.

Although the internal code automations do switch profiles and set temp targets these can be set individually as well if necessary and particularly if there is a concern about overall excess strength causing hypoglycemia or various conditions [such as steroids, hormones and so on] interfering in the last few days or weeks,

It is simple enough to switch the “high and low” profiles used by the program.
Also, setting “MJ” types from List1 [or States] sets either 3, 2 , or 1 day of more gentle criteria for changing profiles, setting Steroid profiles [fitst set Kotlin MJ and Steroid buttons] from List1 sets more aggressive changes during the day.

Introduction: Most significant new features [↑ TOC]

This manual concentrates on the operational changes that most distinguish this AutoISF fork from stock AAPS and that are most likely to affect daily use:

1. Full description of the bolus calculator changes; split and delayed bolus, safer calculation;

   Changes to bolus calculator with addition,s modificationeg
   Initial screen and live calculation., carbs, protein, fat . Protein and fat are not included in the immediate total
   Manual split-bolus if bolus over maximum bolus, remainder split
   Automatic delayed bolus at 50% or recent-50. This is distinct from the manual split.

2. Lists 1 and 2: full descriptions; rapid access to settings

   Double-tap the IOB area to open List 1. **Double-tap the basal-rate area** to open List 2
   The outer list shows setting names ; for setting preferences and some functions, with **confirmation.**
   On AAPSClient preferences are not changed locally, so confirmation creates the matching five-minute custom temporary-target code for the pump to receive, apply and cancel.

3. Settings Import Features — Especially With States.

   Four “Keep current…” checkboxes not present in stock AAPS

Automation States are force-disabled after every import, requiring confirm

Import restores states wholesale — where “states I never made myself” can come from

4. AIV table popup, automatic backup with logs and settings immediately and 6 hourly

Long-pressing the ISF/history control opens the AIV history popupEach row presents grouped glucose/raw-delta, ISF-factor, insulin/carb/HP, request/delivery and activity/status fields

Each row joins the nearest available context to an AIV timestamp and presents grouped glucose/raw-delta, ISF-factor, insulin/carb/HP, request/delivery and activity/status fields

Manual/on-open and Automatic /export ; all export AIV csv, text, setting txt, userentries.csv, logarithmic zips

5. Mechanism For Ignoring Or Invoking Native Automations; new ones now self-healing for the coded automations' own states.

his fork re-implements roughly 100+ of what used to be native AAPS Automation-tab triggers/actions directly as hand-coded Kotlin blocks

each native Automation-tab event's title against that registry every time it decides whether to actually run a native event

cf explanation of automation toggles gor MJ, Steroid buttons, installation check of cose tuned on for aISF running

6. Tier 3 UAM Boost criteria, output limits and duration.

This is not active and only runs on virtual pump at the moment.

Tier3 from Boost variant program with different protection; testing only

7. BolusGiven strong-boost effects.

8. BolusGivenMild and Mild Failsafe behavior.

9. Steroid-management initiation and escalation.

10. The Mounjaro three-day reduced-insulin cycle.

11. Hypoglycaemia warnings and protective overrides.

Contents [↑ TOC]

UKF3426 (live/production branch). Companion to "AutoISF Settings Reference" — this document covers behavior, mechanisms, and gotchas that aren't obvious from the settings screens alone: what's different from stock AndroidAPS, what "Automation States" actually requires, how settings import interacts with them, which day-to-day settings carry outsized real-world effect, the steroid and GLP-1 ("Mounjaro") situations specifically, and how native Automation-tab triggers coexist with this fork's own hand-coded automations.

1. Requirements for “Automation States” [↑ TOC]

“Automation States” is a generic named state-machine store (its own plugin, its own “States” tab in the UI) shared between native Automation-tab triggers/actions and this fork's own coded automations in Kotlin. Two things make it easy to trip over:

a. It has a master on/off switch, and things silently do nothing when it's off [↑ TOC]

BooleanKey.AutomationStatesEnabled defaults to true on a fresh install. Several coded automations — the Steroid Kotlin button chain is the clearest example — explicitly check this flag and return immediately, doing nothing at all, if it's off. There's no error, no notification: the button/automation just quietly has no effect. If a state-dependent automation appears to have “stopped working,” check this toggle first.

A state (e.g. “Steroids”, “MJ”) has a name and a fixed list of allowed value-strings (e.g. “Steroids Off” / “SteroidsON”). AutomationStateService.setState() enforces this at runtime with a hard require() — calling setAutomationState() from code with a state name or value that hasn't been declared first throws an IllegalStateException rather than silently creating it. Historically the only way to declare one was manually, on the in-app “States” tab (AutomationStateFragment, “+” to add a state).

b. Fixed 2026-08-19: this is now self-healing for the coded automations' own states [↑ TOC]

Before the self-healing registration was added, a real, already-live crash risk existed: one coded automation called setState() unconditionally with no pre-check at all, so on any device that had never had that specific state typed into the States tab, it would throw. OpenAPSAutoISFPlugin.onStart() now calls ensureRequiredAutomationStatesDeclared() once at every app start, which pre-declares the value-list for every state this fork's coded automations are known to read or write — currently MJ, MJstate, Steroids, LowBG, Steps, BGLstate, Profile, Sleeping, and query_got_up — but only when hasStateValues() says that state isn't already declared, so it can never overwrite/clobber a list you've customized on the States tab yourself.

Practical effect you may actually see on the States tab: the first time a build containing this fix runs, some of those 9 state names can appear with their value-lists already populated even though nobody manually typed them in — that's this bootstrap doing its job, not a mystery and not anything to undo.

c. A second, related gap: declaring the value list still isn't the same as the state having a value [↑ TOC]

Even with the fix above, a brand-new state's value LIST existing doesn't mean it has a CURRENT value — AutomationStateService only assigns one the first time something calls setState() for that name, historically only via a human opening that state on the States tab and pressing OK (which sets the first list entry as current, but only for a state the dialog itself treats as new/blank), or a coded action explicitly setting one. Until that happens, every gate reading that state (checkAutomationState(name, someValue)) just silently reads false — no crash, but no default either, and nothing on screen tells you this is why an automation isn't firing.

The bootstrap also assigns a safe default CURRENT value for the 5 states where there's real evidence of what “neutral/off” means (never guessed): MJ → NOMJremains, MJstate → MJoff, Steroids → Steroids Off, LowBG → NO50rec, Profile → AllOK. Steps, Sleeping, BGLstate, and query_got_up stay declare-only (each for its own reason -- see the changelog for detail). The default only applies when the state currently has no value at all -- it never overwrites something already set.

Also fixed 2026-08-19, same day: the bootstrap now also assigns a safe default CURRENT value, but only for the 5 states where there's real evidence of what “neutral/off” means (never guessed): MJ → NOMJremains, MJstate → MJoff, Steroids → Steroids Off, LowBG → NO50rec, Profile → AllOK. It deliberately leaves Steps, Sleeping, BGLstate, and query_got_up declare-only — no coded automation in this file was ever seen writing the first two (something else owns them), BGLstate has only one enumerated value with no observed “off” counterpart (defaulting to it would wrongly assert a low just happened on every fresh device), and query_got_up is a one-shot signal, not a toggle.

The default is only applied when the state currently has no value at all — it can never overwrite something you (or the import) already set.

2. Settings Import Features — Especially With States [↑ TOC]

Settings import in this fork is substantially more elaborate than stock AndroidAPS. The underlying mechanism is still "clear every SharedPreferences key, then write the file's contents back in" (ImportExportPrefsImpl.doImportSharedPreferences()) — but several fork-specific safeguards sit around that blunt operation.

a. Automation States are force-disabled after every import — by design, not a bug [↑ TOC]

Immediately after any import (cloud or local), AutomationStatesEnabled is unconditionally written back to false, regardless of what the imported file contained or what this device's setting was beforehand. If the imported file's data does include state definitions/values (detected by checking for the automation_state_service/automation_state_values keys), a dialog then appears with a checkbox — "Enable Automation States now?" — defaulting to UNCHECKED. If you import a backup that relies on states (steroid automations, MJ, etc.) and click OK without ticking that box, every state-dependent automation will silently do nothing after the restart, even though the state data itself was restored correctly underneath. This has been the source of confusing "import didn't restore my automations" reports in the past — the data was fine, the master switch just stayed off. Always tick that box (or re-enable the toggle manually afterward) when importing a file you know used states.

What this warning does NOT tell you, and no other warning covers either: it only appears when the imported FILE contains state data. If the file has none — an older export from before you started using states, or one from a device that never had them — but this device currently does, there is no warning of any kind. sp.clear() still wipes automation_state_service/automation_state_values down to their empty defaults, silently, because nothing in the file exists to restore them. If you're about to import a file you're not sure has current state data, and you care about keeping this device's states, export/back up this device's states first — there's no "keep current states" checkbox in the import dialog the way there is for pump config, patient name, BG source, and sync (§ below).

b. Import restores states wholesale — where "states I never made myself" can come from [↑ TOC]

Both halves of the States system — automation_state_service (which value each state is currently set to) and automation_state_values (each state's declared allowed-value list) — are ordinary exportable SharedPreferences keys, stored as single JSON blobs. An import wipes and replaces both wholesale from whatever the imported file contains, with no per-state merging. So after importing an older backup, or a file exported from a different device in this fork's fleet, it's expected — not a mystery — to see states (and their currently-assigned values) appear that you never typed into the States tab on this device yourself: they were sitting in the file, either from earlier on this same device before you deleted them, or from wherever else that file came from. This is separate from, and in addition to, the self-healing declaration described in §1 — an import can still restore a CURRENT VALUE the §1 bootstrap wouldn't have picked (any state/value combination that was ever manually set on any device, not just the 5 states §1 now defaults), and a value restored by import always reflects whatever was really happening on the exporting device at export time, which the bootstrap's fixed, generic defaults cannot know.

c. Four "Keep current…" checkboxes not present in stock AAPS [↑ TOC]

Because importing an older backup would otherwise silently clobber live, device-specific state, the import summary dialog can show up to four independent "preserve this device's current value instead of the file's" checkboxes:

The Virtual-Pump gating on the last three exists because they matter most for Client-style mirror phones — importing a backup onto a Client shouldn't normally overwrite which real device it's mirroring, its own BG source, or its own NS connection, even though the rest of the imported settings (therapy/dosing preferences) should still apply.

A separate, older precedent for the same "keep one thing across the clear()" pattern: StringKey.AapsDirectoryUri (the AAPS working-folder URI) is always snapshotted and restored automatically, with no checkbox — it's considered device plumbing, never something a backup file should be allowed to move.

3. Settings That Materially Affect Day-to-Day Function [↑ TOC]

The full settings tree (~65 entries) is documented setting-by-setting in the companion "AutoISF Settings Reference". This section instead highlights the handful whose day-to-day real-world effect is outsized relative to how easy they are to change by accident, or how non-obvious their interaction with something else is.

a. The SMB delivery brake [↑ TOC]

• FIXED SMB delivery ratio (smb_delivery_ratio) — the single biggest lever on how much of a calculated insulin requirement gets delivered as one SMB. It is also the silent source value for the (GUI-less) "UAM Boost insulin req %" internal setting — changing this ratio changes UAM Boost's bypass behavior too, even though nothing on the Boost screen says so.

• maxIOB threshold percent for SMB (iob_threshold_percent) — defaults to 100% (effectively off, governed only by the hard max-IOB cap). Lowering it adds a second, earlier IOB-based SMB cutoff that's easy to forget is active once set.

Overnight SMB delivery reduction. From 00:00 inclusive until 08:00 exclusive (00:00–07:59), AutoISF reduces the effective SMB delivery ratio by 0.03 before calculating the SMB: effective ratio = max(0, configured smb_delivery_ratio − 0.03). For example, 0.50 becomes 0.47. This is a ratio reduction, not a fixed subtraction of 0.03 U from the resulting bolus, so its unit effect scales with the calculated insulin requirement. The configured ratio resumes at 08:00. The ordinary maximum-bolus, IOB, recent-delivery, interval, rounding and other shared SMB protections still apply, and the adjustment is recorded in the APS reason/debug output.

b. Tier 3 UAM Boost: criteria, output, and duration [↑ TOC]

Tier 3 UAM Boost is an opt-in, Virtual-Pump-only SMB calculation. It is evaluated only inside the normal SMB pathway, so microboluses must be allowed, SMB must be enabled, and glucose must already be above the ordinary SMB threshold. The Tier 3 switch defaults to off and this feature cannot activate while a real pump is selected.

Direct entry criteria: current delta at least 5 mg/dL per 5 minutes (about 0.28 mmol/L); short-average delta at least 3 mg/dL per 5 minutes (about 0.17 mmol/L); delta divided by short-average delta greater than 1.2; absolute delta divided by long-average delta greater than 2; current BG above 80 mg/dL (about 4.4 mmol/L); eventual BG above target; positive calculated insulin requirement; effective boost scale below 3; current IOB below the configured Tier 3 ceiling; and, added 2026-08-23, bg_acce (a quadratic-fit second-derivative acceleration term, computed once per cycle for the AutoISF ISF-acceleration weight elsewhere) above 0.30mmol -- Tier 3's own entry gate never included any true acceleration term before this; the delta-ratio checks above are unstable near a small short/long-average-delta denominator, which is exactly the failure mode traced to a real 23 Aug over-firing (see the Tier 3 stacking-type criteria paragraph below). A missing or effectively zero short/long delta makes its ratio zero and prevents entry. Tier 3 has no direct HP or HP2 criterion.

The IOB ceiling is percentage-based: profile max_iob multiplied by the configured UAM Boost max-IOB percentage, divided by 100. The default is 10%. Remaining Tier 3 allowance is that ceiling minus current IOB. A new preference key is used so a historical absolute value such as 1.0 U cannot be misread as 1%.

For output, Tier 3 now uses the already-calculated usual current SMB as an explicit assessment input. That usual SMB already includes the live variable SMB-delivery ratio and any active BolusGiven or BolusGivenMild ratio increase. Tier 3 multiplies it by the effective Tier 3 boost scale and compares that result with both its basal-based scaled candidate and its baseline-ratio candidate. In compact form, the provisional Tier 3 amount is the largest of: scaled current basal, baseline-ratio SMB, and usual current SMB × Tier 3 scale. This assessment runs only with the Virtual Pump. Tier 3 is considered to have enhanced the candidate only when the rounded result exceeds the ordinary pre-boost SMB. The result remains limited by the configured maximum boost bolus and the remaining percentage-based IOB allowance; that IOB allowance is enforced again immediately before final pump-increment rounding.

The Tier 3 candidate still passes through all shared SMB protections. These include stacking reductions, recent-low rebound reduction, late-fast-rise tapering, rolling 30-minute and 10-minute SMB limits, sub-7.5 mmol/L heavy-delivery cooldown, offset-based SMB zeroing, final rounding, and the ordinary SMB delivery-interval gate. Any of these can reduce the Tier 3 candidate or cancel delivery.

Reporting now describes delivered output rather than an early criteria match. The UamBst CarePortal note and "UamBoost: SMB boosted" SMS are produced only when Tier 3 genuinely increased the initial candidate, a positive amount survived all final safety modifiers, the normal SMB interval permitted delivery, and the result was written to rT.units.

Duration and permitted hours: Tier 3 applies to one calculation cycle only and may be initiated only from 09:00 inclusive until 21:00 exclusive (09:00–20:59), using the device's local time. Outside that window Tier 3 does not boost the candidate, although the ordinary SMB pathway may still calculate and deliver an SMB under its normal rules. Tier 3 has no persistent boost duration and no separate two-minute Tier 3 throttle. The normal SMB delivery interval still controls when another SMB may be delivered. A concurrent 30- or 60-minute low temporary basal can be produced by the ordinary SMB algorithm, but that is not a Tier 3 duration.

Tier 3 also carries three stacking-type safety criteria borrowed from BolusGivenMild, alongside its other entry criteria: a 10-minute self-throttle (an instance-level last-fire timestamp on DetermineBasalAutoISF, since Tier 3 lives outside the plugin class's shared readyToRun()/markRun() map), the same 2-hour quiet window after a manual bolus or carb entry, and BolusGivenMild's own sub-7.5mmol delivery-rate ceiling.

Reference-comparison tooling (UKFboost branch, later merged to the main working branch). A self-contained, faithful port of the literal Tier 3 "UAM Boost" logic from the original Boost-in-AAPS_3.4 reference project -- confirmed identical across two of that project's own variants, so not a guessed match -- including the fast-carb-rebound detection Tier 3's own body there depends on but this fork never had. It runs alongside the adapted Tier 3 above every cycle boost is active, purely for logging ("[Tier3Ref] ..." lines in the console/reason text), never wired into real delivery. delta_accl (the reference's own acceleration term) is substituted for bg_acce throughout, matching the reasoning above; the two re-derived thresholds for that substitution are explicitly documented in code as unvalidated judgment calls, not calibrated figures, pending a few more real logged cycles to check against.

Relationship to SMBs already delivered. Tier 3's initial maximum boost bolus is an absolute per-SMB cap, not a percentage of recently delivered SMBs. Its new scaled-usual-SMB branch increases the current ordinary SMB by the effective Tier 3 scale; it does not discard an increase already produced by the live SMB-delivery ratio or by BolusGiven/Mild. For example, if the usual current SMB is 0.40 U and the effective Tier 3 scale is 1.5, that branch proposes 0.60 U before the Tier 3 ceilings and shared safety reductions. Delivered SMBs affect current IOB once their bolus records arrive. After Tier 3 proposes an amount, the shared safety path directly uses delivered-SMB history: rolling 10-minute limits are 0.6 U from 00:30–04:00 and 1.5 U otherwise; the rolling 30-minute ceiling is 2.1 U; late fast-rise delivery is reduced to 75% after 1.5 U/30 min and 50% after 1.9 U/30 min; rapid spacing and the sub-7.5 mmol/L heavy-delivery cooldown may further reduce or zero the candidate.

BolusGiven strong boost. This is separate from Tier 3 and is controlled by Enable BolusGiven/BolusGivenMild automations. Branches 1 and 2 react after a normal bolus, using bolus age, COB, glucose rise, raw-Libre confirmation, activity, profile and state gates. Branch 3 is delivery-driven: normally no bolus or carbs for at least 120 minutes, a sufficiently large five-minute IOB rise, strong AAPS/raw glucose rise, BG no higher than 9.5 mmol/L, low activity, not on the Low/MJ profile, and not literally MJ active. Branch 3 also has a raw-signal bypass when recent SMB delivery was suppressed despite a strong continuing rise.

Strong-boost effects. When a BolusGiven branch fires it cancels the current temporary target, sets iob_threshold_percent to 71, selects the Standard Profile, restores acceleration weight to its configured normal value, raises postprandial ISF weight, sets SMB delivery ratio to Mild Boost base + 0.03, applies a 110% profile for two minutes, and applies a 4.2 mmol/L target for two minutes. Branch 3 also arms a 30-minute recent-boost marker that can restore selected fast-rise-capped SMBs to the saved uncapped amount; recent-low and final cumulative-delivery protections run afterward and still take precedence.

BolusGivenMild and Mild Failsafe. Mild covers a lower raw-delta band than strong branch 3 and is designed to be mutually exclusive with it. It uses delivery change, raw/AAPS rise confirmation, no recent bolus/carbs, low activity, profile/MJ gates and cross-cooldowns. It leaves profile percentage and IOB threshold unchanged, but raises smb_delivery_ratio to Mild base +0.05 below 7.5 mmol/L, +0.02 below 9.0 mmol/L, or the base at/above 9.0 mmol/L; it also sets a 5.0 mmol/L target for two minutes and raises postprandial ISF weight. Mild Failsafe applies similar effects when BG is above 6.5 mmol/L, the rise is confirmed across delta horizons, IOB is at most 0.20 U, and no SMB has been delivered for 20 minutes.

Initiation order relative to Tier 3. BolusGiven/Mild automations run first and can change smb_delivery_ratio, profile percentage, target and weights. AutoISF then builds the profile passed into DetermineBasal, where Tier 3 independently evaluates its own Virtual-Pump, delta-ratio, BG, eventual-BG, positive-insulin-requirement and IOB-ceiling criteria. A prior boost can therefore alter Tier 3's ordinary candidate, effective scale or target context, but it does not automatically start Tier 3. Tier 3's separate switch and all of its own gates must still pass; afterward every common SMB safety modifier and the normal SMB interval still apply.

Stacking-type criteria (added 2026-08-23). Tier 3 previously had none of the anti-stacking safeguards BolusGiven/BolusGivenMild already had, despite living right alongside them -- traced directly to a real 23 Aug firing that delivered three consecutive doses (09:32/33/34am, no gap at all).

Three additions, borrowed from BolusGivenMild: a 10-minute self-throttle (implemented as an instance-level last-fire timestamp on the DetermineBasalAutoISF class itself, since the file's shared readyToRun()/markRun() throttle map lives on the plugin class, not reachable from Tier 3's own pure-calculation code); the same 2-hour quiet window after a manual bolus or carb entry BolusGivenMild already requires; and BolusGivenMild's own sub-7.5mmol delivery-rate ceiling (blocks re-arming when recent IOB change is already running hot -- over 0.8U/5min -- relative to a still-modest BG). All three sit alongside the existing criteria above, not replacing any of them.

c. The weight system's Resting/Boosted pairs [↑ TOC]

Several ISF-strengthening weights (acceleration, postprandial) exist as a live value plus a separate “Resting” and “Boosted” pair. Automations write the live value transiently during a boost and restore it from “Resting” afterward — editing the live-value setting directly has no lasting effect once the next boost/restore cycle runs; the “Resting” setting is the one that actually persists day to day.

d. Duration (dura) modifications to the AutoISF algorithm [↑ TOC]

What dura measures. The glucose-status calculator walks backwards through valid CGM records and extends a plateau while each older reading remains within plus or minus 5% of the running average. It stops at the first reading outside that band, an unusable/gap-filled reading, or a gap greater than 13 minutes. The outputs are dura_ISF_minutes and the average glucose over that plateau; this is persistence near one level, not simply time above target and not the temporary-target duration.

Engagement gates. Dura contributes no adaptation until the plateau has lasted at least 10 minutes, and it is bypassed whenever the plateau average is at or below the current target. Once both conditions are satisfied it is treated as prolonged high glucose that the existing profile ISF has not resolved.

Core calculation. Before the IOB safety taper, duraBoost = (dura minutes / 60) x (dura_ISF_weight / target BG) x (plateau-average BG - target BG), and dura_ISF = 1 + duraBoost. BG and target use the same mg/dL scale internally. A larger or longer elevation therefore raises the factor. In this algorithm a larger factor makes the effective ISF numerically smaller (profile ISF divided by the final factor), so dura is an insulin-strengthening/resistance response, not a sensitivity back-off.

IOB-based safety taper. The current implementation applies the taper to duraBoost rather than reducing the configured weight globally. At IOB up to 1.5 U the full boost remains. Between 1.5 U and 3.0 U the multiplier falls linearly from 1.0 to 0.30. At or above 3.0 U the multiplier remains 0.30, so dura is trimmed but never completely disabled. If the current HP2 projection is above 7.0 mmol/L, the taper is deliberately bypassed and the full dura boost is retained. Script Debug records the latest taper time, IOB, multiplier and before/after boost, including on later cycles when no new taper occurs.

Combination with the other AutoISF factors. When any strengthening factor is active, the algorithm initially takes the maximum of dura, BG, acceleration and postprandial factors. If acceleration is already braking below 1.0, that braking factor then weakens the selected maximum. The result passes through the ordinary AutoISF limits and sensitivity-ratio logic. If AutoISF weights are in display-only mode, dura is still calculated and graphed but is not applied to the returned dosing ISF.

Settings and List 1 control. dura_ISF_weight is the live weight used by the calculation. Resting duration ISF weight (default 1.2, permitted range 0.0-3.0) stores the user's normal value. List 1 codes 5.022 and 5.024 change the resting value by -0.1 or +0.1. They also move the live value only when it still equals the former resting value; this prevents a separately changed live value from being overwritten. There is currently no distinct boosted/high dura-weight setting and no coded automation that automatically raises the live duration weight.

Other decisions that consume dura. BolusGiven's manual-bolus branches require dura_ISF to be lower than acceleration ISF, so a duration-dominant rise does not enter those branches. The Virtual-Pump-only pseudo-wizard assessment measures how long dura and acceleration have remained active. OvernightDuraRescue can temporarily select the Standard profile for 60 minutes between 02:00 and 04:00 when the Low profile is active, glucose is above 6.0 mmol/L, dura and final ISF are both above 2.5, dura dominates every other factor and final ISF, no SMB was delivered in 30 minutes, no 50recent state exists, and both short- and long-average deltas are within approximately plus or minus 0.1 mmol/L. Its 60-minute cooldown prevents stacking.

Display and records. Each AIV result stores the calculated dura factor and final factor. The selectable DUR ISF graph shows its history, while Graph 5's pp/acce/du annotation row shows the current duration contribution together with the other principal weights. The target-offset/last-duration-taper annotation and Script Debug text help distinguish an actively tapered dura episode from a large untapered duration factor.

e. TDD factor / sensitivity toggles [↑ TOC]

TDD Sensitivity (ISF/Basal) and TDD Factor (max_iob/insulinReq) both default on and quietly govern how much the blended-TDD ratio adjusts dosing in the background. Turning either off doesn't disable dosing — it just locks that particular ratio to neutral (1.0) or to the TDD Factor fallback value, which most users have never touched since it only matters while the toggle above it is off.

f. The overnight safety-guard architecture [↑ TOC]

Built after two real hypoglycemia incidents (27 Jul and 6–7 Aug 2026), this is the most safety-critical cluster of settings/automations in the fork: NightIobCeiling (00:00–06:00) and EveningIobCeiling (20:00–00:00) cap iobTH/acce weight purely on time + current value; an escalating SMB anti-stacking trim plus a cumulative 10-minute SMB-sum cap in DetermineBasalAutoISF.kt measure delivered amount rather than rate; four sites unconditionally switch to the Low profile 22:00–06:00; and OvernightDuraRescue is a narrow, one-shot 02:00–04:00 exception that can switch back up when genuinely warranted. None of these have their own dedicated GUI toggle beyond the underlying weight/threshold settings they act on — they're time-and-condition-gated automations, not switches, so they can't be "turned off" from a menu even if a value they're capping looks wrong. See CLAUDE.md's own documentation of this architecture for the full mechanism and the known EveningTH/NightIobCeiling flip-flop regression pattern (§ "Other idiosyncrasies" below has the short version).

4. Setting: If Steroids Are Needed [↑ TOC]

Steroids (e.g. prednisone) sharply and progressively raise insulin resistance for as long as they're active. This fork has a dedicated, fully worked-out mechanism for it — the most complete example of a "medical-event ladder" anywhere in the codebase, and a reasonable template if a similar mechanism is ever built for another condition (see §5).

a. Turning it on [↑ TOC]

• GUI: "Steroid Kotlin button" setting (autoisf_steroid_kotlin_button_enabled, default on) shows a direct "Steroids ON" button on the Overview screen.

• Requires: BooleanKey.AutomationStatesEnabled must be on (see §1) — the whole chain silently no-ops otherwise.

• Also reachable via a direct TT-code (a specific temp-target value used purely as a signal, not a real target) for remote/automated triggering without opening the app — see §6 for how TT-codes are used generally in this fork.

b. The escalation ladder [↑ TOC]

Each step requires the previous step's Automation State + profile match to still hold, switches to a named Steroid profile, and bumps iobTH percent and bgAccel_ISF_weight — each step is a deliberate, separate user action, not automatic escalation over time:

Step	Profile switched to	iobTH %	bgAccel weight
START (110%)	Steroid Profile110	71	0.70
INCREASE (130%)	Steroid Profile130	73	0.70
INCREASE (150%)	Steroid Profile150	74	0.70
INCREASE (190%)	Current Profile190Real	75	0.70
INCREASE (250%)	Current ProfileReal250	76	0.70
TURN OFF	Current ProfileReal	71	0.71

Every step re-validates its own preconditions right before executing (profile percentage still 100, the right prior Automation State still active) and silently aborts with an overview refresh if something changed in the meantime — e.g. pressing "increase" after someone already turned steroids off elsewhere does nothing rather than acting on stale intent. Each step also fires a confirmatory SMS and a distinct CarePortal note (SteroidsON, Steroids130, Steroids150, Steroids190, Steroids250, SteroidsOff), and several unrelated automations elsewhere in the fork explicitly check checkAutomationState("Steroids", "Steroids Off") as one of their own preconditions — turning steroids on doesn't just change dosing directly, it also silences a number of other automations that assume a normal (non-steroid) state.

5. Mounjaro (MJ): three-day reduced-insulin cycle [↑ TOC]

MJ means Mounjaro in this fork. It does not mean Morning Jog. The MJ controls implement a reduced-insulin state cycle for the period after a Mounjaro dose, when insulin requirements may be lower and hypoglycaemia risk may be greater. The Overview button describes the intended operating period as three days of the configured low-profile strategy; the implementation uses named states and clock-time transitions rather than an exact 72-hour countdown.

a. Starting the cycle [↑ TOC]

The start button is offered when Automation State MJ is NOMJremains. Automation States must be enabled; otherwise the action returns without changing anything. After confirmation, the code sends the SMS "Injection MJ 0.35_.70", creates an urgent MJ notification and graph announcement, sets bgAccel_ISF_weight to 0.35, sets iob_threshold_percent to 70, switches to the configured Low Profile, changes MJ to MJ active, and writes the CarePortal note "MJ active".

The button wording says "3 days of 80%", but the Kotlin action does not hard-code an 80% temporary profile percentage. It selects the profile named by the Low Profile setting. Whether that profile represents 80% depends on the user's configured profile; the manual must therefore describe it as the configured Low Profile rather than promise a fixed 80%.

b. Day-by-day state progression [↑ TOC]

Day 1 — MJ active: this state begins immediately after the Mounjaro start action. While MJ is literally MJ active, several automations apply their MJ-specific branches or block stronger non-MJ behavior.

Day 2 — MJ2: when MJ is still MJ active, the MJ2 automation can advance it to MJ2 during 02:10–03:10. It sends "MJ2" and records the CarePortal note MJ2.

Day 3 — MJ3: when MJ is MJ2, the MJ3 automation can advance it to MJ3 during 01:05–02:05 on a later pass through that window. It sends "MJ3" and records the CarePortal note MJ3.

These are clock-window transitions, not elapsed-day calculations. The precise wall-clock duration therefore depends on when the start button was pressed and whether the app ran during each transition window. The states represent the intended three-stage course, but the code does not independently prove that exactly three full 24-hour periods have elapsed.

c. What changes while MJ remains active [↑ TOC]

The immediate start action applies the configured Low Profile, acceleration weight 0.35, and IOB-threshold percentage 70. The MJ state is also read throughout the automation layer: it changes eligibility for profile restoration, bolus/post-bolus handling, evening and overnight thresholds, time-of-day offset resets, and several high-glucose or steroid actions. Consequently, MJ is not only a label; it is a shared gate that changes which other automations may run.

Some checks intentionally distinguish MJ active from later MJ2/MJ3 states. In particular, mjActive() is true only for the literal MJ active state. Other checks distinguish NOMJremains from every active-stage value. This means the behavior can change as the cycle advances even before it is fully cleared.

d. Hypoglycaemia warnings and protective overrides [↑ TOC]

Hypoglycaemia warnings do not automatically mean "cancel MJ and restore 100%". GentleHypoRisk, AlarmHypo1, AlarmHypo2, the 50% profile machinery, and related HP/HP2 checks can lower acceleration weighting, lower the IOB threshold, set protective low/recent-low states, or apply a protective profile/temporary target while the MJ cycle remains relevant.

The MoreMJ automation can advance MJ directly to MJ3 when its low-risk conditions are met: very low acceleration weight, low recent activity, glucose below 5.5 mmol/L, a 50% profile, no COB, more than 210 minutes since the last bolus, recent-low state, and at least 0.2 U IOB. A separate path can also set MJ3 when no raw Libre value above 12

mmol/L has been seen in the preceding 48 hours. This advances the cycle toward its exit state; it is not the same as immediately restoring the standard profile.
Because the protective automations and MJ state machine are separate mechanisms, a hypoglycaemia alert should be treated as an override requiring attention, not as evidence that the three-day cycle has already been completely cleared.

e. Automatic exit and manual restoration [↑ TOC]

Once MJ is MJ3, MJoff can set it to NOMJremains either between 00:00 and 01:00, or between 12:00 and 21:04 when glucose is at least 10.5 mmol/L. This clears the MJ state and records an MJoff note. The automatic MJoff action itself changes the state; it does not explicitly restore all profile and weight settings in the same block.
The manual restore button is shown for any MJ value other than NOMJremains and is labelled for a dose four or more days old, or whenever the user wants to restore the 100% strategy. It switches to the configured Standard Profile, sets acceleration weight to 0.50, sets the IOB-threshold percentage to 70, changes MJ to NOMJremains, sends “MJ dose 4+ days old”, and writes the CarePortal note NOMJremains.

6. Mechanism For Ignoring Or Invoking Native Automations [↑ TOC]

This fork re-implements roughly 100+ of what used to be native AAPS Automation-tab triggers/actions directly as hand-coded Kotlin blocks inside OpenAPSAutoISFPlugin.kt (the “coded automations” layer described in CLAUDE.md), so they can react every 1-minute loop cycle instead of on the native automation engine's slower tick. Both systems — native Automation-tab events and coded automations — remain fully present in the app at the same time, which creates a real risk of the same logical action firing twice (once from each system) unless something reconciles them.

a. The reconciliation mechanism [↑ TOC]

CodedAutomationNames.kt keeps a hand-maintained registry of every readyToRun()/markRun() key used by the coded-automation layer (currently 100+ keys, extracted via a documented grep one-liner whenever the list drifts). AutomationPlugin.kt classifies each native Automation-tab event's title against that registry every time it decides whether to actually run a native event:

Match	Meaning	Effect on the native event
EXACT	normalized title equals a coded key exactly	Always suppressed — the coded equivalent is assumed to be running every cycle instead, no user choice involved
CLOSE	one normalized string contains the other, without being exactly equal	Suppressed only if the user hasn't explicitly accepted it via the review popup (below); a prior “accept” decision (persisted per title) lets it run
NONE	no relationship to any coded key	Never touched — runs exactly as stock AAPS would

CLOSE matches surface as a review popup (pendingCodedAutomationReviews(), shown from Overview) listing each ambiguous title for a one-time accept/deny decision, persisted in AutomationStringKey.CodedAutomationDecisions — deliberately simple substring matching rather than fuzzy/edit-distance scoring, on the reasoning that an occasional unnecessary review prompt is harmless, while a missed suppression is not.

b. The suppression only applies while AutoISF is actually the active algorithm [↑ TOC]

The whole mechanism is additionally gated on preferences.get(BooleanKey.ApsAutoIsfCustomAutomationsEnabled) AND the currently active APS algorithm actually being AUTO_ISF — not just the preference toggle being on. This matters: if you ever switch the active APS algorithm away from AutoISF, OpenAPSAutoISFPlugin.invoke() (and therefore every coded automation) simply stops running, but any native events that were being suppressed as EXACT/CLOSE matches go right on being suppressed too, since the suppression check itself doesn't care which algorithm is active — only whether the toggle is on. In that state neither the coded automation nor the native one is doing anything, and it isn't obvious from the UI. If you ever change APS algorithm away from AutoISF, deliberately revisit “Enable Tier 3…”-style toggles and the custom-automations suppression, since “AutoISF-specific automation was silently suppressing a native one” becomes “nothing is running that automation at all.”

7. Other Idiosyncrasies Worth Knowing [↑ TOC]

a. CarePortal note codes as a compact diagnostic language [↑ TOC]

Rather than free-text notes, several export/automation pathways write short fixed codes to CarePortal specifically so their outcome history can be grepped later, the same way this manual's authors diagnosed the UKFcheck read-path bug from note counts alone:

Code family	Meaning
AVLs / AVLf	AIV local export succeeded / failed
ACEs / ACEf	combined-export (combined<Name>.txt) rebuild succeeded / failed
UKCs / UKCn / UKCf	UKFcheck diagnostic export found matches / ran fine but found nothing / genuinely failed (UKCf can ONLY come from an exception — never means “no findings”)
StB On / StB Off	Steroid Kotlin button feature toggled on/off via its TT-code
SteroidsON / Steroids130…250 / SteroidsOff	each step of the steroid ladder (§5)

Grepping a device's combined export for these codes is often faster and more reliable than trying to reproduce a suspected bug live — this is exactly how the exportUkfCheckText() read-path bug was confirmed (15/15 UKCf, zero UKCs/UKCn) before any code was touched.

b. MJ as the fork's other major state machine [↑ TOC]

Beyond Steroids, MJ (Mounjaro) is the fork's other major state machine. Numerous coded automations gate on checkAutomationState("MJ", ...), and several steroid escalation actions require MJ=NOMJremains. The MJ Kotlin buttons setting controls whether the Overview quick-action buttons are shown; disabling those buttons does not remove the underlying Mounjaro states or the automations that read them. See §5 for the complete three-stage cycle, hypoglycaemia protections, and restoration behavior.

c. The AIV/UKFcheck export pipeline has more moving parts than it looks [↑ TOC]

Automatic exports run every 6 hours (KeepAliveWorker) plus on every manual dialog-open, writing several files (CSV/TXT AIV table, settings snapshot, combined rebuild, UKFcheck diagnostic, UserEntries) to a per-device folder scoped by GeneralPatientName. Two details are easy to get wrong when troubleshooting a missing file: (1) UKFcheck_*.txt's own lookback window is only ~6–7 hours of raw log per file, much shorter than the AIV table's 30-hour window, though the underlying raw log itself survives 30 days regardless; (2) the raw log files UKFcheck reads from can silently live in more than one on-device directory (public Documents/aapsLogs vs. the app-specific fallback), and a function that only checks one of them can fail 100% of the time on a device where logging quietly moved to the other — the actual root cause behind a real 2026-08-19 bug fixed the same day this manual was written.

d. A concrete regression pattern worth remembering [↑ TOC]

The EveningTH automation once had its target action value changed (50→45) without re-checking its own self-latch condition against that new value (latch stayed “<50”) — it then flip-flopped with NightIobCeiling every ~5 minutes for hours before being caught via CarePortal notes. The general lesson generalizes well beyond this one automation: whenever an automation's action value changes, always re-check its own re-arm/self-latch condition against the new value, not just against the old one.

8. Manual controls and AIV history [↑ TOC]

a. Bolus calculator: screen, calculation and follow-up dosing [↑ TOC]

Initial screen and live calculation. The Wizard recalculates after each entry or checkbox change. The screen can include BG and target, carbs, protein, fat, correction (units or percentage), carb time, profile, BG trend, COB, bolus and basal IOB, the session-only maximum-bolus override, and the calculation breakdown. Selecting COB also selects IOB; clearing IOB clears COB. The displayed total is the constrained, pump-step-rounded proposal. A calculated 0 U remains confirmable so that the decision can be recorded. The maximum-bolus override affects this Wizard session only and is restored on every exit path.

Profile-percentage behavior. At an active 50% profile, the normal immediate calculation is reduced through the profile's effective IC and the automatic delayed-bolus mechanism may later check whether the remaining gap is justified. A separate LowBG=50recent state, or BG below 5.0 mmol/L and not rising, halves only the carbs-driven component when the active profile is not already 50%; the two reductions cannot stack. The manual carb-split row is displayed only when the feature is enabled, the active profile is exactly 100%, the proposed dose exceeds the maximum allowed bolus, and that maximum is positive. The delivery engine accepts an already-selected split at 100% or higher, but the present screen does not offer the row at a profile above 100%. Protein/fat projections can still be shown independently; their delayed delivery requires a profile of at least 100%.

Component calculation. The immediate proposal combines BG correction against the selected profile or temporary target, a 15-minute trend contribution, carbs divided by IC, included COB divided by IC, the negative of included bolus/basal IOB, a manual correction, and any selected superbolus. Protein and fat are not included in the immediate total: protein is converted as protein/25 divided by IC and fat as fat/11 divided by IC, then held as separate delayed-dose projections. The Wizard percentage is applied to the non-protein/fat immediate total before pump-step rounding and the normal bolus-constraint pipeline.

Fast-rise projected-HP protection. The current BolusWizard path, including Quick Wizard calculations using that engine, applies the retrospective guard only when projected HP is at most 6.5 mmol/L, current BG is below 10.0 mmol/L, both current and short-average rise are at least 0.6 mmol/L per five minutes, positive IOB is at least 2.0 U, and the proposed dose is at least 0.75 U. It first removes the calculated COB-insulin contribution and then reduces the remaining already percentage-adjusted proposal to 75%. The changed dose is recalculated and shown as the proposal; the saved calculator result records original dose, adjusted dose, projected HP and COB insulin removed. This is automatic, not a second optional action in the confirmation dialog.

Manual split-bolus projection and first delivery. When the split row is available it starts checked and uses the configured interval (editable from 1 to 60 minutes). The screen shows the planned number and size of parts: the maximum allowed bolus is the first part, with the residual projected at the selected interval. After confirmation, the first constrained part is delivered immediately. The residual is scheduled only once, and a CarePortal scheduling marker records the projected future total. SMBs remain allowed during this manual split sequence.

Manual split-bolus progress and recalculation. Before each later part the code verifies that the user has not stopped follow-up dosing, no newer bolus/carbs entry has superseded the schedule, profile remains at least 100%, the pump is available, and superbolus is not active. It obtains fresh glucose. Missing data is retried after two minutes. BG below 6.0 mmol/L is unsafe; from 6.0 to below 8.0 mmol/L, a delta or short-average delta at or below -0.05 mmol/L is also unsafe. The first two consecutive unsafe checks defer by a full split interval; the third cancels the residual. The complete soft-retry window is limited to 60 minutes. Each part is reduced by any positive rise in live IOB since the preceding post-dose baseline and rounded to the pump step. A result at or below zero is skipped and retried at the next interval rather than increasing the dose. Success schedules the next residual; failure raises the normal delivery alarm. Cancellation and calculated-zero outcomes are written to treatment/history markers.

Automatic delayed bolus at 50% or recent-50. This is distinct from the manual split. After the initial Wizard/Quick Wizard bolus, the worker checks every 10 minutes from +10 through +80 minutes while temporarily blocking SMBs. It requires glucose data no more than five minutes old, BG above 4.5 mmol/L, current delta above +0.10 mmol/L, short-average delta above +0.20 mmol/L, and long-average delta above +0.05 mmol/L. When the criteria first pass, it calculates the frozen original full-required amount minus the original delivered dose, then delivers 90% of that gap if confirmation occurs by 20 minutes or 50% if it occurs later. It does not recompute the original IOB snapshot. Db10, Db20 and subsequent check markers show dose, wait or end; SMBs are unblocked on delivery, cancellation or final expiry.

A “Walking soon” checkbox offers a third way to enter the same delayed-bolus check, alongside the profile-50% and recent-50 paths, gated on the checkbox itself rather than profile or state. It pre-ticks when recent steps are either very low (anticipating a walk) or already high (exercising now) and BG isn't already rising fast, but can be toggled manually regardless. When checked, the immediate dose is forced to 50% of the normal Wizard percentage without changing that setting; the delayed top-up then reuses the same +10-to-+80-minute check and 90%/50% confirmation-timing split, topping up toward the unreduced percentage, not a full 100% dose.

All four wizard-dialog mechanisms are shared with every Quick Wizard button: the max-bolus constraint clamp and the automatic delayed bolus at 50%/recent-50 apply unconditionally (gated on live profile percentage/automation state, not which screen requested the calculation); the manual split-bolus checkbox/interval and “Walking soon” checkbox are configured per-button on each button's own Edit screen (EditQuickWizardDialog.kt), stored in that entry's JSON (useSplitBolus, splitBolusIntervalMins, useWalkingSoon) and threaded through the same doCalc()/manualSplitBolusEnabled path the main dialog uses.

Protein and fat delayed doses. Protein is scheduled once at +120 minutes and fat once at +180 minutes, independently of the manual carb split. At delivery each requires the schedule still to be current, profile at least 100%, pump available, no superbolus, fresh BG at least 7.0 mmol/L, and both delta measures above -0.05 mmol/L. A positive rise in IOB since the immediate post-bolus baseline is deducted. If the remainder is zero or negative, a 0 U history entry plus cancellation note is recorded; otherwise the rounded remainder is delivered. These fixed-time doses do not retry after a failed BG gate.

Confirmation and visible history. Pressing OK performs one final recalculation, copies the split checkbox/interval into the BolusWizard, opens the standard confirmation/execution flow, and prevents a second OK tap from duplicating it. The BolusCalculatorResult keeps the calculation inputs and notes, including any carb-split projection, protein/fat timing and HP-safety adjustment. CarePortal notes and treatment entries then show scheduling, checks, delivered parts, zero-dose outcomes and cancellation reasons; these entries are the durable status trail after the original dialog closes.

b. Lists 1 and 2: mechanism, current status and confirmation [↑ TOC]

List 1 entry and purpose. Double-tap the IOB area to open List 1. On a pump/Virtual build its title is Direct AutoISF settings and a confirmed selection invokes the matching local AutoISF handler without creating a temporary target. On AAPSClient its title is Settings - relay to pump; preferences are not changed locally, so confirmation creates the matching five-minute custom temporary-target code for the pump to receive, apply and cancel.

List 1 rows and current status. The outer list shows setting names rather than raw relay numbers. It includes paired adjustments for SMB-delivery values, normal/high ISF weights, Libre slope/offset, SMB offset, Wizard percentage, mild-boost ratio, insulin peak, AutoISF limits and eight time-of-day offsets; single actions/toggles include sensor sensitivity, all boost automations, clean graph, Graph2/Graph5, cloud logs, UKFset1 where permitted and SensorAge; named paired controls cover MJ state and Standard/Low profile. Opening a row shows Current on pump/Virtual. A client shows Pump current from the latest mirrored device-status settings snapshot, with its age, or unavailable if the key has not arrived. Dual-key rows show both values. Stateless commands show an Effect description instead of pretending to have a current state.

List 1 offers two related but distinct profile-role rows. “Switch to Standard/Low profile now” immediately switches the ACTIVE profile to whichever one currently fills the role you're not on (via ProfileStandardTT/ProfileLowTT, 5.148/5.150) -- a toggle, not a two-option choice: it shows “Currently: Standard (X). Switch to Low (Y)?” and falls back to offering both explicitly only when neither role is currently active. “Re-pick coded profiles (Standard/Low/Steroid)” is a separate Action-type row (no TT number, local only) that reopens showProfileNamesPopup() to change which profile fills each role -- see “Coded profile roles” below. “Cloud logs upload”'s description lists everything the action exports: AIV data, a UserEntries CSV/TXT export, and a settings-snapshot backup, before the log zip itself.

“Live Libre slope/offset (view only)” is a view-only List 1 row (TtCode.Action) showing the LIVE FslCalSlope/FslCalOffset and ApsAutoIsfOldSensorAdjActive state alongside the configured baseline, with a drift warning if they disagree with no tier marked active -- the two “(Or)” rows above only ever show/nudge the baseline itself, never the live value. See “SensorAge” (§13) for the real discrepancy this surfaced between devices.

List 1 confirmation. A single/toggle row opens its description/current-value popup and requires OK. A stepped row opens mutually exclusive decrease/increase (or named-state) checkboxes labelled with the real change; the selection is not applied until OK. OK with neither direction selected is a no-op. Cancel, back, or outside-dismiss from an inner popup returns to List 1 so another row can be selected; Cancel on the outer list closes to Overview. Client relay commands retain the pump-side repeat guard; direct pump/Virtual actions do not use that relay guard.

OK now also returns to List 1 (previously only Cancel/back/outside-dismiss did), matching how TtCode.Action rows already behaved.

List 2 entry and actions. Double-tap the basal-rate area to open List 2. Its direct-action rows cover MJ start/restore, steroid start and escalation/turn-off, MJ and steroid button toggles, and the appropriate AnyDesk restart choices. On pump/Virtual, confirmed actions are dispatched locally. On AAPSClient, most are carried by their five-minute relay temporary-target code. AnyDesk is different: it always writes an ADesk Note; it adds a relay TT only when no real temporary target is active, preserving an existing target. The local-test AnyDesk row queues the real receiving-side path and records its own diagnostic note.

List 2 also offers three numeric stepped rows for Tier 3 UAM Boost's tuning parameters: “Boost scale (boost_scale)” (±0.1, clamped 0.1-2.9), “Boost max (boost_max, U)” (±0.25U, clamped 0.1-10.0U), and “Boost max IOB (% of max_iob)” (±5%, clamped 1-100%) -- List 2's only numeric stepped rows, using the same checkbox-pair confirmation dialog List 1's stepped rows use.

List 2 also carries “Tier 3 UAM Boost on/off” (5.194) and “AutoISF calcs: UKF1 vs LibreSpecial EMA” (5.196), toggling ApsAutoIsfUamBoostEnabled/ApsAutoIsfUseUkf1ForDosing respectively — the same flags their Settings-screen switches expose; List 2 is just a faster path to the same toggle, not a second independent one. Dispatched the same EventAutoIsfDirectTtCode way as the MJ/steroid Kotlin button toggles.

ApsAutoIsfUkf1DeltaCompensationSlope/Offset (Settings only) rescale UKF1's delta/shortAvgDelta/longAvgDelta/bg_acce while the UKF1-dosing toggle is on, defaulting to 1.0/0.0 (no-op) -- see §11d for the full feature.

List 2 graph controls and current status. The final rows control UKF1, UKF2, UKF3 and the Graph5 panel. Selecting one opens checkboxes showing the current local display state: Graph on plus its calibration option; UKF2 calibration is shown checked but disabled because calibration is already applied upstream. Graph5's second checkbox means BGL only. These settings are display-local and are written only after OK. Direct-action rows provide a confirmation question but do not all expose a universal live state; their resulting automation state, profile, target or note is the authoritative status after execution.

List 2 confirmation and return behavior. Every direct action requires the confirmation popup. Cancel returns to List 2. For a graph row, OK saves the selected checkboxes and refreshes Overview; Cancel, back or outside-dismiss returns to List 2. The outer Cancel closes the list. Client List 2 is therefore a command/relay interface, while its graph rows remain immediate local display settings.

Every BasalDirectAction row shows a “current value” before confirming where one exists: “Current: ON/OFF” for the MJ/steroid Kotlin button, Tier 3 Boost, and UKF1-dosing toggles, and “Current profile: X” for the steroid escalation actions. OK also returns to List 2 (matching List 1's own behaviour above), so e.g. toggling Tier 3 Boost then UKF1-dosing back to back doesn't need a fresh double-tap between each one.

c. AIV table: popup population and backup/export behavior [↑ TOC]

Opening and background population. Long-pressing the ISF/history control opens the AIV history popup. On first creation it reads the preceding six hours of AIV records in descending time order, APS results, step counts, SMB boluses and CarePortal notes. It also reads raw glucose with a 20-minute lead-in for delta calculations and computes the UKF3 comparison series. The work is assembled from the local database; it is not populated by parsing the displayed graph. A frozen Time column remains aligned with the horizontally scrolling header/body. SMB only rebuilds the visible table from records whose delivered-SMB value is positive, and All restores every loaded record; filtering does not change the full-record lookback used for derived values. Rotation preserves the filter and does not start another export.

What the popup represents. Each row joins the nearest available context to an AIV timestamp and presents grouped glucose/raw-delta, ISF-factor, insulin/carb/HP, request/delivery and activity/status fields. A missing six-hour history produces an explicit no-data row. Values such as COBt, IOB change, HP/HP2/HP3 and UKF-derived columns are calculated by the exporter/table helpers from the loaded windows; they should therefore be interpreted with their source timing and availability, not as independent sensor samples.

Manual/on-open export. The first opening of the popup starts an off-UI-thread export immediately. It writes three current AIV files (CSV, text and settings), refreshes the combined export, records success or failure in logs and an AVLs/AVLf CarePortal marker, then attempts cloud upload when cloud storage is configured. After the AIV upload callback it can invoke the same log-send route without exporting AIV a second time. Opening the popup is therefore the manual refresh mechanism, although the file work itself continues in the background and the table remains usable.

Automatic backup/export. KeepAlive checks the AIV export timestamp and normally performs the same last-six-hours three-file export every six hours, also rebuilding the combined file and its dated per-device copy. The files are always written locally first. When cloud storage is active they are additionally uploaded to the configured AIV cloud path, scoped by General patient name when supplied. If automatic cloud log export is enabled and due, KeepAlive reserves that six-hour slot before launching the asynchronous AIV-plus-log sequence so overlapping KeepAlive runs do not duplicate it. Automatic operation is silent: success/failure is logged rather than shown as a popup.

Backup scope and recovery meaning. These AIV files are history/diagnostic exports, not a restorable database backup. Manual popup-open and automatic six-hour runs create equivalent current exports and may upload copies; neither reimports AIV rows into the application. The combined/dated files preserve longer troubleshooting continuity, while each popup view itself remains a six-hour database window. A failed cloud upload does not remove the local files, and absence of cloud configuration simply leaves the local export as the backup copy.

Command	Relay TT (mmol/L)	Current effect
SMBdel base + mild-Bst	5.002 / 5.004	Decrease/increase both SMB-delivery baseline and Mild Boost ratio by 0.01.

Command	Relay TT (mmol/L)	Current effect
Toggle Libre sensor adjustment	5.006	Toggle aging-sensor/Libre adjustment on or off.
Toggle boost automations	5.008	Toggle the whole boost-automation group on or off.
pp ISF weight - normal	5.012 / 5.014	Decrease/increase normal postprandial ISF weight by 0.01.
Acceleration ISF weight - normal	5.016 / 5.018	Decrease/increase normal acceleration ISF weight by 0.05.
Duration ISF weight - normal	5.022 / 5.024	Decrease/increase normal duration ISF weight by 0.1.
Libre slope - original	5.026 / 5.028	Decrease/increase the baseline Libre calibration slope by 0.01.
Libre offset - original	5.032 / 5.034	Decrease/increase the baseline Libre calibration offset by 0.05.
SMB offset	5.036 / 5.038	Decrease/increase the hard SMB-offset override by 0.1.
Clean graph view	5.042	Hide SMB dose labels/arrows and show the plain solid-green view.
Wizard bolus percentage	5.046 / 5.048	Decrease/increase the Wizard percentage by five percentage points.
Mild Boost ratio	5.052 / 5.054	Decrease/increase Mild Boost ratio alone by 0.01.
pp ISF weight - high	5.056 / 5.058	Decrease/increase boosted postprandial ISF weight by 0.01.
Acceleration ISF weight - high	5.062 / 5.064	Decrease/increase boosted acceleration ISF weight by 0.01.
Higher-ISF-range weight	5.068 / 5.070	Decrease/increase high-BG ISF weight by 0.1.
Peak insulin time	5.074 / 5.076	Decrease/increase insulin peak by five minutes.
AutoISF maximum - low BG	5.080 / 5.082	Decrease/increase the low-BG AutoISF maximum by 0.1.
AutoISF maximum - normal	5.086 / 5.088	Decrease/increase the normal AutoISF maximum by 0.1.
TOD offset 00:00-02:00	5.092 / 5.094	Decrease/increase this time-of-day offset by 0.1.
TOD offset 02:00-04:00	5.098 / 5.100	Decrease/increase this time-of-day offset by 0.1.
TOD offset 04:00-06:00	5.104 / 5.106	Decrease/increase this time-of-day offset by 0.1.
TOD offset 06:00-09:00	5.110 / 5.112	Decrease/increase this time-of-day offset by 0.1.
TOD offset 09:00-12:00	5.116 / 5.118	Decrease/increase this time-of-day offset by 0.1.
TOD offset 12:00-18:00	5.122 / 5.124	Decrease/increase this time-of-day offset by 0.1.
TOD offset 18:00-22:00	5.128 / 5.130	Decrease/increase this time-of-day offset by 0.1.
TOD offset 22:00-00:00	5.134 / 5.136	Decrease/increase this time-of-day offset by 0.1.
Toggle Graph2 carb-model curve	5.138	Toggle the empirical/theoretical carb-model curve display.
Cloud logs upload	5.140	Zip and send logs to configured cloud storage, otherwise use email.
Toggle Graph5	5.142	Toggle the Graph5 comparison panel.
MJ state manual override	5.144 / 5.146	Set MJ to NOMJremains or MJ3.
Profile manual override	5.148 / 5.150	Select the configured Standard or Low profile.
Toggle UKFset1 live comparison	5.152	Toggle the Virtual-Pump, non-Client live UKFset1 comparison; hidden elsewhere.
Run SensorAge code	5.156	Toggle the sensor-age code on or off.
Command	**Relay TT (mmol/L)**	**Current effect**
MJ start	5.158	Start the confirmed Mounjaro cycle.
MJ restore	5.160	Run the confirmed Mounjaro restoration action.
Steroid start	5.162	Start the steroid-management sequence.
Toggle MJ Kotlin buttons	5.164	Show/hide the MJ buttons.
Toggle Steroid Kotlin button	5.166	Show/hide the steroid button.
Steroid 110 to 130	5.168	Run the confirmed steroid escalation to 130.
Steroid 130 to 150	5.170	Run the confirmed steroid escalation to 150.
Steroid 150 to 190	5.172	Run the confirmed steroid escalation to 190.
Steroid 190 to 250	5.174	Run the confirmed steroid escalation to 250.
Steroids OFF	5.176	Run the confirmed steroid turn-off action.
AnyDesk restart	5.178	Client writes an ADesk Note and uses a five-minute relay TT only if no real TT is active; otherwise Note-only transport preserves the real TT.
AnyDesk restart - local test	5.180	Local command test through the real receiver; no NS round-trip or relay TT.
Graph: UKF1	No relay TT	Local display toggle plus optional Libre slope/offset calibration.
Graph: UKF2	No relay TT	Local display toggle; calibration is already upstream and remains shown as on.
Graph: UKF3	No relay TT	Local display toggle plus optional Libre slope/offset calibration.
Graph: Graph5 panel	No relay TT	Local panel toggle; optional BGL-only mode hides insulin activity and the three

Command	Relay TT (mmol/L)	Current effect
		carb series.

d. Coded profile roles (Standard/Low): assignment, re-picking, and why it matters [↑ TOC]

What the roles are. Every one of OpenAPSAutoISFPlugin.kt's coded automations addresses two abstract profile roles -- Standard (stronger; used for corrections/highs) and Low (weaker; used to back off/reduce insulin, e.g. overnight and MJ) -- rather than a literal profile name. The role assignment itself lives in two preference keys, StringKey.ApsAutoIsfStandardProfileName and StringKey.ApsAutoIsfLowProfileName, which originally defaulted to hardcoded literals ("Current ProfileReal"/"Current Profile") and are now freely reassignable to whichever of the user's actual configured profiles should fill each role.

Where it's read: 38 live call sites, not cached. All 38 StringKey.ApsAutoIsfStandardProfileName/LowProfileName reads across OpenAPSAutoISFPlugin.kt -- switchProfileIfNeeded() calls plus a handful of onCurrentProfile/onLowProfile/onNormalProfile equality checks -- call preferences.get(...) fresh every time, rather than reading a value cached once at plugin construction or at the start of a cycle. That is what makes re-picking the assignment safe to do at any time: the very next 1-minute loop cycle after a change, every one of those 38 sites is already dispatching to the newly-picked profile -- no restart, no stale reference anywhere in between.

One-time onboarding: showProfileNamesPopup(). On first-ever launch (gated on the OverviewStringKey.ApsAutoIsfProfileNamesReviewed preference, checked from checkOnUpdatePopups()), a not-cancelable dialog presents two spinners populated from the user's actual configured profile list (activePlugin.activeProfileSource.profile.getProfileList()) -- one labelled "Standard profile (stronger -- used for corrections/highs)", one "Low profile (weaker -- used to back off/reduce insulin)" -- each pre-selected to whatever the preference already holds. OK writes both picks and marks the review flag; Cancel leaves the current (possibly still-default) values untouched but still marks the flag reviewed, so an already-seen user is never nagged again on every launch. If no profiles are configured yet (a genuinely first run), the popup skips itself silently rather than marking reviewed on incomplete data, so it re-checks on the next resume instead.

List 1's “Re-pick coded profiles (Standard/Low/Steroid)” row (an Action-type row, no TT number, local only) reopens showProfileNamesPopup() on demand, returning to List 1 afterward -- the only way to change the role assignment once the one-time onboarding popup has already been dismissed, short of clearing the preference directly.

Why the old "Profile (manual override)" List 1 row was removed instead of kept alongside this. That row (via the now-freed ProfileStandardTT/ProfileLowTT relay codes) switched the AAPS active profile straight to whichever profile currently fills the Standard or Low role -- a manual, one-off profile switch achievable identically by switching profiles directly in AAPS's own profile picker. It never touched the role assignment itself. On review the user judged that specific function redundant and asked for it to be removed as such; "Re-pick coded profiles" is a distinct function -- it changes which profile answers to each role, not which profile is active right now -- and was the one actually wanted.

Usefulness of the indirection. Because all 38 sites address the roles abstractly, retiring an old profile in favour of a re-tuned replacement (new basal rates, a new IC ratio, whatever prompted the change) needs exactly one re-pick via this popup -- not 38 individual code edits, not a rebuild, and not a restart. The new profile is live in every coded automation from the very next cycle. This is also what keeps the coded-automation layer stable across the user's own profile-naming and profile-tuning churn over time: the code never hardcodes a profile name after this point, only the two role preferences do.

Two real incidents from live AIV/UserEntries data, both traced the same day, that clarify what this system does and doesn't cover. First: a real BasalUp ("BsUp") firing on 23 Aug correctly switched to "Current ProfileReal" rather than the user's newly-created "Current Profile100" -- not a bug, just timing: the Standard role hadn't been re-picked to point at the new profile yet at that exact cycle, confirmed by tracking autoisf_standard_profile_name across every settings snapshot that day, and the very next coded-automation profile switch (the reinstated "Switch to Standard/Low profile now" List 1 row, "PrSt" note) correctly used the new name once the re-pick had landed.

Second, more important: a LATER switch back to "Current ProfileReal" traced to a source literally named "button1" in the UserEntries log -- not "AutoISF code-based profile switch" or "AutoISF: reset to 100%", which is what every one of THIS file's ported automations always logs. "button1" is a native AAPS Automation-tab trigger, configured entirely outside this codebase (Settings -> Automation), with the old profile name typed directly into its own Profile Switch action. No amount of re-picking the coded roles touches it -- it has to be edited by hand in the Automation tab itself.

e. Steroid escalation as a third coded role (added 2026-08-23) [↑ TOC]

The six steroid-escalation profile switches (110/130/150/190/250 percent, plus the revert-to-baseline “100” step) read StringKey.ApsAutoIsfSteroid100/110/130/150/190/250ProfileName live, the same pattern as Standard/Low, rather than hardcoding a literal profile name -- defaults match the original literals (“Steroid Profile110”, “Current Profile190Real”, “Current ProfileReal250”, etc., an inconsistent naming pattern preserved as-is since these are real profile names, not code identifiers).

Where it's read. Every switchProfileIfNeeded("Steroid ProfileXXX") call in the steroid escalation handler (handleDirectSteroidUserAction()) now reads the matching preference. Two further hardcoded-literal sites, easy to miss and initially missed in this same pass, needed the identical fix: the handler's own pre-execution staleness re-check (conditionStillMatches -- confirms you're still on the PREVIOUS tier's profile before allowing escalation to the next one) and a second, wholly separate copy of the same four checks in OverviewFragment.kt that gate whether the escalation buttons are shown on Overview at all. All ten sites (6 action + 4 pre-check/button-visibility) now agree.

showProfileNamesPopup() uses a data-driven loop over 8 roles (roleSpecs: List<ProfileRoleSpec>) rather than each role's spinner being hand-duplicated. List 1's “Re-pick coded profiles” row is labelled “...(Standard/Low/Steroid)” to match.

classifyCurrentProfileRole() -- new, additive-only classifier. Returns both a strength ("Standard"/"Low") and a steroidTier ("100"-"250") for whatever profile is currently running, with explicit defaults when the running profile matches none of the eight roles: strength defaults to "Low", steroidTier defaults to "100" (steroids-off baseline). Deliberately NOT wired into any of the existing scattered comparisons (onNormalProfile, onCurrentProfileEither, the Mild/BG3 Boost gates, etc.) -- those still do their own direct checks exactly as before. Retrofitting them to use this classifier was considered and explicitly deferred: it would be a real, broad change to live SMB-firing conditions, the same shape of risk as the EveningTH/NightIobCeiling flip-flop regression this manual already documents as a pattern to avoid repeating, and wasn't judged worth doing speculatively.

Do the existing 15 profile-role checks even see steroid tiers? Traced all 11 real gate/check sites (the rest of the ~30 StringKey.ApsAutoIsf*ProfileName hits are switchProfileIfNeeded() writes, not reads). None reference a steroid role name directly, but steroid tiers interact with about half of them by accident of phrasing. Five are "narrow" (must equal Standard or Low exactly) -- BolusGiven's post-bolus block, OffHighProf, a pod/MJ-time block, and the evening/night evB2/evB3 gates -- and these quietly pause themselves while any steroid tier is active, resuming only once back on Standard or Low. Six are "broad" (not-equal-to-Low only) -- all of Mild/BG3 Boost's own gates -- and these treat a steroid tier exactly like Standard, so Boost keeps working normally through a steroid course. Neither behaviour was designed on purpose; it simply falls out of which phrasing each gate happened to use.

Fresh-install/under-configured default risk, worth knowing before relying on any of this. showProfileNamesPopup()'s spinner pre-selection is profileNames.indexOf(current) falling back to index 0 when the stored default doesn't match any real profile. With zero profiles configured the popup skips itself entirely (safe -- switchProfileIfNeeded() then no-ops everywhere, since getSpecificProfile() finds nothing). But with only one profile, or with profiles that don't happen to be named the exact defaults, EVERY unmatched role -- Standard, Low, and all six Steroid tiers alike -- silently pre-selects whatever profile sits at index 0 of the list if the popup is accepted without manually checking each spinner. Not a crash, but a real silent-misconfiguration risk on first setup.

A native AAPS Automation-tab trigger (anything configured in Settings → Automation, not this codebase) that has a profile name typed directly into its own action is completely untouched by any of this -- re-picking a coded role never reaches it; it must be edited by hand in the Automation tab itself. See the changelog for the real incident that surfaced this.

One hardcoded literal remains outside this role system by design: switchProfileIfNeeded("Current Profile50") in Battery1pc/BatteryOver1pc (critical-battery emergency profile switch, real-pump-only), kept as its own separate, narrower mechanism rather than folded into Standard/Low/Steroid.

9. GUI display changes [↑ TOC]

a. Graph types and the triggers for each [↑ TOC]

Main glucose graph (graph 0). This graph is always the primary time-aligned display. Its chart-menu switches select predictions, treatments/therapy events, activity, empirical carb absorption, UAM carb impact, combined carbs, BG parabola, raw BG, smoothed raw/UKF traces and basal data. The separate Graph2 carb-model preference controls the theoretical carb-model curve independently of the empirical absorption switch. Targets, running-mode markers, the current-time line and the principal AutoISF annotations are added automatically.

Secondary graphs (graphs 1-4). The Overview chart menu creates and labels a secondary panel when at least one eligible series is assigned to it. Available series include absolute IOB, IOB, COB, IOB threshold, deviation, -BGI, autosensitivity ratio, variable sensitivity, deviation slope, heart rate, steps, final/acce/BG/postprandial/duration AutoISF factors, raw BG, smoothed raw/UKF data, UAM carb impact and combined carbs. The first active type normally establishes the scale; compatible IOB/threshold, deviation/-BGI and multiple AutoISF-factor selections share aligned scales.

Graph 5. Graph 5 is a dedicated main-sized comparison panel, not another chart-menu secondary graph. It is enabled by the Graph5 preference, the List 1 Graph5 toggle or List 2's Graph: Graph5 panel control. It deliberately ignores graph 0's individual visibility switches and always draws its own BGL comparison set. List 2 also provides BGL only, which hides insulin activity and the three carb-related curves while retaining BGL, basal and annotation rows.

Gesture and remote triggers. A basal-area long-press cycles the main graph's display preset and controls the quick visibility of empirical carb line 1. An IOB-area long-press toggles SMB labels, resets the basal display preset to normal and forces carb line 1 on. The Clean graph command (List 1/relay code 5.042) applies the no-SMB-label/plain BGL preset once and then clears its request flag. Double-tapping the basal area opens List 2, where UKF1/UKF2/UKF3 and Graph5 checkboxes change local graph display settings; UKF2's calibration box is informational because calibration is already applied upstream.

b. Colour changes driven by AutoISF input [↑ TOC]

Dominant AutoISF colour. For each AIV point the display compares how far the acceleration, BG, postprandial and duration ISF factors lie from neutral 1.0. The factor with the greatest deviation supplies its theme colour. That dominant colour can tint the corresponding glucose dot, SMB delivery marker and the AutoISF temp-basal overlay, so the graph shows which AutoISF input was most influential at that time rather than merely whether the final factor was above or below 1.0.

Safety and fallback precedence. A low glucose dot always uses the low-glucose colour and is never masked by an AutoISF tint. Without a usable dominant-factor override, high points use the normal high colour and other points use the original/default BG colour. Prediction dots retain their IOB, COB, UAM or zero-temp prediction colours. When the plain/clean preset is active, BG dots revert to the conventional low, in-range and high colours and the ISF-coloured temp-basal overlay is suppressed.

Other fixed colour conventions. Raw BG is split into colour-banded segments rather than shown as one undifferentiated trace. UKF1, UKF2 and UKF3 retain distinct comparison-line styling. The AIV/ISF index row uses separate colours for final, acceleration, BG, postprandial and duration factors; an SMB value of zero falls back to the ordinary insulin colour, while a delivered SMB can use the dominant AutoISF colour.

c. Information shown on each graph [↑ TOC]

Graph 0 information. The main graph can show measured and bucketed glucose, prediction traces, the shaded target range, bolus/carbohydrate treatments, effective-profile switches, therapy events, basals, temporary target, running modes and the now line. Optional analytical curves include insulin activity, empirical absorption, theoretical carb model, UAM and combined carbs, parabola/prediction, raw Libre and UKF-smoothed comparisons. Fixed annotations show the AAPS one-minute delta, UKF/Libre one-minute delta, HP/hypoglycaemia projection and IOB-peak information. Tapping a BG point or treatment displays its detail label.

Graph 1 information. In addition to its user-selected secondary series, graph 1 carries the target-offset/last-duration-taper row, stacked steps, the extra DR/AW/LS status row, noisy/raw-delta information, and the colour-segmented index row showing final, acceleration, BG, postprandial, duration and SMB values. These rows are fixed additions and therefore may appear even when the graph's principal selectable curve is something else.

The main graph no longer shows two fixed BGL-attached labels that used to be pinned at the current noisy-BG point (“L=.../A1=.../A5=.../MJ”) and at UKF1's own current point (“U=<5-min delta>”) -- removed to declutter the main trace; the underlying computation (used elsewhere, e.g. graph 3's HP2= row) is unaffected.

Graph 2 information. Graph 2 shows its selected secondary series and the per-SMB label/arrow layer. SMB label visibility follows the IOB long-press/Clean graph state, so the underlying numeric curve can remain while dose labels are hidden.

Graph 3 information. Graph 3 shows its selected secondary series plus the total-SMB-per-stack labels at the base of the panel. The older pp/acce/duration annotation row no longer belongs here; it has moved to Graph 5.

Graph 4 information. Graph 4 shows its selected secondary series. When the main Treatments switch is enabled it also shows CarePortal note text and matching arrowheads in the upper part of the panel, separating diagnostic notes from the SMB stack labels now placed on graphs 2 and 3.

Graph 5 information. Graph 5 provides a plain comparison picture: measured/bucketed glucose and predictions, shaded range, BG parabola, raw BG, all three UKF comparison lines, basals, target, running modes and now line. In full mode it also shows insulin activity, theoretical carb model, UAM carb impact and combined carbs. Its fixed annotation rows show target offset/last duration taper and pp/acce/duration weights. It intentionally omits treatments, therapy-event/note text, delta labels and the hypo-prediction row so marker text does not obscure the comparison traces.

d. What the different carbohydrate graphs mean [↑ TOC]

Carbohydrate entries and COB are not absorption-rate curves. A carbohydrate treatment marker records the grams entered at a particular time. COB is the estimated grams from entered carbohydrate that remain unabsorbed and is normally selected on a secondary graph. COB therefore describes a remaining quantity, whereas the curves below describe a rate or inferred effect in grams per five minutes.
Carbs Absorption (C-ABS) is the empirical known-carbohydrate absorption rate. It uses autosens this5MinAbsorption, is exponentially smoothed to reduce five-minute bucket noise, and contains historical points only because future absorption has not yet been observed. It is drawn as a solid measured-data line. On graph 0 its chart-menu checkbox is the master gate, while the basal/IOB long-press display preset can temporarily show or hide it.
Carb model curve is a theoretical forward curve calculated from entered carbohydrate, not a glucose-derived measurement. Each entry uses the two-compartment Gamma(2) form Ra(t) = 5 x carbs x 0.9 x k^2 x t x exp(-k x t), with k=1/90 per minute, a 90-minute peak and a six-hour cutoff. Contributions from overlapping entries are added. Because it can be calculated from known entries, it extends into the future and is drawn dashed. Its List 1 Graph2 toggle (relay code 5.138) is independent of the C-ABS checkbox.
UAM Carb Impact (UAMci) is the deviation-inferred carbohydrate-equivalent rate from the nearest AIV record, accepted when it lies within four minutes of the graph bucket and then exponentially smoothed. It represents glucose behaviour that looks like carbohydrate impact, including possible unannounced carbohydrate. It remains uncapped so an unusually large spike is visible diagnostically. It is historical only and is drawn as a dotted inferred-data line, distinct from the solid C-ABS line and dashed model curve.
Combined Carbs (CmbC) combines known empirical absorption with only the UAM amount above that already-explained absorption. Its implemented calculation is C-ABS + max(UAMci - C-ABS, 0), which is equivalent to max(C-ABS, UAMci). This avoids counting the same known meal response twice while retaining excess impact that may represent unannounced carbohydrate. It does not include the theoretical carb-model curve. It is historical, solid and shown in its own colour.
Scaling and interpretation. Each rate line is normalized against its own peak when placed with differently dimensioned series such as insulin activity. Equal apparent peak heights therefore support comparison of shape and timing—onset, peak and decay—but do not mean the lines have equal gram values. Read the underlying source and line style before interpreting vertical separation as a quantitative difference.
Where the curves appear. Graph 0 can independently show C-ABS, the carb model, UAMci and CmbC. UAMci and CmbC can also be assigned to secondary graphs. Graph 5 full mode shows the carb model, UAMci and CmbC, but not the empirical C-ABS line; its BGL only checkbox hides all three of those Graph 5 carbohydrate curves. Graph 5 ignores graph 0's individual visibility selections, although the global carb-model setting must be on for model data to exist.

10. Factors affecting progress and hypoglycaemia outcomes [↑ TOC]

a. Pod age and pump-type qualification [↑ TOC]

Pod-age-dependent paths use the time since the last cannula/site-change record only after confirming that the active pump reports itself as a patch pump. On a tubed pump or MDI setup the pod-age helper returns no value, so a normal infusion-site age cannot accidentally satisfy a fresh-pod or old-pod condition. For a patch pump, age can affect progression in several distinct ways: a pod older than 60 hours can widen the NewDay2 calibration tier when the other sensor-adjustment gates are present; sustained high glucose for two hours after 60 hours can produce the one-per-pod old-pod warning; and fresh/old-pod automation branches can alter profile, target or weighting only when all their additional glucose, state, bolus-age and timing gates also pass. The pump-type check is therefore required and is already performed centrally before any of these calculations receives a pod age.

b. Projected HP at bolus-calculator initiation [↑ TOC]

The Wizard calculation already includes a projected-HP safety assessment during each live recalculation, including the initial display and Quick Wizard calculations that use the same BolusWizard engine. It is a prospective bolus-specific HP, not the live APS HP2 value: projected HP = current glucose - positive IOB - the percentage-adjusted proposed bolus, plus only negative current and short-average delta contributions at one quarter weight. When projected HP is at most 6.5 mmol/L, glucose is below 10.0 mmol/L, both rise measures are at least 0.6 mmol/L per five minutes, positive IOB is at least 2.0 U, and the proposal is at least 0.75 U, COB-derived insulin is removed and the remainder is reduced to 75%. The recalculated lower dose is shown directly as the proposed dose before confirmation; it is not merely a warning offering a later manual reduction.

c. Effects of 50recent, a 50% profile and a hypo temporary target on progress [↑ TOC]

These are related but not interchangeable states. An active 50% profile changes the effective insulin-to-carbohydrate calculation and can arm the automatic delayed-bolus review. LowBG=50recent independently halves only the newly entered carbohydrate component when the active profile is not already 50%, preventing the two mechanisms from stacking into a quarter dose; it also activates the recent-low rebound protection used by SMB calculations. Glucose below 5.0 mmol/L and not rising can invoke the same carbohydrate-only halving even without the state. A hypo temporary target changes the BG-correction target and can block, postpone or cancel later split/delayed delivery when the follow-up safety gates no longer pass. During delayed progress, the active 50% profile or 50recent state can preserve eligibility for reassessment, but every later dose is recalculated against fresh glucose, profile, IOB, pump availability, newer treatments and the relevant deadline; none of these states guarantees that the remainder will be delivered.

d. Hypoglycaemia warning causes and outcomes [↑ TOC]

Warnings arise from several coded paths rather than one glucose threshold. GentleHypoRisk can be caused by agreement between low AAPS glucose and low/falling UKF raw glucose, by a falling 50% profile condition, or by HP2 at or below 4.1, with the additional requirement that the raw one- and five-minute directions are non-rising and that a low/HP2/activity qualifier is present. AlarmHypo1 covers a slow decline below 4.3 mmol/L, an emergency glucose below 3.0, a daytime steps-associated decline, or agreement between tightened HP2 and HP1 limits. AlarmHypo2 covers non-rising glucose at or below 4.3, glucose at or below 5.5 with substantial recent steps, or the same HP1/HP2 agreement. Outcomes are protective state and dosing-context changes as well as alerts: acceleration weighting is reduced, BGLstate is marked as recently low, AlarmHypo1/2 set LowBG=50recent, GentleHypoRisk lowers the IOB-threshold percentage, graph/CarePortal markers are written, and configured notifications/SMS messages are issued. Alert sound/SMS behavior can be suppressed in quiet hours while the protective state changes still occur, so the absence of an overnight audible warning does not mean that the hypo pathway did not act.

11. UKF: formulas, roles and initial comparison assessment [↑ TOC]

a. The UKF variants and where each is used [↑ TOC]

The labels describe different compositions of two operations - Unscented Kalman filtering and LibreSpecial exponential smoothing - rather than three interchangeable copies of one value. UKF1 is a batch, display-oriented UKF/RTS pass over the raw/noise series. Its graph can optionally apply the Libre slope and offset after filtering. UKFset1 is the live incremental counterpart that can replace the ordinary LibreSpecial path only on the non-Client Virtual-Pump comparison device; it maintains its own current state and short retrospective history. UKF2 applies the normal calibrated LibreSpecial EMA first and then refines that EMA result with UKF; its graph reads the stored post-UKF value actually used when UKF2 was live, not the earlier pre-refinement EMA. UKF3 reverses the order: it starts with UKF1, optionally applies Libre calibration, and then runs the LibreSpecial EMA. UKF3 is a display/diagnostic comparison and is not presently an independently selectable dosing source.

b. Formula and full explanation of each [↑ TOC]

Common UKF model. The two-element state is x = [G, r], where G is glucose in mg/dL and r is its rate in mg/dL per minute. For elapsed time dt, prediction uses G(next) = G + r x dt and r(next) = 0.98 x r. Merwe-scaled sigma points (alpha 1.0, beta 0.0, kappa 3.0) propagate the state and covariance through that model. The observation is the glucose reading itself, z = G + measurement noise. The innovation is z - predicted G; the Kalman gain determines how much of that difference corrects both glucose and rate. The estimated rate is constrained to -4 to +4 mg/dL/min. The full adaptive filter uses process covariance Q = diag(1.0, 0.40), scaled by dt/5, and adapts measurement variance R to sensor quality. The raw comparison paths deliberately use heavier fixed measurement variance R = 225 and Q = diag(1.0, 0.15), producing stronger smoothing without altering the live adaptive filter's learned state.

UKF1 formula and meaning. UKF1 applies the common UKF prediction/update pass to raw readings and then, for the retrospective graph, applies Rauch-Tung-Striebel backward smoothing: x_s(k) = x_f(k) + C(k)[x_s(k+1) - x_pred(k+1)], where C is formed from the filtered covariance, transition model and next predicted covariance. This uses later points to improve earlier displayed estimates, so the retrospective UKF1 graph must not be mistaken for a value that was fully available in real time. The live UKFset1 form performs only the causal incremental prediction/update for the newest point and persists that current state. A gap over seven minutes inflates uncertainty and decays the rate; a non-positive gap or one over 60 minutes starts a fresh state. Optional graph calibration is G_cal = max(40, slope x G_UKF1 + offset), with the offset converted to mg/dL when the displayed preference is in mmol/L.

LibreSpecial EMA formula. Let a be the configured smoothing factor, e the elapsed minutes, c the expected CGM interval (normally one minute; five for the relevant G7 path), and M the maximum smoothing gap. The effective factor is a_eff = min[1, a + (1-a) x {max(0,e-c)/(M-c)}^2]. The new value is S = S_previous + a_eff x (G_calibrated - S_previous); the first point uses G_calibrated directly. As the gap approaches M, a_eff approaches 1, preventing an old smoothed value from dominating after missing data.

UKF2 formula and meaning. UKF2 is UKF(EMA(calibrated Libre)). Each new reading first passes through the Libre calibration and the EMA formula above. That EMA value is appended to a rolling history; a UKF/RTS pass is recomputed over up to 90 minutes/120 points, and the refined value at the current timestamp becomes the UKF2 result. Its separate post-refinement history retains the values actually returned for dosing and supplies the UKF2 graph and delta calculations. Because calibration occurs before the EMA and UKF, UKF2 has no separate graph-calibration checkbox.

UKF3 formula and meaning. UKF3 is EMA(calibrate(UKF1)). Starting with UKF1's oldest-to-newest series, each point is optionally calibrated using max(40, slope x G_UKF1 + offset), then passed through the same gap-adjusted EMA. This is the opposite order from UKF2. Filtering and EMA do not commute: UKF2 lets the EMA shape what the UKF receives, whereas UKF3 lets the UKF shape what the EMA receives. Their different timing and curvature are therefore expected and are precisely what the comparison graph is intended to reveal.

c. Progress of the initial UKF-check assessment and earliest-change question [↑ TOC]

The initial observation was that the UKF1 graph appeared to react to a fall before LibreSpecial. The current code treats that as a hypothesis requiring repeated measurement, not as proof that UKF1 should replace the live source. On the non-Client Virtual-Pump comparison device only, it keeps the alternative histories warm and compares the loop value, UKF2, live-style UKFset1, a shadow LibreSpecial value, retrospective UKF1 and UKF3. Every five minutes it records each type's 5-, 15- and 30-minute deltas plus fitted acceleration, positive/negative parabola deltas, fit window and correlation quality. It separately warns when a type's meaningful acceleration sign disagrees with the live loop.

Earliest-change tracking defines a genuine falling crossing as delta5 at or below -0.2 mmol/L per five minutes. The loop crossing is checked every cycle; the five alternative-type comparisons are normally sampled by their five-minute diagnostic blocks. For each falling episode, the first crossing time for every type is retained in memory and later crossings are reported as seconds behind the first. Recovery above the threshold re-arms that type for the next episode. A dedicated LibreSpecial-versus-UKFset1 race removes interference from the other variants: the leader writes LibreLd or Set1Ld to CarePortal, while both lead/lag times remain in the log. The UKFcheck export gathers these metrics and race records during manual AIV export and the normal six-hour export cycle.

Use beyond the assessment. At present the earliest signal is diagnostic only: it does not automatically change the live glucose source, dose, temporary target or automation. A single early crossing may be genuine lead time, retrospective RTS advantage, or a false lead caused by raw noise. Permanent or external use should therefore wait for enough matched episodes to compare median lead time, missed declines, false alarms, low-point error and recovery lag using only values that were causally available at that moment. In particular, retrospective UKF1/UKF3 graph points must not be credited as real-time warning unless the same advantage is reproduced by the causal UKFset1 history. If those findings remain favorable, the live UKFset1 toggle is the appropriate candidate for a controlled trial; the diagnostic should not itself select whichever line happened to cross first in each episode.

d. UKF1 for live AutoISF dosing (toggle, added 2026-08-23) [↑ TOC]

ApsAutoIsfUseUkf1ForDosing (Settings + List 1 + List 2, TT 5.196). "UKF1" is used for two different things in this codebase, and only one of them is actually safe to run every dosing cycle -- worth distinguishing before anything else. UKFset1 (smoothRawRealtime(), toggled separately by the already-existing FslUseUkfSmoothing) is an incremental, forward-only, live-safe filter with its own persisted state. Literal UKF1 (smoothForDisplay(), used for the graph line and UKFcheck's own comparison) is a stateless BATCH pass with a backward RTS smoother, explicitly documented as display/comparison-only, never meant to run in the live pipeline. Real UKFcheck data showed UKFset1 was actually the WORST-lagging of five compared sources (+1007-2037s behind), while literal UKF1 was consistently fastest (always +0.0s) -- so this toggle wires in literal UKF1, the one the data actually supports, not the one that happened to already have a live toggle.

Single choke point, not two patched call sites. OpenAPSAutoISFPlugin.getGlucoseStatusData() -- which GlucoseStatusProviderImpl.glucoseStatusData calls app-wide whenever this plugin is the active APS -- was overridden (applyUkf1DosingOverride()) rather than patching invoke()'s and autoISF()'s own separate local reads. That means the Wizard/QuickWizard calculator, TriggerBg/TriggerDelta automation triggers, the Overview widget, wear/Tizen, and SMS "bg" replies all agree with the same UKF1-based number while the toggle is on -- a deliberate consequence: those all read the same "what's AutoISF's current glucose status" answer this plugin is the authority for, so silently disagreeing with AutoISF's own dosing math would be worse than being consistent with it.

Scope, deliberately partial. Only glucose/date/delta/shortAvgDelta/longAvgDelta/bgAcceleration/deltaPl/deltaPn/corrSqu are replaced. duraISFminutes/duraISFaverage (the separate 5%-band average-duration calculation) and a0/a1/a2 (the parabola's raw coefficients, read only by autoISF()'s own peak-BG prediction) stay LibreSpecial-EMA-based even with the toggle on -- judged out of scope for a first pass. Reuses the existing ukf1RecentHistory()/ukf1DeltaMetrics()/ukf1AccelerationMetrics() helpers built earlier for the UKFcheck comparison work, rather than new math.

Delta-threshold interpretation changes -- a real disadvantage, not just a data-source swap. Every fixed threshold in this fork (Tier 3's delta>=5/shortAvgDelta>=3/bg_acce>5.4, Mild/BG3 Boost's bands, the Fast-Rise cascade's tiers) was calibrated against real historical LibreSpecial-EMA Delta. UKF1 reacts with less inherent lag than an EMA, so the SAME threshold number trips earlier and more readily once the underlying signal changes -- confirmed directly, not just reasoned about: a real 23 Aug episode showed bg_acce/UKF1 crossing Tier 3's own 5.4 gate about 5-6 minutes before Delta/SDelta crossed theirs on the same rise (see the AIV review checklist's card 07b). This also means Boost-in-AAPS_3.4's reference-comparison logging (tier3BoostReferenceComparison(), still never wired to real delivery) shifts its own numbers the same way once this toggle is on, since it reads the same overridden glucose_status.

ApsAutoIsfUkf1DeltaCompensationSlope/Offset (Settings only, no List row). A slope*x+offset adjustment on delta/shortAvgDelta/longAvgDelta/bg_acce (offset applies to the three deltas only, not bg_acce -- an additive mg/dL-per-5min offset has no dimensionally sound meaning on an acceleration term), applied inside applyUkf1DosingOverride() itself so it reaches every consumer consistently -- Tier 3, Mild/BG3 Boost, autoISF()'s ISF weights, and Fast-Rise (which then still applies its own separate fastRiseSlopeCompensationRatio for LibreSlope drift on top -- the two are orthogonal and stack, not redundant). Default 1.0/0.0 (no-op) until real UKF1-mode AIV data justifies a real value; offset units are already mg/dL per 5min, matching Tier 3's own gate scale directly, unlike FslCalOffset which is auto-converted from mmol.

Reversion is full, immediate and stateless. Every function this override touches -- applyUkf1DosingOverride() itself, ukf1RecentHistory()/ukf1DeltaMetrics()/ukf1AccelerationMetrics() -- recomputes fresh from raw readings every call and never persists anything or touches the live smoothing pipeline's own state. Turning the toggle back off makes the very next cycle's getGlucoseStatusData() call return the ordinary LibreSpecial-EMA object exactly as if the toggle had never existed. The one thing reversion cannot undo is any SMB already delivered while it was on.

12. Battery 1% automation and profile recovery [↑ TOC]

a. Battery1pc: action at 1% or below [↑ TOC]

Battery1pc is checked with a 20-minute floor. It can fire only when the active profile percentage is 100%, Automation State Profile is PP130, C100 or AllOK, the receiving phone reports battery at or below 1%, and the active pump is not Virtual Pump. It deliberately has no pod/cannula-age condition: critical phone battery applies equally to patch and tubed live-pump systems. When it fires, it switches immediately to the named Current Profile50, raises the urgent Batt1% notification, adds the main-graph announcement Batt1% and CarePortal note Bt<1%, sends LowBattery by the ordinary SMS route and to the configured battery-alert numbers, and records the run. The profile switch makes the 100% precondition false on the next loop, providing an additional self-guard against repeated action.

b. BatteryOver1pc: restoration after recovery [↑ TOC]

BatteryOver1pc is checked with a five-minute floor. It requires battery above 1% and the exact current profile name Current Profile50. The profile-name test is intentional: a hypo-created 50% percentage on ProfileReal is not the battery profile and must not be cancelled merely because the battery recovered. The older Profile=Batt1% state test could never succeed because Battery1pc did not set that state. On recovery, the automation switches to the configured Standard Profile, sends AllOK Batt, sets Automation State Profile to AllOK, writes CarePortal note bat>1 and records the recovery run. There is no separate user confirmation; progression is Battery1pc reduction followed by BatteryOver1pc restoration once these exact conditions are observed.

13. SensorAge: aging/new-sensor Libre calibration compensation [↑ TOC]

SensorAge (OldSensorAdj in code) overrides the Libre calibration slope/offset (FslCalSlope/FslCalOffset) during known-unreliable stretches of a sensor's life, tapering back to the user's configured baseline everywhere else. Unlike most of this fork's automations it has no dosing effect of its own — it only reshapes the calibrated glucose value that everything else, including dosing, is computed from. Unlike the reminder cluster in §13e below, the core tier-compensation block carries no Virtual-Pump gate at all: it applies identically regardless of pump type.

a. What it compensates for and how the tiers are built [↑ TOC]

Sensors read less reliably at both ends of their life: very early (0–3 days, settling/compression-artifact inaccuracy) and later on (11–15 days, swinging high/low more easily than the usual slope/offset calibration accounts for). Each tier applies a slope/offset delta computed relative to the user's own configured baseline (ApsAutoIsfLibreSlopeOrig, default 0.72; ApsAutoIsfLibreOffsetOrig, default 1.4) rather than hardcoded absolutes, so retuning the baseline shifts every tier together. There are three severities on each side of the sensor's life: mild (slope −0.02 / offset +0.05), medium (−0.04 / +0.10), and extreme (−0.07 / +0.15).

b. The six age tiers and their symmetry [↑ TOC]

NewDay1 (under 1 day — the whole first calendar day, no separate sub-tier) applies the extreme adjustment; NewDay2 (1–2 days) applies the medium adjustment; NewDay3 (2–3 days) applies the mild adjustment. At the other end, tier "1" (11–13 days) applies the mild adjustment, tier "2" (13–14 days) the medium, and tier "3" (14–15 days) the extreme. The two ends deliberately mirror each other: day 0 matches the most extreme 14–15 day tier, day 1 matches 13–14, day 2 matches 12–13 — a sensor eases away from unreliability in the same shape at both ends of its life, not just the same direction. NewDay2 also fires purely on cannula age (over 60 hours — the same threshold the OldPod/RecentPod automations use for "pod is old"), OR'd with the normal 1–2 day sensor-age window: a fresh sensor on an aging pod is treated the same as a 1–2 day-old sensor even if the sensor itself is older. That OR only widens the NewDay2 window — it never narrows NewDay1, NewDay3, or the 11–15 day tiers, which remain sensor-age-only.

c. Two independent enable switches [↑ TOC]

ApsAutoIsfSensorAgeCodeEnabled (autoisf_sensor_age_code_enabled, default on) is the master switch for the whole routine: when off, the block does nothing but restore the configured baseline if it was left overridden. ApsAutoIsfOldSensorAdjEnabled (autoisf_old_sensor_adj_enabled, default on) is narrower — checked further downstream, once the master switch has already let the block run — and only gates whether a matched age tier is actually allowed to apply. Both can be toggled remotely by setting a signal temporary target: SensorAgeCodeToggleTT (5.156mmol) flips the master switch and writes SACOn/SACOff; SensorAgeToggleTT (5.006mmol) flips the narrower one and writes STgOn/STgOff. Both cancel the signal TT immediately so it never affects a real target (see §7e for the full code glossary).

The block logs sensorAgeDays, oldSensorTier, oldSensorEnabled, recentHighBglSeen (with its underlying raw mmol reading), oldSensorActiveNow and the current live FslCalSlope every cycle it reaches this far (sensorAgeCodeEnabled true, LowRaw24Cal not overriding) -- not only on a successful tier match, so a "should have engaged but didn't" cycle is visible too.

ApsAutoIsfSensorAgeCodeEnabled has a Settings-screen switch (placed directly above ApsAutoIsfOldSensorAdjEnabled, the master gate for the whole block) as well as its List 1 toggle ("SensorAgeCodeToggleTT", TT 5.156).

A real cross-device discrepancy (Virtual showing FslCalSlope stuck at baseline while Client/Live showed it correctly tiered, despite identical toggle states) suggests the tier-entry condition can flicker cycle-to-cycle rather than being cleanly broken on one device -- see the changelog for the full investigation; the diagnostic logging above is what's needed to catch it live.

List 1's “Live Libre slope/offset (view only)” row (§8b) surfaces exactly this live-vs-baseline gap without needing a settings export.

d. Entry/exit gating and interaction with LowRaw24Cal [↑ TOC]

LowRaw24Cal — a separate, unrelated automation that forces a fixed 0.75/1.3 override once raw Libre BGL has been continuously below 10.0mmol for 24 hours — takes priority whenever its own override is active: SensorAge’s entire block returns immediately without touching slope/offset while that’s true. Even when an age tier matches, entry additionally requires that raw Libre BGL has exceeded roughly 12.0mmol at some point in the preceding 24 hours (a rolling “last seen” timestamp, not a continuous-window check) — the tier compensation is calibrated against genuine high-BGL Libre-vs-reference divergence, so without recent evidence of that divergence the tiers are blocked entirely and slope/offset are pulled back to baseline instead, even on a sensor age that would otherwise qualify. The block carries no readyToRun() throttle — it is deliberately checked every cycle, the same as DelOff, so day-boundary transitions revert promptly; both the “apply a tier” and “restore baseline” branches are idempotent, so re-checking every cycle is harmless. Restoration compares the live slope/offset values directly rather than trusting the “active” latch alone, so it self-heals if that latch and the actual live values ever disagree — for example after an app restart or a settings import.

Real cross-device discrepancy found the same day, still not fully explained. Comparing settings snapshots at the same moment: Client and the real pump-driving device ("Live") both showed old_sensor_adj_active=true and a correctly tiered live FslCalSlope (0.70, stepped down from a 0.72 baseline), while Virtual showed old_sensor_adj_active=false and FslCalSlope stuck at the raw, never-adjusted 0.72 baseline -- despite identical toggle states (both enable switches on) on all three.

Live's own data later showed this ISN'T permanently broken -- a later snapshot from Live showed the SAME stuck-at-baseline pattern Virtual had shown earlier -- suggesting the underlying tier-entry condition (recentHighBglSeen/sensorAgeDays/the tier-boundary match) is genuinely flickering cycle-to-cycle on whichever device, not cleanly broken on one device only. The new diagnostic logging (this paragraph, above) is what's needed to catch a live failure with actual field values -- not yet captured, since the logging hadn't been built onto any device as of this writing.

e. Related sensor/pod-age reminders (notification-only) [↑ TOC]

Several informational siblings live in the same part of the file and are gated on similar age math, but they never touch calibration or dosing — they only raise a notification, SMS, or graph announcement, and all of them are live-pump-only (blocked on Virtual Pump), unlike the core compensation block above. PreSoakSensor24hrs reminds to pre-soak a new sensor at roughly 14.0 or 14.5 days into the current one’s life, skipped if the pod is also near end-of-life. SensorS1hr and SensorS2hr remind at roughly 14.9 days (“possible overlap”) and 14 days 22 hours. Pod2 and Pod1 fire at roughly 78 and 79 hours of pod age; Pod2 is the odd one out here, since unlike the other reminders it also performs a real action — switching to the Standard profile and setting automation state Profile=PP130 — rather than being purely informational. All of these use narrow ~0.1-hour match windows, so each fires close to its target age rather than repeatedly.

14. VirtualPseudoWizard: background real-bolus correction (Virtual Pump, testing only) [↑ TOC]

VirtualPseudoWizard is different in kind from Tier 3 UAM Boost, BolusGiven, and BolusGivenMild (§3b), which only ever strengthen the SMB delivery ratio. It places an actual BS.Type.NORMAL bolus, computed by running the same constrained BolusWizard used by the manual calculator, but entirely in the background — no calculator screen is ever shown, and no user confirmation occurs. It is a retrospective-testing tool, not a production dosing feature, and is hard-gated twice specifically so it cannot fire on a real pump: once in its eligibility check, and again immediately before the bolus is queued.

a. Why it is not a real carb treatment [↑ TOC]

The background Wizard call is given a “calculation carbs” figure (10g/15g/20g depending on how high glucose currently is) purely so the constrained Wizard has something proportionate to size the correction against. The bolus record it queues afterward is written with carbs=0 and eventType CORRECTION_BOLUS: no carbohydrate treatment is ever created, only the insulin delivery is real.

b. Entry gating [↑ TOC]

All of the following must hold: Virtual Pump active; UKF raw BGL above 12.0mmol; acceleration factor (bgAcce) above 3.0; short-average delta above 0.5mmol per 5 minutes; every recent BG reading over the last 90 minutes above 8.0mmol; either (dura active over 5 minutes AND acceleration active over 4 minutes) OR (HP2 above 7.5mmol AND current BG above 10.0mmol) as an alternate entry path; zero meal COB; no active temporary target; a steps gate (under 600 steps in the last 60 minutes, or under 1500 if BG is above 11.0mmol); a computed hypo-prediction (HP2) that is both present and above 7.5mmol; no NORMAL, non-SMB bolus in the last 120 minutes; a 90-minute per-action cooldown; and no bolus already queued. Every cycle writes a full READY/blocked diagnostic line naming which of these held or failed — this is what let a real "why didn't it fire" question get answered from the logs alone earlier in this document's own history.

c. Calculation and delivery mechanics [↑ TOC]

The Wizard call uses useBg=true, useCob=false (the calculation-carbs figure is passed as carbs, not COB), includes both bolus and basal IOB, and disables superbolus, temporary target, trend, and alarm inputs. The resulting insulinAfterConstraints is what actually gets queued. Immediately before queueing, the code checks activePlugin.activePump is VirtualPump a second time, independently of the eligibility check above — so even if the first gate were ever satisfied on a real-pump build, this second check still blocks delivery.

d. Outcome reporting [↑ TOC]

An in-memory status string (virtualPseudoWizardLastStatus) records the most recent SUCCESS/FAILED outcome and is echoed into the console log every cycle alongside the READY/blocked diagnostic line, so the current state is always visible without waiting for the next attempt. As of 2026-08-21, SUCCESS, FAILED (pump/callback rejection), and FAILED (bolus queue rejected the request) each also write a CarePortal note — previously this pathway had no durable CarePortal trail at all, only app logs.

15. Fast-Rise SMB Tapering: the Tiered Cascade and the Separate Late Taper [↑ TOC]

a. Where this sits, and the entry gate [↑ TOC]

The whole fast-rise system lives inside DetermineBasalAutoISF.kt's determine_basal(), in a single long if/else if chain that decides how far the ordinarily-calculated microBolus gets trimmed for the current cycle. Only one branch of that chain can act per cycle. Reading it top to bottom, the branches immediately before FAST RISE HANDLING are: Shower/Twilight AM Protection, Low IOB Acceleration Glitch Throttle, Sensor Glitch/Swing Damping, Sensor Glitch2/Swing Damping, Post Carbs/Swing Damping, High TT Protection, Glitch Zero SMB, and finally Sensor Glitch3/Short Spike Without Trend — each covered in §d below. FAST RISE HANDLING itself is the next else if after Sensor Glitch3, gated on libreActive && !profile.temptargetSet: it only evaluates at all when a Libre sensor is active and no temporary target is set.

Before any of this, the cycle computes ThresholForFastRise = TDDfactor * 0.030 * profile.max_iob, rounded to two decimals. This is the recurring yardstick the tiers below use to decide whether a candidate SMB is "still large enough to matter" after the Delta/SDelta bands have already flagged a rise. At a typical max_iob of 9.5 U and TDDfactor of 1, it works out to roughly 0.29 U — a small fraction of max_iob, deliberately: it is meant to catch ordinary-sized SMBs, not just outsized ones. TDDfactor scales it with the user's recent total daily dose, so the same guard tightens or loosens automatically as insulin needs change.

Inside the libreActive && !profile.temptargetSet branch, the tier ladder itself (§b) is gated by its own entry condition: current BG between 6.0 and 12.0 mmol/L; COB no more than 25 g; smoothed Delta at least 0.25 mmol per 5 minutes; smoothed SDelta at least 0.10 mmol per 5 minutes; and — as an OR-clause — either IOB above 0.12× max_iob (widened from 0.10, per an in-code comment, to let slightly more daytime IOB accumulate before this whole capping cascade starts applying) or the current hour is in the 22:00–05:00 night window. On top of all that, the entry gate also requires the 5-minute raw delta (rawDelta5Mgdl) at or above 0.25 mmol per 5 minutes AND the AAPS-processed 1-minute delta (aapsDelta1Mgdl) at or above the same 0.25 mmol.

rawDelta1Mgdl (the single-minute raw, unsmoothed delta) does not appear in any Fast-Rise gate condition; only rawDelta5Mgdl (5-minute raw delta) remains as the raw-signal check. aapsDelta1Mgdl is unrelated and remains part of the entry gate. (Exports taken before 2026-08-22 predate this removal -- see the changelog if reconciling an older export.)

Per an in-code comment, these raw/AAPS delta confirmations apply only at each branch's entry point — the nested BG sub-branches inside the tiers below do not re-check them.

As of 2026-08-22, rawDelta1Mgdl — the single-minute *raw, unsmoothed* delta — was removed from every gate in this cascade, and the code now carries an explanatory comment at the point where the old check used to sit. The removal was driven by real AIV data from two same-day sustained-rise events (01:50–01:58 and 02:50–02:55): at every single sampled minute of both events, rawDelta1Mgdl read negative or flat (as low as -0.56 mmol) even while Delta, SDelta and rawDelta5Mgdl all clearly confirmed a genuine, sustained rise the whole time.

A single unsmoothed one-minute raw reading is simply too noise-prone to be a reliable confirmation on top of signals that already agree — requiring it was silently blocking the entire cascade during exactly the kind of gradual sustained rise it exists to catch. rawDelta5Mgdl (the 5-minute raw delta) was kept, since it passed cleanly at both real events and still gives a raw-signal check independent of the smoothed AAPS delta.

Checking the current source confirms the removal is complete: rawDelta1Mgdl now appears only as a function-parameter declaration (default sentinel 9999.0, meaning "no data supplied") and in this explanatory comment — it is not referenced in any condition anywhere in the file any more. aapsDelta1Mgdl, by contrast, remains part of the entry gate (and several sibling branches' gates) throughout.

b. The tier ladder: Delta/SDelta bands, BG sub-branches, and multipliers [↑ TOC]

Once the entry gate above passes, the cascade tries, top to bottom, four severity tiers (only one fires, whichever matches first):

Extreme fast rise. Delta, SDelta and LDelta all at or above 1.0 mmol per 5 minutes. microBolus is multiplied by 0.2 — the deepest cut in the ladder. Logged as "CHANGED SIZE 0.201 for fast rise 0.201 smb".

Moderate fast rise. Delta between 0.55 and 1.0 mmol, SDelta between 0.30 and 1.0 mmol. This tier then branches again on current BG: above 8.8 mmol, microBolus is multiplied by 0.6 (logged "CHANGED SIZE 0.602 for moderate fast rise 0.602"); between 8.0 and 8.8 mmol, the multiplier is 0.65 (logged "...0.653..."); at or below 8.0 mmol, if microBolus still exceeds ThresholForFastRise OR the hour is 8am or earlier, the multiplier is 0.5 (logged "...0.504..."); otherwise, at or below 8.0 mmol with microBolus already under threshold, nothing changes (logged "smbUn 0.564..."). (Fixed 2026-08-22: the two non-label reason-text lines here used to read a stale "* 0.8"/"* 0.7" that no longer matched the 0.6/0.65 actually applied — both now read correctly.)

Mild fast rise. Delta between 0.25 and 0.55 mmol, SDelta between 0.15 and 0.55 mmol. Again branching on BG: above 8.8 mmol, multiplier 0.9 (logged "CHANGED SIZE 0.900 for mild fast rise 0.900"); between 8.0 and 8.8 mmol, multiplier 0.85 (logged "...0.850..."); at or below 8.0 mmol, if over threshold OR the hour is between 3am and 8am inclusive, multiplier 0.75 (logged "...0.750..."); otherwise unchanged (logged "smbUn 0.707..."). (Fixed 2026-08-22: these three labels used to lag behind an earlier loosening of the real multiplier from 0.85/0.8/0.7 to 0.9/0.85/0.75 — corrected so the logged numeral now matches what's actually applied.)

Early fast rise. A narrower Delta band, 0.25 to 0.35 mmol, with SDelta at or above 0.10 mmol (this branch sits as a sibling else if off the main entry gate rather than nested inside it, so it only evaluates when the main entry gate above did not match — effectively it re-checks rawDelta5Mgdl/aapsDelta1Mgdl at 0.25 mmol directly rather than inheriting the gate's BG/COB/IOB-or-night conditions). If microBolus exceeds threshold OR the hour is 8am or earlier, multiplier 0.7 (logged "CHANGED SIZE 0.700 for early fast rise 0.700"). Otherwise unchanged (logged "smbUn 0.608...", a separate no-change branch whose own label was not part of this fix). (Fixed 2026-08-22: the CHANGED SIZE label here used to read a stale 0.608, reflecting a pre-loosening 0.6 multiplier — corrected to 0.700, matching the 0.7 actually applied.)

Across the ladder, the "was X" comments show these four tiers (moderate's two upper-BG branches, plus all three mild branches, plus the early tier) each had their real multiplier loosened at least once. As of 2026-08-22, all six affected reason-text lines (moderate's two non-label lines, mild's three labels, and the early tier's label) have been corrected so every logged numeral matches the multiplier actually applied — see §f for how this changes what the AIV export shows going forward, and for the exact pre-fix values you'll still see in any export taken before that date.

c. Other branches inside FAST RISE HANDLING: Higher BG, early-morning guard, twilight hours [↑ TOC]

Three further else if branches sit alongside the tier ladder, inside the same libreActive && !profile.temptargetSet block, each with its own independent entry conditions rather than inheriting the main entry gate:

Higher BG Fast Rise. Fires when Delta is at least 0.9 mmol, SDelta at least 0.7 mmol, BG is between 11.5 and 13.5 mmol, COB no more than 25 g, rawDelta5Mgdl and aapsDelta1Mgdl both at least 0.9 mmol, and current IOB exceeds ThresholForFastRise. (Fixed 2026-08-22: this gate used to read "IOB > ThresholForFastRise * profile.max_iob", multiplying max_iob into the comparison a second time — ThresholForFastRise already has one factor of max_iob baked in — yielding a gate roughly 10x higher than intended, about 2.7 U instead of about 0.29 U at typical settings. Git history at commit 320.010 confirmed this was an accidental artifact rather than intended design: the line originally read "IOB > lower_SMB * 0.30 * max_iob", and ThresholForFastRise was substituted in for "lower_SMB * 0.30" without dropping the now-redundant trailing "* max_iob". The extra factor has been removed, restoring the same ~0.29 U meaning ThresholForFastRise carries everywhere else it's used — this branch will fire considerably more readily than it did before the fix.) When it fires, microBolus is multiplied by 0.75 (logged "CHANGED SIZE 0.759 for fast rise 0.759 smb").

Early-Morning Extra Fast-Rise Guard (rev 2, raw-delta based). Active 6am–10am, requires IOB above 0.075×max_iob, COB no more than 25 g, immediateRawDelta5Mgdl (the always-unsmoothed 5-minute raw delta) at least 0.5 mmol, and either rawDelta15Mgdl is at its 9999 "no data" sentinel, or rawDelta15Mgdl is at least 0.2 mmol AND immediateRawDelta5Mgdl exceeds 1.5× rawDelta15Mgdl. An in-code comment explains the history: an earlier smoothed-Delta-based version (plus an unconditional pre-4am IOB-only branch) was disabled because the IOB-only branch capped SMB on IOB alone with no rise required — much blunter than intended. This revival dropped that branch and re-triggers on raw 5-minute delta instead, because a real incident (22 July, ~9:41am) showed smoothed Delta peaking at only 0.26 mmol — never enough to fire the old 0.3 trigger — while raw 5-minute delta hit 0.83 mmol and raw 1-minute hit 1.69 mmol in the same window; smoothed Delta was structurally too damped to catch a brief, real bump. The rawDelta15Mgdl check is corroboration that the move is sustained rather than one noisy raw reading, and the "immediate 5-min exceeds 15-min × 1.5" comparison is meant to catch a fresh reversal (e.g. off a preceding fall) specifically. The window was widened from an original 8–9am draft to 6–10am because the Shower/Twilight block (which covers 5am–10am, see below and §d) requires low step counts, and ordinary movement around a real shower can exceed that — this guard has no step gate, so it backstops Shower's blind spot across most of its window. When it fires, microBolus is multiplied by 0.8 (logged "CHANGED SIZE 0.810b early-AM immediate-raw-rise guard: rawD5 ... rawD15 ...").

Twilight / Other Hours SMB Limiting. 6am–8am inclusive, BG under 9.0 mmol, Delta and SDelta both under 1.0 mmol, Steps60M under 10, COB no more than 25 g. Two independent caps apply here rather than a single multiplier: if microBolus exceeds 0.02×max_iob, it's capped to exactly that (logged "CHANGED SIZE 0.211 Twilight fast rise 0.211..."); separately, if microBolus + IOB would exceed 0.075×max_iob, microBolus is reduced to 0.075×max_iob − IOB (logged "CHANGED SIZE 0.7512 fast rise 0.7512...", plus a trailing "CHANGED SIZE SMB other hours" marker). Both checks can fire in the same cycle since they are independent ifs, not an if/else.

If none of the tier ladder or these three branches match, the cascade falls through to an explicit no-op branch, logged simply "NOT CHANGED SIZE SMB".

d. The preceding sibling branches, including Sensor Glitch3 [↑ TOC]

As noted in §a, FAST RISE HANDLING is one link in a longer if/else if chain, and several earlier links in that same chain are mutually exclusive with it — if one of these fires, none of §b/§c can also fire that cycle. In source order: Shower/Twilight AM Protection (5am–10am, two escalating Delta/SDelta bands, hard-caps microBolus to 0.02×max_iob and IOB+microBolus to 0.075×max_iob, reusing the "fast rise" phrase in its own reason text at labels 0.713/0.715/0.716); Low IOB Acceleration Glitch Throttle (bg_acce above 0.30 mmol, Delta at least 0.50 mmol, IOB under 0.70 U → ×0.7, label 0.712); Sensor Glitch/Swing Damping (Delta at least 0.40 mmol with SDelta at most half of Delta and LDelta at most a tenth of Delta, COB≤20, bg<10.0 mmol → ×0.5, label 0.511); Sensor Glitch2/Swing Damping (Delta≥0.50 mmol, SDelta≥0.40 mmol, LDelta≤0.15 mmol, bg<10.0 mmol → ×0.5, label 0.512); Post Carbs/Swing Damping (Delta between 0.50 and 0.95 mmol, evening hours 18:00+, COB>10 → ×0.7, label 0.713); High TT Protection (Delta≥0.25 mmol, SDelta≥0.20 mmol, bg<10.0 mmol, an active temp target with target_bg≤4.1 mmol, plus the same rawDelta5Mgdl/aapsDelta1Mgdl≥0.25 mmol confirmations used in the main fast-rise entry gate → ×0.5, no "fast rise" phrase in its label); Glitch Zero SMB (Delta>1.0 mmol but LDelta<−0.05 mmol and bg<162 mg/dL → zeroed outright); and finally Sensor Glitch3 / Short Spike Without Trend immediately before FAST RISE HANDLING: SDelta≥0.25 mmol, LDelta≤0.08 mmol, and SDelta at least 3× (LDelta+0.01) — a short-term spike with no supporting longer-term trend behind it → ×0.5 (label "CHANGED SIZE fast rise 0.513 short spike no trend"). Only after all of these fail to match does the chain reach the libreActive && !profile.temptargetSet branch containing FAST RISE HANDLING itself.

e. The separate late taper and cumulative SMB caps [↑ TOC]

Further down determine_basal() — after the tiered cascade above, after the anti-stacking trim, and after the "recent delivery boost" restore that undoes the fast-rise caps when a BolusGiven/BolusGivenMild event fired in the last 30 minutes — sits a completely different mechanism, explicitly commented as running "after smbBoostRecent restoration so that boost cannot undo it." This is NOT part of the tiered ladder in §b/§c: it does not use ThresholForFastRise, LDelta, or any BG-banded sub-branching, and it keys on cumulative delivered amount rather than the current cycle's entry criteria.

A boolean fastRiseNow is computed: libreActive (note: no !profile.temptargetSet check here, unlike the tiered cascade's outer gate) AND bg between 6.0–12.0 mmol AND COB≤25 AND Delta≥0.25 mmol AND SDelta≥0.10 mmol AND rawDelta5Mgdl≥0.25 mmol AND aapsDelta1Mgdl≥0.25 mmol. This mirrors the tiered cascade's own entry-gate thresholds almost exactly, but omits both the temp-target check and the IOB-or-nighttime OR-clause — it is evaluated independently, not reused from the earlier gate's result.

When fastRiseNow is true and microBolus>0, a lateFastRiseFactor is chosen purely from smbSum30Min (total SMB units delivered in the trailing 30 minutes, tracked by smbSum30Min() in OpenAPSAutoISFPlugin.kt): at or above 1.9 U delivered, factor 0.50; at or above 1.5 U, factor 0.75; otherwise 1.0 (no change). If the factor is under 1.0, microBolus is multiplied by it, logged as "late FastRise SMB30 <sum>U: x<factor> <before> -> <after>". Immediately after, a hard 30-minute cumulative cap of 2.1 U is applied regardless of fastRiseNow: smbAllowance30Min = (2.1 − smbSum30Min).coerceAtLeast(0.0), and if microBolus exceeds that remaining allowance it is trimmed down to it (logged "30min SMB cap: ..."). An in-code comment explains this is deliberately not gated on fastRiseNow staying true: at the end of a real event (8 Aug) the raw rise flattened for two cycles while SMB continued, exactly when the already-accumulated insulin still needed protecting.

Separately again — positioned last, after every other modifier including the smbBoostRecent restore, so nothing can bypass it — sits the rolling 10-minute cumulative cap already documented elsewhere in this manual's overnight safety-guard material: smbCap10Min is 0.6 U inside the 00:30–04:00 deep-night window (inDeepNightSmbWindow) or 1.5 U outside it, trimming microBolus to whatever remains of that window's allowance (smbAllowanceLeft = smbCap10Min − smbSum10Min). This 10-minute cap is unconditional — it does not check fastRiseNow, libreActive, or any Delta/SDelta band at all — so it is a general anti-stacking backstop rather than a fast-rise-specific mechanism, even though it sits physically adjacent to the 30-minute late taper described above.

In short: the tiered cascade in §b/§c decides, per cycle, from live rate-of-change and BG-band evidence, how hard to trim a fresh SMB candidate; this late mechanism instead looks backward at how much insulin has already gone out over the last 30 (or 10) minutes and trims or caps regardless of which — if any — tier fired that cycle.

f. What actually reaches the AIV "FastRise" column [↑ TOC]

AutoIsfHistoryExporter.kt's exactFastRiseStr() populates the AIV table/CSV "FastRise" column by finding the nearest APSResult within 15 minutes of a given timestamp and running two regexes, in order, against its reason text: fastRiseFullRegex = Regex("""fast\s*rise\s*([0-9]+\.[0-9]+)""", RegexOption.IGNORE_CASE) is tried first; if it matches, the captured decimal (e.g. 0.507) is reformatted by stripping the decimal point and leading zeros (0.507 → "507"). If it doesn't match, fastRiseFactorRegex = Regex("""=\s*microBolus\s**\s*([0-9.]+)\s*;""") is tried as a fallback, and its captured factor is multiplied by 10 and rounded (e.g. a bare "* 0.8 ;" → "8"). If neither matches anywhere in the nearest result's reason text, the column shows "--".

Because IGNORE_CASE is set and \s* allows zero whitespace, fastRiseFullRegex matches the literal word "FastRise" (no space) just as readily as "fast rise" — but it still requires a decimal number to immediately follow (after only whitespace), so the late taper's own text ("late FastRise SMB30 1.95U: x0.5 ...") does NOT match: what follows "FastRise" there is "SMB30", not a digit. Neither does it match the fallback pattern, since the late-taper code assembles its factor with *= and a different phrasing ("x0.5"), never the literal "= microBolus * 0.5 ;" substring the fallback regex looks for. This confirms the prompt's expectation directly: the late 30-/10-minute mechanism in §e genuinely does not feed the FastRise column, however close its name looks.

What the two regexes *do* pick up is broader than just the tiered cascade in §b/§c, however. Every "CHANGED SIZE ... fast rise X.XXX" label anywhere in the file matches fastRiseFullRegex regardless of which top-level branch produced it — which means the sibling branches in §d (Shower/Twilight AM Protection's 0.713/0.715/0.716, Low IOB Acceleration Glitch Throttle's 0.712, Sensor Glitch's 0.511, Sensor Glitch2's 0.512, Post Carbs's 0.713, and Sensor Glitch3's 0.513) all also populate the FastRise column, as do the tiered ladder's own labels in §b (0.201, 0.602/0.653/0.504, 0.855/0.806/0.707, 0.608) and the §c branches (Higher BG's 0.759, the early-morning guard's fallback-matched "8" via the second regex since its own label text reads "raw-rise guard" rather than "fast rise", and Twilight/Other Hours's 0.211/0.7512). By contrast, High TT Protection and Glitch Zero SMB do not use the phrase "fast rise" at all and their reason text does not match the fallback pattern either, so a cycle where only one of those two fired shows "--" in the FastRise column (unless a nearby cycle within 15 minutes produced a match instead). So the precise statement is: the FastRise column reflects any branch's own logged text that happens to contain "fast rise" followed by a decimal (or, failing that, the generic "= microBolus * factor ;" phrasing) — which in practice covers most, but not literally all, of the branches that trim microBolus for a rise-like reason, and never the late 30-/10-minute mechanism.

All FastRise reason-text/label numerals match the multiplier actually applied as of 2026-08-22 -- see the changelog if reconciling an export taken before that date, since several branches' labels were stale relative to their real multiplier before the fix.

One further wrinkle worth flagging for anyone reading exported FastRise values from before 2026-08-22: for four of the tiered-ladder branches — mild fast rise's three BG sub-branches (0.9/0.85/0.75) and the early fast-rise tier (0.7) — the actual multiplier had been loosened from an earlier value via an inline "was X" comment, and the *first* reason-text line for each was updated to match the new multiplier, but the "CHANGED SIZE X.XXX for ... fast rise X.XXX" label's own numeral was not — so any AIV export taken before 2026-08-22 shows the *pre-loosening* value (0.855, 0.806, 0.707, 0.608) rather than the multiplier genuinely applied that cycle (0.9, 0.85, 0.75, 0.7 respectively) for those four branches.

The moderate tier's two upper-BG branches had the opposite mismatch — their labels (0.602, 0.653) already tracked their current 0.6/0.65 multipliers correctly, but the *other*, non-label reason-text line in those two branches was stale (still reading "* 0.8" and "* 0.7"). All six were corrected 2026-08-22 so every logged numeral now matches what's actually applied — but if you're reconstructing historical FastRise values from an export taken before that date, the pre-fix numbers above are what you'll actually see in the file, not the current code's behavior.

g. Libre-calibration slope compensation (added 2026-08-22) [↑ TOC]

The cascade's mmol-based Delta/SDelta gates (§a-§c above) are computed from calibrated BG (BG = raw × FslCalSlope + FslCalOffset); since a constant offset cancels in any delta, Delta_calibrated ≈ FslCalSlope × Delta_raw. A real preferences pull from the live phone on 12 Jul 2026 showed fslCal_Slope = 0.75 -- the era these thresholds were tuned against -- while a mirrored snapshot on 22 Aug 2026 showed the baseline itself had since moved to fslCal_Slope = 0.72 (and it drops further, to baseline−0.07, during OldSensorAdj's Day-0/Day-1 tiers -- as low as 0.65 at a 0.72 baseline). The same real rate of BG rise therefore now reads a measurable amount smaller than it did when these thresholds were assessed, silently under-powering the calibrated-delta gates.

determine_basal() now takes a fastRiseSlopeCompensationRatio parameter (default 1.0, i.e. no change, for existing callers/tests). OpenAPSAutoISFPlugin.kt's fastRiseSlopeCompensationRatio() computes it live each cycle as 0.75 ÷ current fslCal_Slope, except on a true Virtual pump (VirtualPump and not AAPSCLIENT) where it is forced to 1.0 -- Virtual never ran a real Libre calibration at all (FslCalSlope sits at its neutral 1.0 passthrough there), so 0.75 isn't a meaningful reference to compensate against; Client is NOT excluded, since it mirrors a real device's already-calibrated data. Inside DetermineBasalAutoISF.kt, this ratio divides a local shadowed copy of Delta/SDelta/LDelta declared immediately before the SHOWER/TWILIGHT branch and valid only for the lifetime of that one if/else-if cascade (ending at the "DEFAULT: NO SMB SIZE CHANGE" branch, just before the separate HIGH STEPS gate resumes using the real, uncompensated values) -- every bare Delta/SDelta/LDelta reference already inside that cascade picks the compensated value up automatically, and nothing outside it (ISF weights, TT handling, the later independent late-taper mechanism in §d, etc.) is affected.

Deliberately NOT extended to basal rate or SMB delivery ratio: the mmol thresholds measure glucose kinetics (rate of BG change), a property independent of how much insulin the pump delivers, and ThresholForFastRise already scales dynamically with TDDfactor/max_iob wherever the cascade compares against an insulin amount rather than a rate of rise. Likewise microBolus -- what the tiered multipliers act on, and what ThresholForFastRise itself is compared against -- is already computed downstream of ApsAutoIsfSmbDeliveryRatio, so both the taper's proportional reduction and that comparison already scale correctly with whatever delivery-ratio setting is active, with no extra compensation needed.

Flavor	Application ID	Icon	Role
full	info.nightscout.androidaps	standard AAPS	The real, pump-driving install. This is what "Regan" / the primary phone runs.
pumpcontrol	info.nightscout.aapspumpcontrol	pumpcontrol icon	Stock AAPS pump-control-only variant; not actively used by this fork's own device fleet, but still built.
aapsclient	info.nightscout.aapsclient	yellow owl	"Client"-style mirror. Runs its own full loop calculation every cycle (same as full)

Flavor	Application ID	Icon	Role
			but with VirtualPump as its selected pump, so nothing it computes is ever actually delivered — it's a live shadow-loop, not just a passive NS viewer.
aapsclient2	info.nightscout.aapsclient2	blue owl	A second, independent Client mirror — same mechanism as aapsclient, distinct app/prefs, lets one phone run two separate mirror instances.

Appendix C — Contents [↑ TOC]

The section-by-section outline used to be duplicated here. Removed 2026-08-24 as redundant -- use the main Contents near the start of this document for the full 1-16 outline: [Main Contents]

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Auto ISF — top level TOC

Setting	Default (range) [type]	Detailed description	Current setting (Client)
MJ Kotlin buttons	on [switch]	Shows a direct "Start MJ" / "Restore MJ" button on the Overview screen instead of requiring the native Automation tab. MJ ("Morning Jog"-style state machine, per this fork's naming) is a coded state that several other automations key off (checkAutomationState("MJ", ...)) to gate profile-switch and boost behaviour. Purely a UI convenience toggle — turning it off does not disable the underlying MJ automations, only the quick-access buttons.	on
Steroid Kotlin button	on [switch]	Shows a direct "Steroids ON" button on Overview instead of requiring the native Automation entry. Feeds the steroid-adjustment automations (which progressively raise iobTH/ISF stringency while steroids are active) via the same EventSteroidUserAction path the native automation trigger would use. UI convenience only — the steroid automations themselves are controlled elsewhere.	on
Half basal exercise target	160 (120–200) [number (unit-aware)]	Defines the exercise-sensitivity curve's steepness: at this TT level (with "High temptarget raises	124 mg/dL / 6.90 mmol/L

Setting	Default (range) [type]	Detailed description	Current setting (Client)
		sensitivity" on), basal runs at exactly 50%. Other TT levels interpolate along the same curve — e.g. 120 → ~75% basal, 140 → ~60% basal — so lowering this value makes the sensitivity response kick in more aggressively at lower TT levels.	

Activity Monitor TOC

Setting	Default (range) [type]	Detailed description	Current setting (Client)
Scale factor for final activity sensitivity	1.0 (0.0–1.5) [number]	Multiplies activity's default 15–30% sensitivity contribution up or down. Setting to 0.0 disables activity's contribution entirely while leaving inactivity's contribution (below) untouched.	Not in snapshot
Scale factor for final inactivity sensitivity	1.0 (0.0–1.5) [number]	Same idea for inactivity's default 10–20% contribution. 0.0 disables just the inactivity side.	Not in snapshot

AutoISF settings → Libre special settings TOC

Setting	Default (range) [type]	Detailed description	Current setting (Client)
Offset of Libre calibration versus finger (fslCal_Offset)	0.0 (-50.0–50.0) [number]	The OFFSET term in this fork's own Libre-vs-reference calibration formula: Libre_calibrated = OFFSET + SLOPE × Libre_raw. Same style of two-point calibration xDrip+ itself does, applied here instead.	1.35
Slope of Libre calibration versus finger (fslCal_Slope)	1.0 (0.5–1.5) [number]	The SLOPE term in the same formula. Together with the offset above, this is the LIVE calibration pair actually applied to raw Libre values — it's what "Aging-sensor calibration adjustment" below temporarily overrides during its tiered windows.	0.74
Libre 1st order exponential smoothing factor	0.3 (0.1–1.0) [number]	Controls how aggressively raw Libre readings are exponentially smoothed before use, when not relying on xDrip+'s own 1-minute smoothing. 0.0 = maximum smoothing effect, 1.0 = minimum (closest to raw).	Not in snapshot
Base Libre cal slope (OldSensorAdj)	0.72 (0.5–1.5) [number]	The baseline all of Aging-sensor calibration's tiers subtract from: −0.02 (mild, days 2-3 fresh / 11-13 aging), −0.04 (medium, day 1-2 fresh / 13-14 aging), −0.07 (extreme, day 0 fresh / 14-15 aging). Retuning this one value shifts every tier together, rather than needing to update several independent hardcoded numbers.	0.74

Setting	Default (range) [type]	Detailed description	Current setting (Client)
Base Libre cal offset (OldSensorAdj)	1.4 (-50.0–50.0) [number]	The equivalent baseline for offset: the same tiers add +0.05/+0.10/+0.15 to this value for the mild/medium/extreme windows.	1.35

AutoISF settings → weight system TOC

Setting	Default (range) [type]	Detailed description	Current setting (Client)
Max autoISF ratio (autoISF_max)	1.0 (1.0–3.0) [number]	Ceiling on how strong the combined adjustment can make ISF. 1.0 disables the strengthening side. Values above 2.5 are explicitly flagged as requiring extreme caution — a large ISF strengthening directly increases how much insulin a given dose delivers.	2.60
Weight while BG accelerates (bgAccel_ISF_weight)	0.0 (0.0–1.0) [number]	Strengthens ISF while glucose is accelerating (rate-of-rise increasing), which potentially increases SMB sizing. 0.0 disables this contribution. This is the LIVE value multiple automations write to transiently (during boosts) and restore afterward — see the two settings immediately below.	0.50
Resting acceleration ISF weight	0.70 (0.0–1.0) [number]	The value RecentPodOff restores bgAccel_ISF_weight to once its own trigger conditions clear — your normal/baseline acceleration weight, distinct from the live value above which gets overwritten transiently.	0.70
Boosted acceleration ISF weight	0.95 (0.0–1.0) [number]	The value OldPod2/RecentPod set bgAccel_ISF_weight to while their own boost conditions are active, and what the AcceUp0.5 automation now also targets. Previously a hardcoded literal in three separate places in the code; now one live, user-tunable setting.	0.95
Higher ISF-range weight (higher_ISFrange_weight)	0.0 (0.0–2.0) [number]	Strengthens ISF while glucose is above target, increasing delivery the further above target BG runs. 0.0 disables it.	0.80
Postprandial ISF weight (pp_ISF_weight)	0.0 (0.0–0.15) [number]	Strengthens ISF while glucose is actively rising (the "pp" = postprandial signal), increasing SMB sizing during a rise. 0.0 disables it. Like the acceleration weight, this is a LIVE value multiple boost automations overwrite transiently and restore afterward — see the two settings immediately below.	0.10
Resting postprandial ISF weight	0.08 (0.0–0.15) [number]	The value boost/recovery automations restore pp_ISF_weight to once a boost window ends.	0.10
Boosted postprandial ISF weight	0.15 (0.0–0.15) [number]	The value the fast-rise boost automations (BolusGiven, BolusGivenMild, High6PP, HighOldPod, PodChangeHighPP130, OldPod2,	0.15

Setting	Default (range) [type]	Detailed description	Current setting (Client)
		RecentPod) set while actively boosting. Previously hardcoded; now a live setting shared by all of them.	
dura_ISF weight (dura_ISF_weight)	0.0 (0.0–3.0) [number]	Strengthens ISF during longer-lasting elevated glucose plateaus ("duration" of a high, not just its level). 0.0 disables it.	2.50
Resting duration ISF weight	1.2 (0.0–3.0) [number]	Your normal dura_ISF_weight value. Unlike the acceleration/pp pairs above, nothing currently writes to the live dura_ISF_weight, so this setting isn't yet used as an active restore target — recorded here for when/if that changes.	2.50
maxIOB threshold percent for SMB (iob_threshold_percent)	100% (10–100) [number]	SMB delivery is disabled once current IOB exceeds this percentage of your profile's max IOB. 100% (the default) effectively disables this specific throttle — SMB then remains governed only by the hard max-IOB cap itself.	70%

AutoISF settings → SMB delivery settings TOC

Setting	Default (range) [type]	Detailed description	Current setting (Client)
FIXED SMB delivery ratio (smb_delivery_ratio)	0.2 (0.1–1.0) [number]	The share of total calculated insulin requirement that can actually be delivered as a single SMB — a core OpenAPS safety cap, since delivering the full requirement at once would risk stacking. This is also the source value for "UAM Boost insulin req %" internally — that Tier-3 setting has no GUI entry of its own and is instead calculated as this ratio × 100.	0.15
Simple SMB offset override	off [switch]	When on, replaces the ratio-based SMB derivation entirely with a much simpler rule: SMB is zeroed outright whenever BG is below target + the fixed offset below, full stop. The 8 time-of-day offsets still layer on top either way. Simpler to reason about than tuning the ratio/range pieces above, at the cost of losing their gradation.	on
SMB offset (mmol/L above target)	0.5 (0.0–2.0) [number]	The fixed offset the override above uses. Only has any effect while "Simple SMB offset override" is on.	0.80
Time-of-day offset, 00–02h	0.0 (-2.0–2.0) [number]	Added on top of the effective SMB-blocking threshold during 00:00–02:00 only, regardless of which offset mode (ratio-based or simple override) is active. Positive = threshold raised = less SMB allowed in this window; negative = more SMB allowed.	0.00
Time-of-day offset, 02–04h	0.0 (-2.0–2.0) [number]	Same mechanism, 02:00–04:00 window.	0.00

Setting	Default (range) [type]	Detailed description	Current setting (Client)
Time-of-day offset, 04–06h	0.0 (-2.0–2.0) [number]	Same mechanism, 04:00–06:00 window.	0.00
Time-of-day offset, 06–09h	0.0 (-2.0–2.0) [number]	Same mechanism, 06:00–09:00 window.	0.00
Time-of-day offset, 09–12h	0.0 (-2.0–2.0) [number]	Same mechanism, 09:00–12:00 window.	0.00
Time-of-day offset, 12–18h	0.0 (-2.0–2.0) [number]	Same mechanism, 12:00–18:00 window.	0.00
Time-of-day offset, 18–22h	0.0 (-2.0–2.0) [number]	Same mechanism, 18:00–22:00 window.	0.00
Time-of-day offset, 22–00h	0.0 (-2.0–2.0) [number]	Same mechanism, 22:00–00:00 window (the overnight boundary).	0.00
Resting SMB delivery ratio	0.14 (0.1–0.5) [number]	The delivery ratio the boost/recovery automations (DelOff and related) restore to once a boost window ends — a separate, lower baseline from the general SMB delivery ratio above, which those same automations also write into transiently while a boost is active.	0.15
Enable boost automations (BolusGiven + Mild)	on [switch]	Single master switch for BOTH the BolusGiven family (bg1/bg2/bg3, escalating-severity fast-rise responses) and BolusGivenMild together. Turning it off disables both, not just one.	on
Mild boost delivery ratio	0.20 (0.1–0.5) [number]	Base ratio for BolusGivenMild's own boost — its BGL tiers apply +0.05/+0.02/+0 bumps on top of this value depending on severity. BolusGiven bg3's "strong" tier ratio is this same value +0.03, not configured independently.	0.23
Enable Tier 3 UAM Boost	off [switch]	Master switch for a separate, newer boost condition in DetermineBasalAutoISF.kt (strong glucose acceleration combined with a positive delta) — distinct from "Enable boost automations" above, which is BolusGiven/BolusGivenMild's own unrelated feature. Also requires VirtualPump as the active pump driver — this Tier 3 logic cannot fire on a real pump build regardless of this toggle, by design, since it's still under real-world evaluation.	Not in snapshot
UAM Boost max bolus (boost_max)	2.5 U (0.1–10.0) [number]	Hard cap on the size of a single UAM Boost SMB.	Not in snapshot
UAM Boost max IOB (boostMaxIOB)	1.0 U (0.1–12.0) [number]	UAM Boost stops firing once current IOB reaches this level, independent of whatever the other trigger conditions say.	Not in snapshot
UAM Boost scale (boost_scale)	1.0 (0.1–3.0) [number]	Multiplier applied to the pre-boost insulin requirement before it's capped at the max bolus above. This is the raw, unscaled preference value — the live SMB size is additionally scaled by the	Not in snapshot

Setting	Default (range) [type]	Detailed description	Current setting (Client)
		current profile percentage on top of this at dosing time.	
Enable alternative activation of SMB depending on active target being even/odd	off [switch]	An alternate SMB-gating rule: when enabled, SMB is only active while the current target (mmol/L with an even decimal digit, or mg/dL as an even whole number) meets that parity condition — otherwise SMB is inactive. Off = normal behaviour, ungated by target parity.	Not in snapshot
Split Bolus	off [switch]	When no profile percentage override is active and a calculated SMB would hit the max-bolus cap, this delivers the remainder as a second part after the split interval below, rather than waiting a full SMB interval for it.	on
Show Carb Model Curve	on [switch]	Overlays a dashed, calculated two-compartment carb-absorption curve (90-minute peak, truncated at 240 minutes) on the Carbs Absorption graph, alongside the normal empirical (measured) absorption line — a visual comparison tool, no dosing effect.	on
Split Bolus Interval (min)	7 (1–10) [number]	Minutes between the first and second parts of a split bolus. Only takes effect when Split Bolus is on, profile% = 100%, and the SMB in question is exactly at maxBolus.	7
TDD Sensitivity (ISF/Basal)	on [switch]	Uses the blended-TDD sensitivity ratio to adjust ISF and basal specifically while Autosens (top-level setting) is off. Disabling this keeps sensitivity neutral (ratio locked to 1.0) instead.	on
TDD Factor (max_iob/insulinReq)	on [switch]	Scales both max_iob and the calculated insulin requirement by the same blended-TDD ratio. When off, the fixed fallback value below is used in its place instead.	on
TDD Factor fallback value	1.0 (0.8–1.2) [number]	The fixed TDDfactor used only while "TDD Factor" above is disabled — has zero effect while that's enabled, since the live TDD ratio overrides it every cycle. Range matches the live calculation's own internal clamp.	1.00

Notes TOC

• "UAM Boost insulin req %" is used internally by Tier 3 UAM Boost's bypass path but has no GUI entry of its own — it is calculated as (FIXED SMB delivery ratio × 100) rather than being independently tunable, so it always tracks that setting.

• Settings marked as a live/transient pair (e.g. "Weight while BG accelerates" plus its "Resting" and "Boosted" counterparts) share one underlying live preference that several automations write to and restore — the "Resting"/"Boosted" settings are the values those automations use, not independent effects of their own.

Not Yet In This Snapshot (added after build 650) TOC

• Enable Tier 3 UAM Boost — default off, no live value available from this build's export.

• UAM Boost max bolus (boost_max) — default 2.5 U (0.1–10.0), no live value available from this build's export.

• UAM Boost max IOB (% of max_iob) — default 10% (1–100%), no live value available from this build's export.

• UAM Boost scale (boost_scale) — default 1.0 (0.1–3.0), no live value available from this build's export.

Related Settings Present In The Export, Outside This Document's Tables TOC

Real values from the device, for settings this document's earlier tables don't individually list — included for completeness rather than silently dropped.

• Aging-sensor calibration adjustment (master enable) (autoisf_old_sensor_adj_enabled): true. Referenced in the Libre section's description text but has no row of its own in this document's table.

• Weight while BG decelerates (bgBrake_ISF_weight): 0.25. Sibling of "Weight while BG accelerates"; not currently listed as its own row in this document.

• Enable alternative activation of SMB depending on target parity (Enable alternative activation of SMB always): true. Export label text doesn't exactly match this document's setting name ("...always" vs "...depending on active target being even/odd") — verify on-device before treating as the same setting.

• Min autoISF ratio / low-BG max ratio (autoISF_min / autoISF_max_low): 0.30 / 1.00. Not present in this document's tables — likely governed elsewhere in the AutoISF weight system code, or scoped out.

• Enable ISF adaptation by glucose behaviour (openapsama_enable_autoISF): true. Plugin-level master switch, outside this document's per-setting tables.

• SMB delivery ratio — BG range / max / min, range extension (openapsama_smb_delivery_ratio_bg_range / _max / _min / smb_max_range_extension): 0.00 / 0.50 / 0.95 / 4.00. Related sibling settings to "FIXED SMB delivery ratio", not individually tabled in this document.

• High/low temptarget sensitivity toggles (high_temptarget_raises_sensitivity / low_temptarget_lowers_sensitivity): true / true. Stock AAPS sensitivity settings, not part of the AutoISF-specific screens this document covers.

Internal / Non-Setting Keys In This Export TOC

Present in the raw export file but not GUI settings — internal state, other plugins, or device metadata.

Preference key	Current value	Why it's not a setting
autoisf_clean_graph_requested	false	One-shot internal flag, not a setting.
autoisf_custom_automations_enabled	true	Not exposed via addPreferenceScreen; read directly by code.
autoisf_fslcal_offset_normal / _slope_normal	1.40 / 0.72	Internal backup of fslCal_Offset/Slope taken before OldSensorAdj overrides them.
autoisf_low_profile_name / autoisf_standard_profile_name	Current Profile / Current ProfileReal	String-type profile-name settings, outside this document's numeric/switch scope.
autoisf_low_raw_24_override_active / _armed	false / false	Internal LowRaw24Cal state. Note: _armed no longer exists in the current codebase (removed the same day this export predates).
autoisf_low_storage_notified / autoisf_old_pod_notified	false / false	Internal notify-once latches.
autoisf_mild_offset_zero_active / autoisf_old_sensor_adj_active	false / false	Internal live-state flags reflecting whether a tier is currently applying.
autoisf_sensor_age_code_enabled	true	Related but separate gate from "Aging-sensor

Preference key	Current value	Why it's not a setting
		calibration adjustment” above; not confirmed to have its own row.
automation_state_MJ	MJ3	MJ state-machine's current value, not a preference.
boluswizard_percentage / insulin_oref_peak	70 / 45	Belong to the Bolus Wizard / Insulin plugins, not AutoISF.
fsl_use_ukf_smoothing / graph5_bgl_only / show_graph5	false / false / true	Overview double-tap-gesture toggles, not part of the AutoISF preference screens.
configuration_* / source / main_phone_snapshot_time	(device metadata)	Export/device metadata, not settings.

List 1 and List 2 command/temporary-target reference

These mmol/L values are command signals, not clinical targets. Pump/Virtual builds execute the selected handler locally without creating a temporary target. AAPSClient uses a five-minute relay TT so the pump can receive the command; the pump applies the action and cancels the relay. List 2 graph rows are local display controls and never create relay targets.

a. List 1 - settings and relay commands

Command	Relay TT	Effect	Current-setting behavior
SMBdel base + mild-Bst	5.002 / 5.004	Decrease/increase both SMB-delivery baseline and Mild Boost ratio by 0.01.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Toggle Libre sensor adjustment	5.006	Toggle aging-sensor/Libre adjustment on or off.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Toggle boost automations	5.008	Toggle the whole boost-automation group on or off.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
pp ISF weight - normal	5.012 / 5.014	Decrease/increase normal postprandial ISF weight by 0.01.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Acceleration ISF weight - normal	5.016 / 5.018	Decrease/increase normal acceleration ISF weight by 0.05.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Duration ISF weight - normal	5.022 / 5.024	Decrease/increase normal duration ISF weight by 0.1.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Libre slope - original	5.026 / 5.028	Decrease/increase the baseline Libre calibration slope by 0.01.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Libre offset - original	5.032 / 5.034	Decrease/increase the baseline Libre calibration offset by 0.05.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
SMB offset	5.036 / 5.038	Decrease/increase the hard SMB-offset override by 0.1.	Pump/Virtual: local value shown and changed after confirmation. Client:

Command	Relay TT	Effect	Current-setting behavior
			mirrored pump value shown when available; local Client preference is not changed.
Clean graph view	5.042	Hide SMB dose labels/arrows and show the plain solid-green view.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Wizard bolus percentage	5.046 / 5.048	Decrease/increase the Wizard percentage by five percentage points.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Mild Boost ratio	5.052 / 5.054	Decrease/increase Mild Boost ratio alone by 0.01.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
pp ISF weight - high	5.056 / 5.058	Decrease/increase boosted postprandial ISF weight by 0.01.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Acceleration ISF weight - high	5.062 / 5.064	Decrease/increase boosted acceleration ISF weight by 0.01.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Higher-ISF-range weight	5.068 / 5.070	Decrease/increase high-BG ISF weight by 0.1.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Peak insulin time	5.074 / 5.076	Decrease/increase insulin peak by five minutes.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
AutoISF maximum - low BG	5.080 / 5.082	Decrease/increase the low-BG AutoISF maximum by 0.1.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
AutoISF maximum - normal	5.086 / 5.088	Decrease/increase the normal AutoISF maximum by 0.1.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
TOD offset 00:00-02:00	5.092 / 5.094	Decrease/increase this time-of-day offset by 0.1.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
TOD offset 02:00-04:00	5.098 / 5.100	Decrease/increase this time-of-day offset by 0.1.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
TOD offset 04:00-06:00	5.104 / 5.106	Decrease/increase this time-of-day offset by 0.1.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
TOD offset 06:00-09:00	5.110 / 5.112	Decrease/increase this time-of-day offset by 0.1.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
TOD offset 09:00-12:00	5.116 / 5.118	Decrease/increase this time-of-day offset by 0.1.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
TOD offset 12:00-18:00	5.122 / 5.124	Decrease/increase this time-of-day offset by 0.1.	Pump/Virtual: local value shown and

Command	Relay TT	Effect	Current-setting behavior
			changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
TOD offset 18:00-22:00	5.128 / 5.130	Decrease/increase this time-of-day offset by 0.1.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
TOD offset 22:00-00:00	5.134 / 5.136	Decrease/increase this time-of-day offset by 0.1.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Toggle Graph2 carb-model curve	5.138	Toggle the empirical/theoretical carb-model curve display.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Cloud logs upload	5.140	Zip and send logs to configured cloud storage, otherwise use email.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Toggle Graph5	5.142	Toggle the Graph5 comparison panel.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
MJ state manual override	5.144 / 5.146	Set MJ to NOMJremains or MJ3.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Profile manual override	5.148 / 5.150	Select the configured Standard or Low profile.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Toggle UKFset1 live comparison	5.152	Toggle the Virtual-Pump, non-Client live UKFset1 comparison; hidden elsewhere.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.
Run SensorAge code	5.156	Toggle the sensor-age code on or off.	Pump/Virtual: local value shown and changed after confirmation. Client: mirrored pump value shown when available; local Client preference is not changed.

b. List 2 - actions, relays and local graph controls

Command	Relay TT	Effect	Transport/status
MJ start	5.158	Start the confirmed Mounjaro cycle.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
MJ restore	5.160	Run the confirmed Mounjaro restoration action.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
Steroid start	5.162	Start the steroid-management sequence.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
Toggle MJ Kotlin buttons	5.164	Show/hide the MJ buttons.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.

Command	Relay TT	Effect	Transport/status
Toggle Steroid Kotlin button	5.166	Show/hide the steroid button.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
Steroid 110 to 130	5.168	Run the confirmed steroid escalation to 130.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
Steroid 130 to 150	5.170	Run the confirmed steroid escalation to 150.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
Steroid 150 to 190	5.172	Run the confirmed steroid escalation to 190.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
Steroid 190 to 250	5.174	Run the confirmed steroid escalation to 250.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
Steroids OFF	5.176	Run the confirmed steroid turn-off action.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
AnyDesk restart	5.178	Client writes an ADesk Note and uses a five-minute relay TT only if no real TT is active; otherwise Note-only transport preserves the real TT.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
AnyDesk restart - local test	5.180	Local command test through the real receiver; no NS round-trip or relay TT.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
Graph: UKF1	No relay TT	Local display toggle plus optional Libre slope/offset calibration.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
Graph: UKF2	No relay TT	Local display toggle; calibration is already upstream and remains shown as on.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
Graph: UKF3	No relay TT	Local display toggle plus optional Libre slope/offset calibration.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.
Graph: Graph5 panel	No relay TT	Local panel toggle; optional BGL-only mode hides insulin activity and the three carb series.	Graph rows: local only. Clinical actions: direct on Pump/Virtual; five-minute relay on Client after confirmation. AnyDesk preserves an existing real TT by using Note-only transport.

c. Confirmation popups, current values and checkboxes

Opening the lists. Double-tap the IOB graph area for List 1. Double-tap the basal-rate icon area for List 2. On AAPSClient the list titles identify that commands will be relayed to the pump; on Pump/Virtual they identify direct local operation.

List 1 single commands. Selecting a toggle or one-shot row opens a separate confirmation popup. Where a live setting exists, the popup shows its current value before OK is pressed. For clean-graph and cloud-log commands it shows the effect instead, because there is no persistent value to display. OK applies the command; Cancel, the Android Back action, or tapping outside returns to List 1 without applying it.

List 1 stepped commands. The confirmation popup shows the current value and two checkboxes labelled with the actual alternatives, such as -0.1/+0.1, -5%/+5%, or NOMJremains/MJ3. The boxes are mutually exclusive: checking one clears the other. The user may correct the selection before pressing OK. OK applies only the checked choice; OK with neither box checked is a no-op. Cancel, Back, or outside dismissal returns to List 1.

Which current value is displayed. Pump and Virtual builds read the local preference or state. AAPSClient displays the most recent settings snapshot received from the pump, identifies it as Pump current, and includes the age of that snapshot. If the pump has not supplied the relevant value, the popup reports Pump current: unavailable. The paired SMB-delivery row shows both baseline and mild-boost values; profile and MJ rows show their current named state rather than implying a numeric setting.

List 2 action commands. Selecting an MJ, steroid or AnyDesk action opens an OK/Cancel confirmation before execution; these action confirmations do not use setting checkboxes. Cancel returns to List 2. For AAPSClient AnyDesk, the confirmation also states whether an existing temporary target will be preserved and the command sent by NS Note only, or whether an NS Note plus five-minute relay TT will be used.

List 2 graph controls. Selecting a graph row opens a popup with checkboxes already set to the current local display preferences. Graph on controls visibility. UKF1 and UKF3 also offer Use libre slope & offset. UKF2 shows that calibration box checked but disabled because calibration is already applied upstream. Graph5 instead uses the second box for BGL only (hide IA/carbs x3). OK saves the displayed choices and refreshes the Overview; Cancel, Back, or outside dismissal returns to List 2 without saving.

Confirmation and delivery status. Choosing and confirming a List 1 direction or List 2 action is the authorization step; the outer list itself does not execute a row. On Pump/Virtual, the corresponding handler is queued locally. On AAPSClient, dosing-setting/action commands use the documented relay route where applicable, while List 2 graph checkboxes remain local display settings and are never sent as relay temporary targets.

AutoISF Coded Automations - Current Code Registry

Current-code audit dated 2026-08-21. This document lists all 118 keys in CodedAutomationNames.KEYS, in registry order. The summary column is extracted from the comments attached to each current readyToRun()/markRun() handler in OpenAPSAutoISFPlugin.kt; it replaces the obsolete 56-row snapshot.

Registry meaning. EXACT and CLOSE matching are used only to decide whether a native Automation-tab item plausibly duplicates a coded automation. Non-matching native automations are not suppressed. Command/setting-relay rows are included because they are coded automation handlers even though their trigger is a List/TT command rather than an autonomous glucose rule. Disabled handlers remain listed and are labelled by their current source comments.

#	Current coded key	Type	Current code definition / criteria and action
1	50SetRecent	Operational automation	50SetRecent: flags LowBG=50recent whenever profile drops to 50% while in NO50rec state 5-min floor throttle added on top of the state guard (see readyToRun() usage note). Hysteresis: also locked out for 15 min after a PP50.Off recovery (readyToRun on the "PP50Off" timestamp), so it cannot immediately re-flag 50recent for PP50.Off to exit again — this breaks the observed ~5-min 50Rec<->50ff flap where the two automations toggled the state each cycle.
2	50pcMakes5.7	Operational automation	50pc makes5.7: safety TT when on 50% profile and BGL dropping below 5.0 mmol Precondition: no TT active. Was purely self-guarding (relying on activeTtMgdl() != null next cycle), but the TT write can lag behind the next loop cycle's read, letting this re-fire within the same 5-min cycle before the guard takes effect. Added an explicit 10-min readyToRun() throttle on top of the self-guard to cover that race.
3	AcceUp0.5	Operational automation	Code port of "AcceUp0.5": raises acce weight to its boosted level (ApsAutoIsfBgAccelWeightHigh, not the resting baseline) when currently very low, BGL is stable/rising, no TT, and either MJ is clear or steroids are on. No live-pump gate: the original's Note field was empty.
4	AcceWeightDownTT	Command / setting relay	AcceWeightDownTT: manually setting a TT of 5.16 mmol is used as a remote -0.05 nudge on ApsAutoIsfBgAccelWeightNormal (acceISFwt_orig), clamped to a min of 0.55 — not a real target. Same pattern/tight 0.001mmol tolerance as the other settings-nudge TTs above.
5	AcceWeightHighDownTT	Command / setting relay	AcceWeightHighDownTT: manually setting a TT of 5.062 mmol is used as a remote -0.01 nudge on ApsAutoIsfBgAccelWeightHigh (acceISFwt_high — the boosted value OldPod2/RecentPod/AcceUp0.5 set, previously a hardcoded 0.95 literal in each), clamped to its own min (0.0) — not a real target. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above. No dual-update: unlike AcceWeightNormal/AcceWeight, this only ever gets written into the live weight transiently when a boost automation actually fires, not continuously.
6	AcceWeightHighUpTT	Command / setting relay	AcceWeightHighUpTT: manually setting a TT of 5.064 mmol is used as a remote +0.01 nudge on ApsAutoIsfBgAccelWeightHigh, clamped to its own max (1.00). Same pattern as AcceWeightHighDownTT above.
7	AcceWeightUpTT	Command / setting relay	AcceWeightUpTT: manually setting a TT of 5.18 mmol is used as a remote +0.05 nudge on ApsAutoIsfBgAccelWeightNormal, clamped to a max of 1.00. Same pattern as AcceWeightDownTT above.
8	ActivityOff	Operational automation	Code port of "ActivityOff 70_0.70 0.35": exit criteria for ActivityProf50%'s 50% profile drop — 3 branches (still on the 6.8mmol Activity TT but BGL climbed too high; TT already ended with BGL stabilising/rising and cannula not brand-new; or a bolus was just given). Restores ProfileReal, acce to 0.35, iobTH to 70%. Original action list had no CarePortal note, kept minimal to match exactly.

#	Current coded key	Type	Current code definition / criteria and action
9	ActivityProf50	Operational automation	ActivityProf50%: sets 50% profile for 180 min during the 6.8mmol Activity TT, BGL stable/falling. Corrected per user: the TT check is an exact match on 6.8mmol (not the 7.3-7.9mmol range this previously guarded on, which never matched the real Activity TT at all), using fuzzyEquals/mmolToMgdl. Also guards against Bolus2/BolusGiven71's own brief 6.8mmol post-bolus TT (same numeric value) via the shared Profile=Bolus state, so a 5-minute post-bolus TT can't be misread as an Activity session and trigger a 180-minute 50% profile drop. Note/SMS text corrected to match the screenshot exactly ("Ac50" / "ActivityProf50% Acce", not the old dynamic g=-value SMS text). 5-min floor throttle added on top of the precondition (see readyToRun() usage note).
10	ActivityTTReversal	Operational automation	Activity TT reversal block: TT6.0New1/TT6.0New2 — exit criteria for the manually-set "Activity" temp target (GUI default ~6.8 mmol, sits inside this 6.5-7.6 mmol band). Nothing in this file creates a TT in this range; it's set by hand via the GUI. Self-guarding like the 5.7 TT block above: cancelling the TT fails the range guard next cycle. Original action lists were just "Send SMS" + "Stop temp target" — kept minimal to match exactly. 5-min floor throttle added on top of the TT-existence self-guard (see readyToRun() usage note).
11	AlarmHypo1	Hypoglycaemia protection	Code port of "AlarmHypo1 0.700.35": hypo alarm — 3 OR-branches (time-gated slow decline, an unconditional emergency floor at 3.0mmol with no time gate, and a steps-driven variant), drops acce weight to 0.10. Per user correction, sets BOTH BGLstate=BGLlastLOW and LowBG=50recent (screenshot only showed the former, but both AlarmHypo automations should set both).
12	AlarmHypo2	Hypoglycaemia protection	Code port of "AlarmHypo2 0.700.35": hypo alarm — single AND group, catches either G<=4.3mmol outright or G<=5.5mmol with recent exercise (Steps30>=1000) likely accelerating the drop. Sets both BGLstate=BGLlastLOW and LowBG=50recent, feeding the existing 50%-profile state machine (50SetRecent/Not50%Recently/Extra50%/PP50.Off).
13	AutoIsfMaxLowDownTT	Command / setting relay	AutoIsfMaxLowDownTT: manually setting a TT of 5.080 mmol is used as a remote -0.1 nudge on ApsAutoIsfMaxLow (autoISF_max_low — not yet wired into any calculation, see DoubleKey.kt), clamped to a min of 1.0 — not a real target. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
14	AutoIsfMaxLowUpTT	Command / setting relay	AutoIsfMaxLowUpTT: manually setting a TT of 5.082 mmol is used as a remote +0.1 nudge on ApsAutoIsfMaxLow, clamped to a max of 3.0. Same pattern as AutoIsfMaxLowDownTT above.
15	AutoIsfMaxNormalDownTT	Command / setting relay	AutoIsfMaxNormalDownTT: manually setting a TT of 5.086 mmol is used as a remote -0.1 nudge on ApsAutoIsfMax (autoISF_max — the existing, already-live setting; "Normal" here just distinguishes it from AutoIsfMaxLow above), clamped to a min of 1.0 — not a real target. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
16	AutoIsfMaxNormalUpTT	Command / setting relay	AutoIsfMaxNormalUpTT: manually setting a TT of 5.088 mmol is used as a remote +0.1 nudge on ApsAutoIsfMax, clamped to a max of 3.0. Same pattern as AutoIsfMaxNormalDownTT above.
17	BasalUp	Operational automation	Code port of "BasalUp": raises acce weight back to neutral (0.50) and switches to Current ProfileReal when glucose is stable/rising outside a pod/MJ/afternoon guard window, low activity, and currently on the Low profile. No live-pump gate: the original's Note field was empty.
18	Battery1pc	Device/pump/sensor	Battery1%: when phone battery drops to <=1%, switch to Current Profile50 and alert Guard: profile=100% (precondition). State Profile must be PP130, C100, or AllOK (normal running states). Self-guarding: switching to Profile50 makes profile_percentage=50 next cycle, failing the precondition. Live-pump-only: skip entirely on Virtual Pump (model() == GENERIC_AAPS is how this codebase identifies it elsewhere, e.g. TriggerPumpBatteryLevelTest). No cannula-age gate — that threshold (80h) was pod-specific and doesn't generalize to tubed pumps with independent infusion-set lifespans; critical phone battery should react regardless. 20-min floor throttle added on top of the state guard (see readyToRun() usage note).
19	BatteryOver1pc	Device/pump/sensor	BatteryOver1%: when battery recovers above 1%, restore Current ProfileReal Keys on being on the named "Current Profile50" (the battery-drop profile) rather than a state flag: that name uniquely marks the battery state, since a hypo 50% is a *percentage* on ProfileReal (different name, profile_percentage==50). This replaced the old guard which required checkAutomationState("Profile","Batt1%") — a flag Battery1pc never set, so recovery never fired — and a "profile_percentage == 50" branch that would have wrongly matched (and cancelled) a hypo 50%. 5-min floor throttle added on top of the profile guard (see readyToRun() usage note).
20	Bolus2	Bolus/post-bolus	Code port of "Bolus2": starts a brief 6.8mmol temp target after a bolus when BGL is falling fast post-bolus (branch 1), or when carbs are up and MJ is active shortly after a bolus (branch 2). Precondition: no TT active. No live-pump gate: the original's Note field was the user's own uncertainty comment ("not sure if auto needed"), not a virtual-pump restriction. SMS text kept literal ("BolusTTfor10mins") despite the action's actual 5 min duration — that mismatch is in the source automation, not introduced here. setAutomationState("Profile","Bolus") is an addition beyond the original screenshot's action list, per user instruction: this TT shares the exact same 6.8mmol value as the manually-set Activity TT, so without this marker ActivityProf50% could misread a brief 5-minute post-bolus TT as an Activity session and trigger its 180-minute 50% profile drop.
21	BolusGiven	Bolus/post-bolus	BolusGiven71_0.70: post-bolus response — boosts iobTH to 71%, acce 0.70, 110% for 10 min 10-min throttle. Three trigger blocks; only outer precondition is profile=100% + no TT. COB>=9 / acce>=0.20 / dura<acce are per-block (postBolusGate) for the manual-bolus branches (bg1/bg2); the SMB-driven branch (bg3) does NOT require them. ApsAutoIsfBoostAutomationsEnabled is the combined master switch for this WHOLE block (bg1/2/3) and BolusGivenMild below — one toggle disables both, per user request.
22	BolusGivenBg3	Bolus/post-bolus	BolusGivenMild: gentle sibling of BolusGiven bg3. Strengthens SMB delivery on a *mild* delivery-driven rise (much lower gates), WITHOUT boosting the profile — it

#	Current coded key	Type	Current code definition / criteria and action
			holds the daytime 5.0 target with a 2-min TT (which doubles as the timer for the delivery-ratio reset above) and raises the SMB delivery ratio to the configurable mild-boost base. Leaves iobTH / acce / ppWeight / profile untouched. Own 10-min throttle, independent of bg1/2/3. rawDelta5 capped BELOW bg3's 0.8mmol threshold so the two are mutually exclusive (one delta value can't satisfy both), preventing a same-loop double-fire (the no-TT precondition can't catch bg3's async-inserted TT within the same loop). ApsAutoIsfBoostAutomationsEnabled: same combined master switch as BolusGiven bg1/2/3 above. While SMBs are stacking (avg gap <=70s), shift the whole band up 10% (x1.10) rather than blocking.
23	BolusGivenMild	Bolus/post-bolus	Safety is NOT reopened by this: the cumulative 30-min/10-min SMB caps in DetermineBasalAutoISF.kt (built from the Aug 6-7 hypo incident) trim/zero microBolus AFTER this fires regardless of time of day -- they're what actually prevents stacking, not this window. Deliberately NOT extended into the 00:00-06:00 overnight guard territory (NightIobCeiling, MJrec, etc.) -- see isTimeBetween()'s own doc comment for why endH=0/ endM=0 here means "runs to end of day," not "wraps past midnight." Cross-cooldown with mild: lastBolusMin/lastCarbMin only track manual boluses (BS.Type.NORMAL), never SMB — so mild firing doesn't touch that gate and couldn't otherwise block bg3 here. Without this, bg3's own IOB bump from a mild fire ~minutes ago could sit inside bg3's next 5-min iobChange5 window and look like fresh evidence for a second (redundant) boost.
24	BolusGivenMildFailsafe	Bolus/post-bolus	BolusGivenMildFailsafe: safety-net sibling of BolusGivenMild for when SMBs simply haven't been delivering during a clearly-confirmed rise (sensor dropout, pod disconnect, etc.) -- not a delivery-ratio tuning scenario, a "the loop isn't acting at all" one. Requires ALL of Delta, SDelta, and AAPS's own longAvgDelta to independently confirm the rise (stricter than BolusGivenMild's own OR-based gate), near-zero IOB (nothing already in the pipe), and genuine SMB silence over a longer window (smbCount20Min() == 0, not the 5-min signal deliverySuppressedMild uses -- that shorter window can just be a normal gap between doses).
25	BoostToggleTT	Command / setting relay	BoostToggleTT: manually setting a TT of 5.08 mmol is used as a remote toggle for ApsAutoIsfBoostAutomationsEnabled (the combined master switch for BolusGiven bg1/2/3 and BolusGivenMild) — not a real target. Same pattern and same tight 0.001mmol tolerance as SensorAgeToggleTT above (see its comment for why).
26	CarbsStopTT1	Operational automation	carbsStopTT1ok4.4: cancel very-low TT (<=4.4 mmol) when carbs active and rising Self-guarding: cancels TT so tt<=79.3 is false next cycle. 5-min floor throttle added on top of the self-guard (see readyToRun() usage note).
27	CarbsStopTT57	Operational automation	CarbsStopTT5.7: cancel 5.7 mmol TT when carbs up or recent bolus and BGL stable/rising Self-guarding: cancels TT so TT range checks fail next cycle. 5-min floor throttle added on top of the self-guard (see readyToRun() usage note).
28	CarbsTHoff	Operational automation	CarbsTHoff: sets iobTH to 70% when post-carb BGL is falling or mid-range anomaly (was 50%; raised to 70 to match the daytime floor so it isn't just bumped up again). 5-min floor throttle added on top of the preconditions (see readyToRun() usage note).
29	CleanGraphTT	Command / setting relay	CleanGraphTT: manually setting a TT of 5.042 mmol is used as a remote trigger for the "clean main graph" display combo (no SMB labels, no BGL arrowheads, solid uniform-green line) — the same combo as long-pressing the IOB icon then the Basal icon. Not a real target. Since this plugin has no dependency on core:graph (where the actual display state — showSmbLabels/ basalToggleIndex — lives), it can't set those companion fields directly; it just raises a one-shot flag that OverviewFragment's updateGraph() picks up and clears on its next run. Same tight 0.0001mmol tolerance/cancel/notify pattern as the other settings-nudge TTs above.
30	CloudLogsUploadTT	Command / setting relay	CloudLogsUploadTT: manually setting a TT of 5.140 mmol remotely triggers the same log upload as the "Send logs" button on the Maintenance screen (zips logs, sends to cloud storage if configured, else email) — for AAPSClient users who have no settings access to trigger it directly. Next value after Graph2ToggleTT's 5.138. Same activeTtNear()/cancel/notify pattern as the other remote-signal TTs above; no toggle state, just fires once per TT-set.
31	ConnectPod	Device/pump/sensor	Code port of "ConnectPod": alerts when the pump hasn't reported in >=20 min during the day while the pod is still within its normal wear window. Mirrors TriggerPumpLastConnection (activePump.lastDataTime). Live-pump-only: "last connection to pump" is meaningless on the Virtual Pump.
32	DuraWeightDownTT	Command / setting relay	DuraWeightDownTT: manually setting a TT of 5.22 mmol is used as a remote -0.1 nudge on ApsAutoIsfDuraWeightNormal (duraISFwt_orig), clamped to a min of 0.00 — not a real target. Same pattern/tight 0.001mmol tolerance as the other settings-nudge TTs above.
33	DuraWeightUpTT	Command / setting relay	DuraWeightUpTT: manually setting a TT of 5.24 mmol is used as a remote +0.1 nudge on ApsAutoIsfDuraWeightNormal, clamped to a max of 3.00. Same pattern as DuraWeightDownTT above.
34	EveningTH	Operational automation	NightIobCeiling: caps iobTH at 18% from midnight to 06:00, independent of BGL Companion to EveningIobCeiling above, same ceiling-not-setter shape (only ever lowers), but a far tighter value because overnight there are no carbs to cover -- IOB that would be routine after a meal is dangerous at 03:00. Motivated by the night of 6->7 Aug 2026, where BGL reached 3.5 and stayed there 04:00-05:30. Two SMB bursts caused it: 23:10-23:18 (~1.50U, IOB 0.36 -> 2.00) and 01:08-01:15 (~1.30U, IOB 1.43 -> 2.71), both at ~60s intervals with COB already decayed to 0 and BGL only 7.2-8.6. NightAcce covers 01:00-06:00 and sets iobTH to 18 already, but could not fire for either burst because it requires BGL <= 6.0mmol -- and BGL was 8.5. That BGL gate is the reason the overnight protection was unreachable, which is why this block deliberately has NO BGL condition at all.
35	ExerciseLimitAcce	Operational automation	Code port of "Exercise limit Acce": alerts when a fast rise coincides with high recent step activity. No live-pump gate: the original's Note field was empty.
36	ExportSettingsPodActivation	Device/pump/sensor	Code port of "ExportSettingsPodActivation": triggers a real encrypted settings backup (exportSettingsFor, faithfully mirroring ActionSettingsExport) when a new pod has been

#	Current coded key	Type	Current code definition / criteria and action
			activated since the last export — mirrors TriggerPodChange's own "Pod activated" check (last CANNULA_CHANGE more recent than last SETTINGS_EXPORT). Live-pump-only, matching the original's note.
37	Extra50	Hypoglycaemia protection	Extra50%: deepens hypo protection (acce→0.07, iobTH→50%) when BGL falling on 100% profile 5-min floor throttle added on top of the state guard (see readyToRun() usage note).
38	GentleHypoRisk	Hypoglycaemia protection	GentleHypoRiskOver4.5: escalates from prepare50 state (weight 0.07) to Skittles state (0.02) Guard: acce weight 0.03–0.08 (only fires when prepare50 is active; Skittles weight 0.02 falls below). 30-min throttle via readyToRun/markRun. Uses UKF-smoothed raw/noise CGM for additional safety checks.
39	Graph2ToggleTT	Command / setting relay	Graph2ToggleTT: manually setting a TT of 5.138 mmol is used as a remote toggle for ApsAutoIsfShowCarbModelCurve (the "graph2" carb model curve display) — not a real target. Next value after TodOffset2200UpTT's 5.136 in the settings-nudge TT cluster. Same activeTtNear()/cancel/notify/toggle pattern as SensorAgeToggleTT/BoostToggleTT above.
40	Graph5ToggleTT	Command / setting relay	Graph5ToggleTT: manually setting a TT of 5.142 mmol is used as a remote toggle for ApsAutoIsfShowGraph5 (the 5th graph — a fixed clone of the main graph, minus basal) — not a real target. Next value after Graph2ToggleTT's 5.138 / CloudLogsUploadTT's 5.140. Same activeTtNear()/cancel/notify/toggle pattern as Graph2ToggleTT above. Independent switch — deliberately does not touch graphs 2/3/4's own settings.
41	High6PP	Operational automation	Code port of "High6PP": brief 120% profile boost across 3 condition blocks when BGL is high with controlled delta and a low/no temp target tolerance. Note: "TT now <=4.2 tolerant". Precondition: profile=100%. Daytime-only outer gate (09:00-21:00) — each branch below still has its own (narrower or equal) window, so this just bounds all three to the same daytime range without changing their individual per-branch windows. 30-min floor throttle. Also requires recentLibreOver12(48): this boost increases insulin delivery in response to a perceived high, so it's gated on genuine confirmation (raw Libre >12.0mmol) within the last 48h -- without that, a borderline/moderate reading isn't enough on its own to warrant it.
42	High6PPoff	Operational automation	Code port of "High6PPoff": exits the 120% boost when either no TT exists at 120%, or BGL/delta have stabilised in the evening window at 120%. No live-pump gate: the original's Note field was empty.
43	HighNight00AM	Operational automation	HighNight00AM: overnight high BGL (>=7.0 mmol) — switch to ProfileReal, set hypo TT 4.2 Fires 01:00–05:45 when glucose elevated and gently rising/flat, iobTH<=50 OR old cannula + MJ state. Threshold lowered from 9.0 to 7.0 mmol (least-invasive first step on the overnight 6-6.5mmol plateau): NightAcce backs off at <=6.0mmol and OffHighProf reverts this automation's own ProfileReal switch once BG falls under 7.5mmol falling, so 7.0 sits with working room on both sides of that existing pair -- low enough to actually engage before BG settles into the 6.0-9.0mmol dead zone where nothing else in this file corrects, high enough to leave NightAcce's own floor and OffHighProf's own revert gate alone. Revisit the exact number after a few nights. 60-min floor throttle added on top of the preconditions (see readyToRun() usage note). DISABLED.
44	HighOldPod	Device/pump/sensor	HighOldPod: sets a brief TT 5.0 + 110% profile when cannula is stale (>=60h) or fresh (<=6h) Preconditions: no TT, profile=100%. Guards: MJ=NOMJremains state + bolus >=90 min ago. 5-min floor throttle added on top of the preconditions (see readyToRun() usage note).
45	HighPP130Off	Operational automation	Code port of "HighPP130Off": exits the 130%/110% boost across 3 exit conditions (stabilised no-TT, falling with TT, or simply no-TT at all while boosted). Note: "or 110%".
46	HigherIsfRangeWeightDownTT	Command / setting relay	HigherIsfRangeWeightDownTT: manually setting a TT of 5.068 mmol is used as a remote -0.1 nudge on ApsAutoIsfHighBgWeight (higher_ISFrange_weight), clamped to a min of 0.0 — not a real target. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
47	HigherIsfRangeWeightUpTT	Command / setting relay	HigherIsfRangeWeightUpTT: manually setting a TT of 5.070 mmol is used as a remote +0.1 nudge on ApsAutoIsfHighBgWeight, clamped to a max of 2.0. Same pattern as HigherIsfRangeWeightDownTT above.
48	LibreOffsetDownTT	Command / setting relay	LibreOffsetDownTT: manually setting a TT of 5.032 mmol is used as a remote -0.05 nudge on ApsAutoIsfLibreOffsetOrig (LibreOffset_orig), clamped to a min of 1.20 — not a real target. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above. If FslCalOffset currently equals its own baseline, nudge both together. Note suffix uses takeLast(4) (not the usual 2) since this setting is >=1.0 — the one outlier allowed to run to 6 total chars ("Ld1.25") rather than dropping the leading digit, so the note stays unambiguous on its own.
49	LibreOffsetUpTT	Command / setting relay	LibreOffsetUpTT: manually setting a TT of 5.034 mmol is used as a remote +0.05 nudge on ApsAutoIsfLibreOffsetOrig, clamped to a max of 1.60. Same pattern as LibreOffsetDownTT above.
50	LibreUkf1ToggleTT	Command / setting relay	LibreUkf1ToggleTT: 5.152 is the "Tog UKFset1" row in the IOB double-tap popup. Its old sibling LibreUkf2ToggleTT (5.154) is removed -- UKFset2 toggling moved to list2's "Graph: UKF2" entry (see GraphToggleEntry.syncedLiveKey in OverviewFragment.kt). The mutual-exclusion write here (turning UKFset1 on used to also force FslUseUkfLibreSpecialSmoothing off) was removed 2026-08-15 for the same reason it was dropped from list2's popup: that flag no longer has any dosing effect (XdripSourcePlugin.kt/NsIncomingDataProcessor.kt call smoothLibreSpecialRealtime() only for its graph-history side effect and discard the return value) -- it now only gates whether UKF2's graph history accumulates. Silently clearing it here every time UKFset1 got toggled on was the actual cause of UKF2's graph never accumulating points: the two settings are independent now, so toggling one must not touch the other.
51	LibreUkf2ToggleTT	Command / setting relay	Retired/unassigned legacy relay key. Code 5.154 was freed when UKF2 graph history was made continuously current without a graph checkbox changing the live dosing engine. No

#	Current coded key	Type	Current code definition / criteria and action
			active List/TT handler presently uses it; it remains in the coded-name registry so an old native automation title can still be classified for review.
52	LibreSlopeDownTT	Command / setting relay	LibreSlopeDownTT: manually setting a TT of 5.026 mmol is used as a remote -0.01 nudge on ApsAutoIsfLibreSlopeOrig (LibreSlope_orig), clamped to a min of 0.60 — not a real target. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above. If FslCalSlope (the live setting) currently equals its own baseline, nudge both together.
53	LibreSlopeUpTT	Command / setting relay	LibreSlopeUpTT: manually setting a TT of 5.028 mmol is used as a remote +0.01 nudge on ApsAutoIsfLibreSlopeOrig, clamped to a max of 1.00. Same pattern as LibreSlopeDownTT above.
54	MJ2old	Mounjaro state	MJ2 old: advances MJ state from "MJ active" → MJ2 at 02:10–03:10 AM
55	MJ3old	Mounjaro state	MJ3 old: advances MJ state from MJ2 → MJ3 at 01:05–02:05 AM
56	MJ4	Validation/test	Code port of the "Test" automation (MJ=MJ4). Self-guarding: state change prevents re-fire. 5-min floor throttle (see readyToRun() usage note near its declaration).
57	MJ5	Validation/test	Code port of the "Test2" automation (MJ=MJ5): also switches to Current ProfileReal for 30 min.
58	MJoff	Mounjaro state	MJoff old: exits MJ cycle when MJ3 active Block 1: 12:00–21:04 with BGL >= 10.5 mmol. Block 2: midnight 00:00–01:00.
59	MJrecentCurrProfAcce	Mounjaro state	Code port of "MJ recent or Steps CurrProf Acce" (title kept the stale "Steps" reference, dropped here since the actual condition tree has no steps-count check at all — confirmed against the current automation dialog). No live-pump gate: the original's Note field was empty (no "not virtual pump" restriction specified for this one). Throttle raised 5 -> 480 min (8h) to serve as this block's SELF-LATCH, replacing the removed "! onCurrentProfile" gate. That gate was doing double duty: the switchProfileIfNeeded call below used to satisfy it, which is what stopped the block re-firing every 5 min for the rest of the 00:00-08:00 window. Dropping the profile switch (see below) removes that latch with it.
60	MildBoostDownTT	Command / setting relay	MildBoostDownTT: manually setting a TT of 5.052 mmol is used as a remote -0.01 nudge on ApsAutoIsfMildBoostRatio alone (unlike SmbDeliveryDownTT/5.002, which nudges it together with ApsAutoIsfSmbDeliveryBaseline), clamped to its own min of 0.1 — not a real target. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
61	MildBoostUpTT	Command / setting relay	MildBoostUpTT: manually setting a TT of 5.054 mmol is used as a remote +0.01 nudge on ApsAutoIsfMildBoostRatio alone, clamped to its own max of 0.5. Same pattern as MildBoostDownTT above.
62	MoreMJ	Mounjaro state	Code port of "MoreMJ": advances MJ state to MJ3 when acce weight is very low, activity is low, glucose is falling low on the 50% profile with no carbs and IOB still present. No live-pump gate: the original's Note field was empty. Outer window 06:00-00:00 (isTimeBetween(6,0,0,0) = 6am through end of day): MoreMJ should not apply overnight 00:00-06:00 at all -- this is the structural fix for the flip against MJoff's midnight block (00:00-01:00), since MoreMJ now simply cannot fire in that window regardless of the noRecentHighTrigger OR-path below (which has no time-of-day/BG gate of its own). Hysteresis: also locked out for 65 min after MJoff last fired (readyToRun on the "MJoff" timestamp) -- kept as a second line of defense for MJoff's OTHER block (12:00-21:04, high BG), which is inside this 06:00-00:00 window and so isn't covered by the gate above.
63	NightAcce	Operational automation	Code port of "NightAcce_0.35TH18": overnight iobTH/acce reset to 18%/0.35 (plus a settings-export backup, reusing exportSettingsFor) when iobTH sits outside the 18% band, no carbs are active, and BGL is low. No live-pump gate: the original's Note field was empty.
64	Not50Recently	Operational automation	Not50%Recently: clears 50recent flag once profile is back at 100% and BGL rising
65	OffHighProf	Operational automation	OffHighProf: overnight BGL falling on non-standard profile → drop to acce 0.18 / iobTH 18% Fires when NOT on the Low profile (i.e. on a named high/steroid profile), Steroids Off, no TT. 30-min floor throttle matches the temporary low-profile duration and prevents another profile-changing block from re-arming OffP-1/OffP-2 within that protection window.
66	OldPod2	Device/pump/sensor	Code port of "RecentPodOff": relaxes acce weight back to baseline once the RecentPod/OldPod2 safety TT has ended and acce weight is still at its boosted (ApsAutoIsfBgAccelWeightHigh) level. No live-pump gate: the original's Note field was empty. fuzzyEquals, not ==: acceHigh has no exact float representation at values like 0.95, so "acceW == acceHigh" could read false after the round-trip and this recovery would never fire (acce stuck at its boosted level). Same float trap as the DelOff reset. Requires that a pod boost ACTUALLY happened, which this block never checked despite its name and the comment above both describing it as undoing RecentPod/OldPod2. Its only real condition was "acce is at High and no TT is running" -- a state AcceUp0.5 produces exactly (it sets acce to High and starts no TT), so this fired immediately after every AcceUp0.5 and undid it.
67	PP50Off	Operational automation	PP50.Off: replaces "PP50.Off CurrProfReal 70_0.70 0.35" automation Exits the 50% prepare state: restores full profile, resets acce weight, clears LowBG=50recent. Guard: LowBG=50recent state must be set; firing sets LowBG=NO50rec so it won't re-trigger. 5-min floor throttle added on top of the state guard (see readyToRun() usage note).
68	PeakInsulinTimeDownTT	Command / setting relay	PeakInsulinTimeDownTT: manually setting a TT of 5.074 mmol is used as a remote -5 nudge on InsulinOrefPeak (peak insulin activity time, minutes), clamped to its own min (35) — not a real target. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
69	PeakInsulinTimeUpTT	Command / setting relay	PeakInsulinTimeUpTT: manually setting a TT of 5.076 mmol is used as a remote +5 nudge on InsulinOrefPeak, clamped to its own max (120). Same pattern as PeakInsulinTimeDownTT above.
70	Pod1	Device/pump/sensor	Code port of "POD 79 hours": narrow ~0.1h match window during 07:00 AM-11:59 PM. Live-pump-only, matching the original's note. (Original had a 2nd OR-branch, "Time

#	Current coded key	Type	Current code definition / criteria and action
			BTW 12:03 AM & 12:03 AM", which is a zero-width/dead window under isTimeBetween's semantics — dropped per instruction, this now matches Pod2's single-AND-group structure.)
71	Pod2	Device/pump/sensor	Code port of "POD 78 hours": narrow ~0.1h match window during 08:00 AM-11:59 PM, permanently switches to Current ProfileReal and flags Profile state PP130. Live-pump-only, matching the original's note.
72	PodChangeHighPP130	Device/pump/sensor	Code port of "PodChangeHighPP130": brief 130% profile boost around a pod change (fresh <=2h or stale >=78h) when BGL is high and rising, 10am-6pm. Live-pump-only, matching the original's note. Precondition: profile=100%. Also requires a genuine recent high (raw Libre >12.0mmol within 48h) -- without that recent confirmation the boost isn't warranted.
73	PpWeightDownTT	Command / setting relay	PpWeightDownTT: manually setting a TT of 5.12 mmol is used as a remote -0.01 nudge on ApsAutoIsfPpWeightNormal (ppISFwt_orig), clamped to its own min (0.0) — not a real target. Same pattern/tight 0.001mmol tolerance as the SmbDelivery*/SensorAge/Boost toggle TTs above.
74	PpWeightHighDownTT	Command / setting relay	PpWeightHighDownTT: manually setting a TT of 5.056 mmol is used as a remote -0.01 nudge on ApsAutoIsfPpWeightHigh (ppISFwt_high — the boosted value the 7 fast-rise automations set, previously a hardcoded 0.15 literal in each), clamped to its own min (0.0) — not a real target. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above. No dual-update: unlike PpWeightNormal/PpWeight, this only ever gets written into the live weight transiently when a boost automation actually fires, not continuously.
75	PpWeightHighUpTT	Command / setting relay	PpWeightHighUpTT: manually setting a TT of 5.058 mmol is used as a remote +0.01 nudge on ApsAutoIsfPpWeightHigh, clamped to its own max (0.15). Same pattern as PpWeightHighDownTT above.
76	PpWeightRevertUnder8_5	Command / setting relay	PpWeightRevertUnder8_5: genuine automatic (not TT-triggered) counterpart to the boost automations that set ApsAutoIsfPpWeight to 0.15 (BolusGiven, BolusGivenMild, High6PP, HighOldPod, PodChangeHighPP130, OldPod2, RecentPod) — always-on, fires whenever the live PP weight currently differs from its own baseline (ApsAutoIsfPpWeightNormal) AND either BG is back under 8.5mmol OR recent activity is high enough — same recentSteps5Minutes/ recentSteps30Minutes thresholds (100/200) already used as the movement-guard BLOCKER in BolusGiven/BolusGivenMild/bg3's own fire conditions, plus bg3's own extra recentSteps60Minutes (300) gate, all reused here as POSITIVE triggers instead: active movement is independent evidence the elevated PP weight is no longer needed.
77	PpWeightUpTT	Command / setting relay	PpWeightUpTT: manually setting a TT of 5.14 mmol is used as a remote +0.01 nudge on ApsAutoIsfPpWeightNormal, clamped to its own max (0.15). Same pattern as PpWeightDownTT above.
78	PreSoakSensor24hrs	Device/pump/sensor	Code port of "PreSoakSENSOR24hrs": reminds to pre-soak a new sensor ~14.0 or ~14.5 days into the current sensor's life. Two narrow ~0.1h match windows, each gated on cannula age <=80h (skip if the pod is also near end of life). Live-pump-only, matching the original's note.
79	PrepareSet50	Hypoglycaemia protection	prepare Set50%: replaces "prepare Set50%0.07 50%" automation Precondition guard: profile_percentage == 100. Once the 50% profile switch fires, profile_percentage becomes 50 on the next loop cycle and the block stops running. All 4 blocks also check Profile pct = 100 in the original; the outer guard handles that. 5-min floor throttle added on top of the precondition (see readyToRun() usage note).
80	RecentPod	Device/pump/sensor	Code port of "RecentPod": brief 130% profile boost + 4.2mmol TT when cannula is very fresh/stale, carbs are up, and BGL is rising with headroom on IOB. Live-pump-only, matching the original's note. Preconditions: profile=100%, no TT. Also requires a genuine recent high (raw Libre >12.0mmol within 48h) -- without that recent confirmation the boost isn't warranted.
81	RecentPodOff	Device/pump/sensor	Code port of "RecentPodOff": relaxes acce weight back to baseline once the RecentPod/OldPod2 safety TT has ended and acce weight is still at its boosted (ApsAutoIsfBgAccelWeightHigh) level. No live-pump gate: the original's Note field was empty.
82	SemiTwilightAcce	Operational automation	Code port of "SemiTwilightAcce_0.50TH16": drops acce weight to 0.50 and iobTH to 16% across 3 morning branches. Branch 3's iobTH check was screenshotted as AND(>17, <=15) — impossible as an AND (self-contradictory, always false). Per user confirmation the actual condition was a single "iobTH percent = 16" equality check, misread off the screenshot as two separate threshold rows. No live-pump gate: the original's Note field was empty.
83	SensorAgeCodeToggleTT	Command / setting relay	SensorAgeCodeToggleTT: master execution toggle for the complete SensorAge calibration routine, separate from the existing 5.006 aging-sensor adjustment preference toggle.
84	SensorAgeToggleTT	Command / setting relay	SensorAgeToggleTT: manually setting a TT of 5.06 mmol is used as a remote toggle for the OldSensorAdj enable switch (ApsAutoIsfOldSensorAdjEnabled) — not a real target. Flips the switch, then immediately cancels the TT so it never actually affects dosing. activeTtNear() (not a hand-rolled mg/dL literal) is deliberate here — see its own doc comment on why the exact mmol->mg/dL conversion constant matters for matching a manually-set TT. Tight 0.001mmol tolerance (not the old 0.02) — this now sits in a cluster with SmbDeliveryDownTT (5.02), SmbDeliveryUpTT (5.04), and BoostToggleTT (5.08), all only 0.02mmol apart from their neighbors, so a loose tolerance would make adjacent windows overlap. Small throttle purely as a defense against double-toggling if the cancel hasn't fully propagated by the next cycle — the cancel itself is what actually prevents re-firing in normal operation.
85	SensorS1hr	Device/pump/sensor	Code port of "SENSOR at 14.9 days": narrow ~0.1h match window, gated on cannula age <=80h. Live-pump-only, matching the original's note.
86	SensorS2hr	Device/pump/sensor	Code port of "SENSOR at 14 days 22 hours due": narrow ~0.1h match window, gated on

#	Current coded key	Type	Current code definition / criteria and action
			cannula age <=80h. Live-pump-only, matching the original's note.
87	Shower12	Operational automation	Shower12: drops iobTH to 12% during quiet morning rise (no steps, no COB, long post-bolus) 5-min floor throttle added on top of the preconditions (see readyToRun() usage note).
88	SkittlesHypoRisk	Hypoglycaemia protection	Skittles hypo-risk: replaces SkittlesTT3CurrP02, SkittlesA3ok8.0,5.0,6.0, Skittles3ok2BG9.0 Primary guard: startTempTargetIfNeeded() no-ops when a TT is already active, so the 7 condition blocks are evaluated every loop cycle but actions only fire once per TT period. All glucose/delta thresholds in mg/dL; originals in mmol/L noted in comments. 5-min floor throttle added on top of the TT-existence self-guard (see readyToRun() usage note).
89	SmbDeliveryDownTT	Command / setting relay	SmbDeliveryDownTT: manually setting a TT of 5.02 mmol is used as a remote -0.01 nudge on both SMB delivery settings (ApsAutoIsfSmbDeliveryBaseline, ApsAutoIsfMildBoostRatio) — not a real target. Clamped to each key's own min (0.1 for both). Same activeTtNear()/cancel/notify pattern as SensorAgeToggleTT/BoostToggleTT above. Tight 0.001mmol tolerance (not the usual 0.02) — 5.02 and 5.04 (SmbDeliveryUpTT) are only 0.02mmol apart, so 0.02 tolerance would make their windows overlap each other (and creep into 5.0's own "own boost TT" territory).
90	SmbDeliveryUpTT	Command / setting relay	SmbDeliveryUpTT: manually setting a TT of 5.04 mmol is used as a remote +0.01 nudge on both SMB delivery settings, clamped to each key's own max (0.5 for both). Same pattern and same tight 0.001mmol tolerance as SmbDeliveryDownTT above.
91	SmbOffsetDownTT	Command / setting relay	SmbOffsetDownTT: manually setting a TT of 5.36 mmol is used as a remote -0.1 nudge on ApsAutoIsfSmbOffsetOverride (SMBoffset), clamped to a min of 0.50 — not a real target. Same pattern/tight 0.001mmol tolerance as the other settings-nudge TTs above.
92	SmbOffsetUpTT	Command / setting relay	SmbOffsetUpTT: manually setting a TT of 5.38 mmol is used as a remote +0.1 nudge on ApsAutoIsfSmbOffsetOverride, clamped to a max of 1.50. Same pattern as SmbOffsetDownTT above.
93	StepsSteroidsOff	Steroid state	Code port of "Steps Steroids OFF": asserts Steroids=Steroids Off based on sustained activity with moderate IOB, controlled glucose, and no carbs. Per user confirmation, this is the automation that actually sets the Steroids state (previously only ever read as a precondition elsewhere in this file, never set) — Steroids ON is a manual user action, no automated counterpart needed.
94	T80Off3ok	Operational automation	T8.0Off3ok: exit criteria for the manually-set "hyp" temp target (8.0mmol for 20 min via GUI, per user confirmation — a separate TT from the 6.5-7.6mmol Activity band above). The screenshot's "Temp Target" bound checks were a misread per user correction: the real check is glucose reaching the TT's own 8.0mmol value, so this guards on the TT itself (~8.0mmol, same ±1.8mg/dL tolerance style as the 5.7mmol block). The inner OR keeps its original two-branch tradeoff — a lower glucose floor (6.5mmol) needs a steeper rise (0.3mmol) to compensate, vs a higher floor (7.0mmol) only needing a gentler rise (0.2mmol) — with no extra top-level glucose gate stacked on top of it. Original action list was just "Send SMS" + "Stop temp target". 5-min floor throttle added on top of the TT-existence self-guard (see readyToRun() usage note).
95	TT57Reversal	Operational automation	TT 5.7 reversal block: replaces TToff2/3/4/5 and HypoTTOff1 automations, plus TT5.8New1/2/3 — screenshotted as "TT=5.8mmol" but ported here against the same 5.7 mmol guard per user confirmation (nothing in this file ever sets a 5.8 mmol TT; CarbsStopTT5.7's own cb3 branch already treats 102.7-104.5 mg/dL, i.e. 5.7-5.8 mmol, as the same TT). Primary guard: activeTtMgdl() must be ~5.7 mmol/L. Once TT is cancelled the guard fails, so these conditions self-prevent re-firing without needing readyToRun() throttle. All glucose/delta thresholds in mg/dL; originals in mmol/L in comments. 5-min floor throttle added on top of the TT-existence self-guard (see readyToRun() usage note).
96	Test3	Validation/test	Code port of the "Test3" automation (MJ=MJ6): validates the state-check -> notification -> state-set mechanics, including the graph-announcement fix (addGraphAnnouncement). Original screenshot's "Min interval (min): 1" is below the 5-min floor now applied across all ported automations that lack their own interval, so this uses 5 rather than 1. Self-guarding: the state-set action moves MJ away from MJ6, so it fires once per cycle that reaches MJ6.
97	TodOffset0002DownTT	Command / setting relay	TodOffset*TT (8 pairs, T1-T8): manually setting a TT of 5.092-5.136 mmol is used as a remote +/-0.1 nudge on one of the 8 fixed time-of-day varOffset windows (ApsAutoIsfTodOffset*, see DetermineBasalAutoISF.kt), clamped to +/-2.0 — not a real target. T1=00-02h, T2=02-04h, T3=04-06h, T4=06-09h, T5=09-12h, T6=12-18h, T7=18-22h, T8=22-00h. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
98	TodOffset0002UpTT	Command / setting relay	TodOffset*TT (8 pairs, T1-T8): manually setting a TT of 5.092-5.136 mmol is used as a remote +/-0.1 nudge on one of the 8 fixed time-of-day varOffset windows (ApsAutoIsfTodOffset*, see DetermineBasalAutoISF.kt), clamped to +/-2.0 — not a real target. T1=00-02h, T2=02-04h, T3=04-06h, T4=06-09h, T5=09-12h, T6=12-18h, T7=18-22h, T8=22-00h. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
99	TodOffset0204DownTT	Command / setting relay	TodOffset*TT (8 pairs, T1-T8): manually setting a TT of 5.092-5.136 mmol is used as a remote +/-0.1 nudge on one of the 8 fixed time-of-day varOffset windows (ApsAutoIsfTodOffset*, see DetermineBasalAutoISF.kt), clamped to +/-2.0 — not a real target. T1=00-02h, T2=02-04h, T3=04-06h, T4=06-09h, T5=09-12h, T6=12-18h, T7=18-22h, T8=22-00h. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
100	TodOffset0204UpTT	Command / setting relay	TodOffset*TT (8 pairs, T1-T8): manually setting a TT of 5.092-5.136 mmol is used as a remote +/-0.1 nudge on one of the 8 fixed time-of-day varOffset windows (ApsAutoIsfTodOffset*, see DetermineBasalAutoISF.kt), clamped to +/-2.0 — not a real target. T1=00-02h, T2=02-04h, T3=04-06h, T4=06-09h, T5=09-12h, T6=12-18h, T7=18-22h, T8=22-00h. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge

#	Current coded key	Type	Current code definition / criteria and action
			TTs above.
101	TodOffset0406DownTT	Command / setting relay	TodOffset*TT (8 pairs, T1-T8): manually setting a TT of 5.092-5.136 mmol is used as a remote +/-0.1 nudge on one of the 8 fixed time-of-day varOffset windows (ApsAutoIsfTodOffset*, see DetermineBasalAutoISF.kt), clamped to +/-2.0 — not a real target. T1=00-02h, T2=02-04h, T3=04-06h, T4=06-09h, T5=09-12h, T6=12-18h, T7=18-22h, T8=22-00h. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
102	TodOffset0406UpTT	Command / setting relay	TodOffset*TT (8 pairs, T1-T8): manually setting a TT of 5.092-5.136 mmol is used as a remote +/-0.1 nudge on one of the 8 fixed time-of-day varOffset windows (ApsAutoIsfTodOffset*, see DetermineBasalAutoISF.kt), clamped to +/-2.0 — not a real target. T1=00-02h, T2=02-04h, T3=04-06h, T4=06-09h, T5=09-12h, T6=12-18h, T7=18-22h, T8=22-00h. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
103	TodOffset0609DownTT	Command / setting relay	TodOffset*TT (8 pairs, T1-T8): manually setting a TT of 5.092-5.136 mmol is used as a remote +/-0.1 nudge on one of the 8 fixed time-of-day varOffset windows (ApsAutoIsfTodOffset*, see DetermineBasalAutoISF.kt), clamped to +/-2.0 — not a real target. T1=00-02h, T2=02-04h, T3=04-06h, T4=06-09h, T5=09-12h, T6=12-18h, T7=18-22h, T8=22-00h. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
104	TodOffset0609UpTT	Command / setting relay	TodOffset*TT (8 pairs, T1-T8): manually setting a TT of 5.092-5.136 mmol is used as a remote +/-0.1 nudge on one of the 8 fixed time-of-day varOffset windows (ApsAutoIsfTodOffset*, see DetermineBasalAutoISF.kt), clamped to +/-2.0 — not a real target. T1=00-02h, T2=02-04h, T3=04-06h, T4=06-09h, T5=09-12h, T6=12-18h, T7=18-22h, T8=22-00h. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
105	TodOffset0912DownTT	Command / setting relay	TodOffset*TT (8 pairs, T1-T8): manually setting a TT of 5.092-5.136 mmol is used as a remote +/-0.1 nudge on one of the 8 fixed time-of-day varOffset windows (ApsAutoIsfTodOffset*, see DetermineBasalAutoISF.kt), clamped to +/-2.0 — not a real target. T1=00-02h, T2=02-04h, T3=04-06h, T4=06-09h, T5=09-12h, T6=12-18h, T7=18-22h, T8=22-00h. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
106	TodOffset0912UpTT	Command / setting relay	TodOffset*TT (8 pairs, T1-T8): manually setting a TT of 5.092-5.136 mmol is used as a remote +/-0.1 nudge on one of the 8 fixed time-of-day varOffset windows (ApsAutoIsfTodOffset*, see DetermineBasalAutoISF.kt), clamped to +/-2.0 — not a real target. T1=00-02h, T2=02-04h, T3=04-06h, T4=06-09h, T5=09-12h, T6=12-18h, T7=18-22h, T8=22-00h. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
107	TodOffset1218DownTT	Command / setting relay	TodOffset*TT (8 pairs, T1-T8): manually setting a TT of 5.092-5.136 mmol is used as a remote +/-0.1 nudge on one of the 8 fixed time-of-day varOffset windows (ApsAutoIsfTodOffset*, see DetermineBasalAutoISF.kt), clamped to +/-2.0 — not a real target. T1=00-02h, T2=02-04h, T3=04-06h, T4=06-09h, T5=09-12h, T6=12-18h, T7=18-22h, T8=22-00h. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
108	TodOffset1218UpTT	Command / setting relay	Member of the eight TodOffset TT pairs. A five-minute relay guard recognizes its exact 5.092-5.136 mmol command, changes the corresponding fixed time-of-day varOffset by 0.1 within the +/-2.0 clamp, cancels the command TT and records/notifies the result.
109	TodOffset1822DownTT	Command / setting relay	Member of the eight TodOffset TT pairs. A five-minute relay guard recognizes its exact 5.092-5.136 mmol command, changes the corresponding fixed time-of-day varOffset by 0.1 within the +/-2.0 clamp, cancels the command TT and records/notifies the result.
110	TodOffset1822UpTT	Command / setting relay	Member of the eight TodOffset TT pairs. A five-minute relay guard recognizes its exact 5.092-5.136 mmol command, changes the corresponding fixed time-of-day varOffset by 0.1 within the +/-2.0 clamp, cancels the command TT and records/notifies the result.
111	TodOffset2200DownTT	Command / setting relay	Member of the eight TodOffset TT pairs. A five-minute relay guard recognizes its exact 5.092-5.136 mmol command, changes the corresponding fixed time-of-day varOffset by 0.1 within the +/-2.0 clamp, cancels the command TT and records/notifies the result.
112	TodOffset2200UpTT	Command / setting relay	Member of the eight TodOffset TT pairs. A five-minute relay guard recognizes its exact 5.092-5.136 mmol command, changes the corresponding fixed time-of-day varOffset by 0.1 within the +/-2.0 clamp, cancels the command TT and records/notifies the result.
113	TwilightTH15Acce	Operational automation	Code port of "TwilightTH15Acce0.50": drops iobTH to 15% and acce weight to 0.50 during a quiet early morning (low steps, flat/falling BGL) while iobTH sits in the 17-39% band. No live-pump gate: the original's Note field was empty.
114	Usual2forTH	Operational automation	Usual2forTH70: restores iobTH=70 and acce weight=0.50 when BGL has recovered Replaces "Usual2forTH70 CurrProfReal0.35" automation. Guard: profile at 100% (not in 50% state), iobTH still reduced (<70), no TT, BGL >= 5.5mmol. Self-guarding: sets iobTH=70 so condition iobThresholdPercent<70 fails next cycle. 5-min floor throttle added on top of the self-guard (see readyToRun() usage note).
115	VirtualPseudoWizard	Bolus/post-bolus	Code port of "Steps Steroids OFF": asserts Steroids=Steroids Off based on sustained activity with moderate IOB, controlled glucose, and no carbs. Per user confirmation, this is the automation that actually sets the Steroids state (previously only ever read as a precondition elsewhere in this file, never set) — Steroids ON is a manual user action, no automated counterpart needed. Experimental correction bolus for retrospective testing on Virtual Pump only. This is intentionally hard-gated twice (eligibility and immediately before queueing) so installing this branch on a real-pump build cannot deliver it. It runs the normal constrained Wizard in the background with a pseudo-COB input, but records carbs=0: no fake carb treatment is made.
116	WizardPctDownTT	Command / setting relay	WizardPctDownTT: manually setting a TT of 5.046 mmol is used as a remote -5 nudge on OverviewBolusPercentage (Wizard bolus %), clamped to a min of 50 — not a real target. Same pattern/tight 0.0001mmol tolerance as the other settings-nudge TTs above.
117	WizardPctUpTT	Command / setting relay	WizardPctUpTT: manually setting a TT of 5.048 mmol is used as a remote +5 nudge on

#	Current coded key	Type	Current code definition / criteria and action
			OverviewBolusPercentage, clamped to a max of 100. Same pattern as WizardPctDownTT above.
118	iobTHDaytimeFloor	Hypoglycaemia protection	iobTHDaytimeFloor: enforces a daytime iobTH% floor so it can't stay stranded at a back-off level (as seen: stuck at 18% → SMBs throttled, low IOB). During the day, in a clearly-normal state, restore iobTH to the usual 70% whenever it has dropped to 50% or below. rescues from <= 50% (bumps 12/15/16/18/…/50 up to 70; leaves intentional 51+ alone) requires profile 100%, LowBG=NO50rec (not in a back-off), no active TT, Steroids Off, runs 08:00-22:00 only; EvCap owns 22:00-midnight, so the two cannot raise/lower iobTH against each other. BGL >= 6.5 mmol and not falling. 30-min throttle.

Appendix A — Installation and explanatory information [↑ TOC]

This fork builds as four separate Android apps ("product flavors"), each with its own application ID, its own icon, and its own independent SharedPreferences store. They install side by side on the same device without conflicting — a real device commonly runs two or three of these at once (e.g. the main pump-driving app plus one or two Client mirrors on other phones/tablets).

Because each flavor is a genuinely separate app, a settings export/import done inside one flavor never touches another flavor's preferences, even on the same physical device — "import into AAPS" and "import into AAPSClient" are two unrelated operations.

a. Version tag & build identification [↑ TOC]

Every console/history line and the on-device log stamp the running build as the last 13 characters of Config.VERSION_NAME (e.g. "aisf321UK_654"), derived automatically from the actual APK version rather than a hand-maintained literal. Known caveat: on the aapsclient/aapsclient2 flavors, VERSION_NAME gets an extra "-aapsclient"/"-aapsclient2" suffix appended, which shifts the fixed 13-character window — the printed tag can look slightly off on Client builds specifically. Not yet fixed as of this writing.

b. Per-device backup/export folder naming [↑ TOC]

The optional "GeneralPatientName" setting scopes where this device's AIV/log exports land — both on-device folder structure (Documents/AAPS/aapsLogs/<Name>/) and the PC-side backup tree pulled off the phone (aiv_<Name>, logs_<Name>, aapsLogs\<Name>\output\). Leaving it blank falls back to a single shared, unscoped folder — fine for one device, ambiguous the moment a second device is added. Set a distinct name per physical/virtual device before its exports start accumulating, since retroactively renaming only affects future exports, not the ones already written under the old name.

Appendix B — Changelog [↑ TOC]

Concise dated index of significant changes. Detailed explanations have been integrated into the relevant topical sections so each feature is described where it is used.

2026-08-19 [↑ TOC]

Automation-state bootstrapping was made self-healing and given evidence-based neutral defaults. Full operational detail is now incorporated into Section 1.

2026-08-22 [↑ TOC]

Bolus and Quick Wizard controls, Fast-Rise signal gating, and FastRise reporting were updated. The current behaviour and historical-export cautions are incorporated into Sections 8 and 15.

2026-08-23 [↑ TOC]

Tier 3 safeguards, List controls, coded profile roles, UKF1 options, SensorAge diagnostics, graph labels, battery recovery, and LowBG recovery were updated. Details are incorporated into their corresponding topical sections and Appendix F.

Appendix D — Alphabetical CarePortal Note Code Reference [↑ TOC]

Compiled directly from every addCarePortalNote()/addGraphAnnouncement() call site in OpenAPSAutoISFPlugin.kt (current-code audit dated 2026-08-21; cross-checked, not superseded, by this manual's own prose above). On/Off toggle pairs are merged into one entry; codes built with a numeric or block-identifying suffix (e.g. "MJoff-N", "compactSettingNote" prefixes like "D<val>") are collapsed to their base code, with the suffix's meaning explained in the entry itself. The free-text "VirtualPseudoWizard SUCCESS/FAILED" notes (§7b's sibling addition) are intentionally excluded — they are not part of this compact-code language.

- **130Off** — HighPP130Off: exits a 130%/110% profile boost (BGL settled with no TT, falling with a TT active, or simply no TT while still boosted) and restores the standard profile.
- **50ff-N (N=1–6)** — PP50.Off: exits the 50%-profile hypo-protection state once BGL has recovered; suffix identifies which of 6 recovery blocks matched (daytime stabilisation, fast rise with verified cannula, treated-hypo carbs+bolus, IOB resolved, overnight recovery, or the acce-weight-gate catch-all).
- **50pcTT** — "50pc makes5.7": while on the 50% profile with BGL dropping under 5.0mmol and still falling, sets a 150-minute 5.7mmol safety temp target.
- **50Rec** — 50SetRecent: flags LowBG=50recent whenever the profile drops back to 50% while previously marked recovered, so downstream low-BG logic knows a 50%-profile episode is (again) active.
- **A1** — Test-automation ports "MJ4"/"MJ5" (direct code-test triggers, fired via a manually-driven MJ automation state): advances the MJ state to NOMJremains (MJ5 additionally forces the standard profile for 30 min).
- **__________A4** — AlarmHypo2's first graph-announcement marker (TE.asAnnouncement mechanism, renders on the main graph): logs a hypo-alarm event — BGL ≤4.3mmol, or ≤5.5mmol with high recent activity, or a hypo-prediction match.
- **AC<val>** — AcceWeightDownTT/UpTT: remote ±0.05 nudge (via a manually-set 5.016/5.018mmol temp target) on the acce-ISF-weight baseline (ApsAutoIsfBgAccelWeightNormal); suffix is the new value.
- **Ac50** — ActivityProf50%: drops to a 50% profile for 180 minutes while the manually-set 6.8mmol "Activity" temp target is active and BGL is stable/falling.
- **Acce** — AcceUp0.5: raises the acce-ISF weight to its boosted level (ApsAutoIsfBgAccelWeightHigh) when it's currently low, BGL is stable/rising, no TT is active, and MJ is clear or steroids are on.
- **AcLT<role> / AcNS<role>** — AnyDesk remote-restart receipt notes (role = C/V/R for Client/Virtual Pump/Real pump): logs that this device accepted an AnyDesk-restart command, either a same-device local test (AcLT) or one relayed via the secondary-NS "ADesk" Note channel (AcNS); does not itself confirm Tasker/AnyDesk actually completed.
- **ACrv / PArv / PPrv** — PpWeightRevertUnder8_5: automatically reverts a boosted ppISF weight and/or acce weight back to baseline once BG < 8.5mmol, activity is high, or no genuine recent high (raw Libre >12.0mmol/48h) was seen; the code picks which of the two settings (or both, PArv) was reverted.
- **ActOff** — "ActivityOff": exit criteria for ActivityProf50%'s 50% profile drop — restores the standard profile, acce 0.35, iobTH 70% once the Activity TT ends/BGL stabilises or a bolus was just given.
- **ActTToff1** — Activity TT reversal branch "TT6.0New1" (6.5–7.6mmol Activity TT band): cancels the TT on an active-meal recovery (high/rising BG with COB and IOB headroom).
- **ActTToff2** — Activity TT reversal branch "TT6.0New2": cancels the same TT on a quiet/settled recovery (moderate BG, very low IOB, flat-or-rising).
- **AcTT<role>** — AnyDeskRestartActionTT: second, redundant AnyDesk-restart trigger channel via a dedicated 5.178mmol temp target (Client's list2 button); logs receipt of that TT alongside its own independent Tasker broadcast.
- **AH<val>** — AcceWeightHighDownTT/UpTT: remote ±0.01 nudge (via a 5.062/5.064mmol TT) on the boosted acce-ISF weight (ApsAutoIsfBgAccelWeightHigh); suffix is the new value.
- **bat>1** — BatteryOver1%: phone battery recovered above 1% while on the battery-drop profile ("Current Profile50") → restores the standard profile and sets automation state Profile=AllOK.

- **Batt1%** — graph-announcement companion to Bt<1% (see below): fires in the same Battery1% block, rendered on the main graph via the announcement mechanism.
- **BMild** — BolusGivenMild: strengthens the SMB delivery ratio (tiered by BGL) on a mild delivery-driven BG rise without touching profile/iobTH, holding a brief 5.0mmol target for 2 minutes.
- **BMildFS** — BolusGivenMildFailsafe: safety-net sibling of BolusGivenMild for when SMBs simply haven't been delivering at all during a strongly-confirmed rise (near-zero IOB, 20-minute SMB silence) — same delivery-ratio boost response.
- **Bol2** — "Bolus2": sets a brief 6.8mmol temp target after a bolus, either on a fast post-bolus BG fall (with COB/IOB present) or when carbs are up while MJ is active.
- **BsUp** — "BasalUp": raises acce weight back to neutral (0.50) and switches from the Low to the Standard profile once BGL is stable/rising outside an overnight/MJ guard window and activity is low.
- **Bt<1%** — Battery1%: phone battery ≤1% while on a normal running profile (PP130/C100/AllOK state) → switches to Current Profile50 and alerts (live pump only, not Virtual Pump).
- **BTgOn / BTgOff** — BoostToggleTT: manually setting a 5.08mmol TT remotely toggles ApsAutoIsfBoostAutomationsEnabled, the master switch for the BolusGiven/BolusGivenMild boost automations.
- **CGrph** — CleanGraphTT: manually setting a 5.042mmol TT remotely requests the "clean main graph" display combo (no SMB labels/arrowheads, solid green line) on the next OverviewFragment refresh.
- **CLup** — CloudLogsUploadTT: manually setting a 5.140mmol TT remotely triggers the same log upload as the Maintenance screen's "Send logs" button — for AAPSClient users with no direct settings access.
- **COff1-N (N=1–2)** — CarbsTHoff: restores iobTH to 70% (and acce to 0.50) when a post-carb BGL is falling or shows a mid-range anomaly; suffix identifies which of 2 condition blocks matched.
- **Coff2-N (N=1–3)** — CarbsStopTT5.7: cancels the 5.7mmol hypo-protection TT once carbs/IOB/BGL evidence shows recovery; suffix identifies which of 3 condition blocks matched.
- **Coff4** — "carbsStopTT1ok4.4": cancels a very-low (≤4.4mmol) TT once carbs are active (COB≥10) and BGL is rising with low IOB, 10:00–22:00.
- **D<val>** — DuraWeightDownTT/UpTT: remote ±0.1 nudge (via a 5.022/5.024mmol TT) on the duraISF-weight baseline (ApsAutoIsfDuraWeightNormal); suffix is the new value, formatted to 3 trailing characters since this setting's range runs to 3.00.
- **______deadpod2hrs?** — "OldPod" graph announcement: fires once a pod is >60h old AND BGL has been continuously high (raw Libre >11.0mmol OR AAPS BGL >10.0mmol) for 2+ hours — reads as pod failure, never touches dosing.
- **DelOff** — SMB-delivery-ratio reset: restores the resting SMB delivery ratio baseline once no boost TT is active (fires once as a boosted ratio drops back to baseline).
- **DuraRsc** — OvernightDuraRescue: a narrow, one-shot 02:00–04:00 switch back UP to the Standard profile when duraISF genuinely dominates a plateaued overnight high on the Low profile, with no stacking and no recent low.
- **EvCap** — EveningIobCeiling: caps iobTH at 45% from 22:00–00:00 whenever it's currently higher, independent of BGL direction (built after the 27 Jul 2026 evening SMB-stacking incident).
- **Eve** — "EveningTH": lowers acce weight to 0.45 and caps iobTH at 45% in the evening (and switches to the Low profile when a low is predicted, or unconditionally 22:00–06:00) while MJ is still clear.
- **G2On / G2Off** — Graph2ToggleTT: manually setting a 5.138mmol TT remotely toggles the "graph2" carb-model-curve display.
- **G5On / G5Off** — Graph5ToggleTT: manually setting a 5.142mmol TT remotely toggles graph5, a fixed clone of the main graph minus basal.
- **Giv-N (N=1–3)** — BolusGiven71_0.70: post-bolus response that boosts iobTH to 71%, restores acce to normal, and sets 110% profile for 2 minutes; suffix identifies which of 3 trigger blocks matched (manual-bolus rise, high-BGL-with-COB rise, or delivery-driven rise with no recent manual bolus/carbs).
- **Gntl5** — GentleHypoRisk: escalates hypo protection from the 50%-prepare state (acce 0.07) to the deeper "Skittles" state (acce 0.02, iobTH 50%) using UKF-smoothed raw CGM signals; also fires a graph announcement ("________________Gentle5") for the same event.

• **H4** — AlarmHypo2's second graph-announcement marker, fired alongside __________A4 in the same block (see below for AlarmHypo1's own "_____H4").

• **_____H4** — AlarmHypo1's graph-announcement marker: logs a hypo-alarm event (slow decline, an unconditional emergency floor at 3.0mmol, or a steps-driven variant) that drops acce weight to 0.10 and marks a recent low.

• **HardStackDelOff** — forced SMB-delivery-ratio reduction (baseline−0.03, not a full reset) whenever genuine SMB stacking is detected (short average gap AND high count in the last 5 minutes), independent of TT state.

• **HI<val>** — HigherIsfRangeWeightDownTT/UpTT: remote ±0.1 nudge (via a 5.068/5.070mmol TT) on the higher-ISF-range weight (ApsAutoIsfHighBgWeight); suffix is the new value.

• **HnAM** — "HighNight00AM" — currently DISABLED in code (highNight00AmEnabled = false): would raise iobTH to 51%/acce to 0.50 and set a 4.2mmol TT to correct a genuine overnight high, but is turned off because it fights NightIobCeiling.

• **IP<val>** — PeakInsulinTimeDownTT/UpTT: remote ±5-minute nudge (via a 5.074/5.076mmol TT) on the configured insulin peak-activity time (InsulinOrefPeak); suffix is the new value in minutes, zero-padded to 3 digits.

• **L<val>** — LibreOffsetDownTT/UpTT: remote ±0.05 nudge (via a 5.032/5.034mmol TT) on the Libre calibration-offset baseline (ApsAutoIsfLibreOffsetOrig); suffix is the new value, allowed to run longer than the usual 5 characters since this setting is ≥1.0.

• **L10Off** — LowRaw24Cal ending: reverts the temporary Libre slope/offset override (0.75/1.3) once raw Libre BGL is no longer continuously below 10.0mmol for 24h.

• **L10On** — LowRaw24Cal starting: applies a temporary Libre slope/offset override (0.75/1.3) once raw Libre BGL has been continuously below 10.0mmol for 24h.

• **LibreLd / Set1Ld** — LibreVsSet1Race: logs which of the LibreSpecial/UKFset1 smoothing types is first to cross a -0.2mmol/5min drop threshold in a given episode (only the leader fires, once per episode); distinct 5-char prefixes keep them distinguishable under the graph's note-label truncation.

• **LS<val>** — LibreSlopeDownTT/UpTT: remote ±0.01 nudge (via a 5.026/5.028mmol TT) on the Libre calibration-slope baseline (ApsAutoIsfLibreSlopeOrig); suffix is the new value.

• **MB<val>** — MildBoostDownTT/UpTT: remote ±0.01 nudge (via a 5.052/5.054mmol TT) on the mild-boost SMB delivery ratio alone (ApsAutoIsfMildBoostRatio), distinct from the paired SB/SM nudge below; suffix is the new value.

• **MJ** — graph announcement fired alongside "MJ active" (below) when the direct MJ-cycle START user action runs; renders on the main graph via the TE.asAnnouncement mechanism instead of a plain note.

• **MJ active** — direct MJ-cycle START user action: begins an MJ (Mounjaro) dosing cycle — acce 0.35, iobTH 70%, switches to the Low profile — and sets automation state MJ=MJ active.

• **MJ2** — old MJ-cycle port: advances MJ state from "MJ active" to MJ2 during 02:10–03:10.

• **MJ3** — old MJ-cycle port: advances MJ state from MJ2 to MJ3 during 01:05–02:05.

• **MJBOn / MJBOff** — MjKotlinButtonsToggleTT: manually setting a 5.164mmol TT remotely toggles whether the Kotlin-ported MJ direct-action buttons are shown/enabled.

• **MJoff-N (N=1–2)** — old MJ-cycle port "MJoff": exits the MJ cycle from MJ3 back to NOMJremains, either on high BGL during the day (block 1, 12:00–21:04, BGL≥10.5mmol) or at midnight (block 2, 00:00–01:00).

• **MJrec** — "MJ recent or Steps CurrProf Acce": overnight (00:00–08:00) tuning applied while MJ is still active and the pod is fresh (<72h), setting acce 0.50/iobTH 70% and (HP2-gated) switching to the Low profile.

• **MJs3** — MjStateMj3TT: manually setting a 5.146mmol TT forcibly sets the MJ automation state directly to MJ3 (manual override, paired with MJsNO as a "set it now" control).

• **MJsNO** — MjStateNoMjTT: manually setting a 5.144mmol TT forcibly sets the MJ automation state directly to NOMJremains (manual override).

• **ML<val>** — AutoIsfMaxLowDownTT/UpTT: remote ±0.1 nudge (via a 5.080/5.082mmol TT) on autoISF_max_low (ApsAutoIsfMaxLow — not yet wired into any calculation); suffix is the new value.

• **MN<val>** — AutoIsfMaxNormalDownTT/UpTT: remote ±0.1 nudge (via a 5.086/5.088mmol TT) on the live autoISF_max ceiling (ApsAutoIsfMax); suffix is the new value.

• **MoreMJ** — "MoreMJ": advances MJ state to MJ3 when acce weight is very low, activity is low, BGL is falling on the 50% profile with no carbs and IOB still present (or, on the OR-path, no genuine recent high has been seen while MJ still allows it).

• **Night** — "NightAcce_0.35TH18": overnight (01:00–06:00) reset of acce to 0.35 and iobTH to 18% (plus an automatic settings-export backup) when iobTH is out of the 18% band, no carbs are active, and BGL is low.

• **No50** — "Not50%Recently": clears the LowBG=50recent flag once the profile is back at 100% and BGL is rising (delta ≥0.1mmol).

• **NOMJremains** — direct MJ-cycle RESTORE user action: ends the MJ dosing cycle (restores standard profile, acce 0.50, iobTH 70%), sets automation state MJ=NOMJremains.

• **NtCap** — NightIobCeiling: caps iobTH at 18% and acce at 0.35 from 00:00–06:00, independent of BGL, whenever either is currently looser (built after the 6→7 Aug 2026 hypo incident).

• **off120** — "High6PPoff": exits the 120% profile boost once no TT exists at 120%, or BGL/delta have stabilised in the evening window at 120%.

• **OffP-N (N=1–2)** — "OffHighProf": overnight, when NOT on the Low profile and BGL is falling with a predicted low, drops to acce 0.18/iobTH 18% (and switches to the Low profile unless a rescue is active); suffix identifies which of 2 condition blocks matched.

• **Old** — "HighOldPod": sets a brief 5.0mmol TT + 110% profile boost when the cannula is stale (≥60h) or very fresh (≤6h) and BGL is high, rising, with a bolus ≥90 min ago.

• **OldSensor<tier>** — OldSensorAdj: tiered Libre slope/offset compensation for an aging (11–15 day) or very-new (0–3 day) sensor when raw Libre BGL has recently been >12.0mmol; tier suffix (NewDay1/NewDay2/NewDay3 for sensor days 0–3, or 1/2/3 for days 11–15) identifies which age band matched.

• **OldSensorOff** — OldSensorAdj reverting: restores the configured Libre slope/offset baseline once outside all the OldSensorAdj age tiers (or the latch was stale relative to the live values).

• **P120** — "High6PP": brief 120% profile boost (3 condition blocks) when BGL is high with controlled delta and low/no TT tolerance, gated on a genuine recent high (raw Libre >12.0mmol/48h).

• **PH<val>** — PpWeightHighDownTT/UpTT: remote ±0.01 nudge (via a 5.056/5.058mmol TT) on the boosted ppISF weight (ApsAutoIsfPpWeightHigh); suffix is the new value.

• **Pod130** — "PodChangeHighPP130": brief 130% profile boost around a pod change (fresh ≤2h or stale ≥78h) when BGL is high and rising, 10:00–18:00, gated on a genuine recent high.

• **_____POD2** — "POD 78 hours": pod age reaches ~78 hours during 08:00–23:59 → switches to the Standard profile (Current ProfileReal), sets automation state Profile=PP130, and sends an SMS alert; unlike its sibling POD 79 h, this one is a real profile-switching action, not just a reminder (live pump only).

• **POD 79 h** — "POD 79 hours" reminder (graph announcement, no dosing action): pod age has reached ~79 hours, 07:00–23:59.

• **PP<val>** — PpWeightDownTT/UpTT: remote ±0.01 nudge (via a 5.012/5.014mmol TT) on the ppISF-weight baseline (ApsAutoIsfPpWeightNormal); suffix is the new value.

• **PreSoak24hrs** — "PreSoakSENSOR24hrs" reminder (graph announcement): reminds to pre-soak a new sensor at ~14.0 or ~14.5 days into the current sensor's life (cannula age permitting).

• **PrfL** — ProfileLowTT: manually setting a 5.150mmol TT remotely switches straight to the Low profile ("set it now", not a lock — other automations can still change it afterwards).

• **PrfS** — ProfileStandardTT: manually setting a 5.148mmol TT remotely switches straight to the Standard profile.

• **PstRs** — "PersistentRiseRelease": clears the way for normal dosing (cancels any active TT, forces the mild-offset-zero flag) once a genuinely sustained mild rise (delta ≥+0.10mmol, zero SMB delivered) has held continuously for ≥10 minutes.

• **pTTOff** — "RecentPodOff": relaxes acce weight back to baseline (and switches to the Standard profile) once a RecentPod/OldPod2 safety boost has ended and acce is still at its boosted level.

• **RecPod** — "RecentPod": brief 130% profile boost + 4.2mmol TT + boosted acce weight when the cannula is very fresh or stale, carbs are up, and BGL is rising with IOB headroom, gated on a genuine recent high (live pump only).

• **SACOn / SACOff** — SensorAgeCodeToggleTT: manually setting a 5.156mmol TT remotely toggles the master enable switch for the whole SensorAge Libre-calibration routine (distinct from STgOn/STgOff's narrower preference toggle).

• **SB<val>** — SmbDeliveryDownTT/UpTT: remote ±0.01 nudge (via a 5.002/5.004mmol TT) on the SMB delivery ratio baseline (ApsAutoIsfSmbDeliveryBaseline), written together with the paired SM nudge (see below) by the same TT; suffix is the new value.

• **Semi** — "SemiTwilightAcce_0.50TH16": drops acce weight to 0.50 and iobTH to 16% across 3 early-morning branches (sustained activity, very-high BGL with low acce, or a stable rising mid-morning window).

• **SensorAgeCodeOff** — OldSensorAdj master-disable path: restores the Libre baseline (and clears the active flag) if the whole SensorAge routine is disabled while an override was still active.

• **Set50-N (N=1–4)** — "prepare Set50%": drops into the 50%-profile hypo-protection state (acce 0.07, iobTH 50%, Low profile for 6h) on falling BGL; suffix identifies which of 4 condition blocks matched (dual-delta confirmation, exercise+IOB risk, native low-BGL fallback, or pre-sleep falling).

• **_____S1hr** — "SENSOR at 14.9 days" reminder (graph announcement): the current sensor is nearing its ~15-day replacement point (possible new-sensor overlap warning).

• **_____S2hr** — "SENSOR at 14 days 22 hours due" reminder (graph announcement): a slightly earlier sensor-age checkpoint than _____S1hr.

• **Shwr12** — "Shower12": drops iobTH to 12% during a quiet morning rise (no steps, no COB, ≥180 min since last bolus) between 05:30–08:30.

• **SM<val>** — SmbDeliveryDownTT/UpTT (second value written alongside SB): remote ±0.01 nudge on ApsAutoIsfMildBoostRatio, made together with the SB baseline nudge by the same 5.002/5.004mmol TT.

• **SO<val>** — SmbOffsetDownTT/UpTT: remote ±0.1 nudge (via a 5.036/5.038mmol TT) on the SMB-offset override (ApsAutoIsfSmbOffsetOverride); suffix is the new value.

• **_____ST601k** — "Exercise limit Acce" alert (graph announcement): fires when a fast BG rise (≥0.4mmol/5min) coincides with very high recent step activity (≥1000 steps/60min).

• **StBOn / StBOff** — SteroidKotlinButtonToggleTT: manually setting a 5.166mmol TT remotely toggles whether the Kotlin-ported Steroid direct-action buttons are shown/enabled.

• **Steps Steroids OFF** — "Steps Steroids OFF": asserts Steroids=Steroids Off based on sustained activity (≥1000 steps/60min) with moderate IOB and controlled glucose — the only automated writer of the Steroids-off state; Steroids ON is always a manual action.

• **Steroids130** — direct Steroid-button INCREASE_130 action: steps the steroid dosing ladder up to the 130% tier (iobTH 73%, acce 0.70, switches to Steroid Profile130).

• **Steroids150** — direct Steroid-button INCREASE_150 action: steps the ladder up to 150% (iobTH 74%, switches to Steroid Profile150).

• **Steroids190** — direct Steroid-button INCREASE_190 action: steps the ladder up to 190% (iobTH 75%, switches to Current Profile190Real).

• **Steroids250** — direct Steroid-button INCREASE_250 action: steps the ladder up to 250% (iobTH 76%, switches to Current ProfileReal250).

• **SteroidsOff** — direct Steroid-button TURN_OFF action: exits the steroid dosing ladder, restores Current ProfileReal, iobTH 71%, acce 0.71, sets automation state Steroids=Steroids Off.

• **SteroidsON** — direct Steroid-button START_110 action: enables the steroid dosing ladder at its 110% tier (iobTH 71%, acce 0.70, switches to Steroid Profile110).

• **STgOn / STgOff** — SensorAgeToggleTT: manually setting a 5.006mmol TT remotely toggles the OldSensorAdj enable preference (ApsAutoIsfOldSensorAdjEnabled) — a narrower switch than SACOn/SACOff's master routine toggle.

• **StLow <free text>** — low-storage warning: the real-pump phone writes "StLow X.XGB free" as a plain CarePortal note once free space drops under 10GiB (re-arms above 10.5GiB); the Virtual Pump phone relays that same note as a graph announcement once it sees it via Nightscout (its relay logic also recognises a legacy "StorageLow " prefix, though nothing in this file currently writes that prefix itself).

• **Sub75Cap** — "Sub75HeavyDelivery": arms a 10-minute SMB cooldown once >1.0U has been delivered within a rolling 10 minutes while BGL is still under 7.5mmol.

• **Sub75Clr** — Sub75HeavyDelivery early release: clears that cooldown early if BGL has already recovered above 7.5mmol and is climbing fast (delta >0.7mmol).

• **T<sign><val>** — TodOffset*DownTT/UpTT (8 window pairs spanning T1–T8, the 8 fixed time-of-day windows 00-02h/02-04h/04-06h/06-09h/09-12h/12-18h/18-22h/22-00h): remote ±0.1 nudge on one of the 8 ApsAutoIsfTodOffset* preferences via its own dedicated TT (5.092–5.136mmol); the note text carries only the resulting signed value, not which of the 8 windows changed.

• **_____Test3** — "Test3" validation port: exercises the state-check → notification → graph-announcement mechanics end-to-end via a test-only MJ=MJ6 automation state (no real dosing effect).

• **THfloor** — "iobTHDaytimeFloor": restores iobTH to 70% during the day (08:00–22:00) whenever it's stuck at ≤50% with no active low/falling BGL/TT reason to be there.

• **TOD0 <reason>** — automatic clearing of one or more time-of-day ISF offsets, once a sustained (≥10min) condition (MJ state change, HP2 crossing 4.0/6.0, or a Steps-state trigger) makes the existing offset direction no longer appropriate; the reason text names which trigger fired.

• **TT3-<block>** — "Skittles" hypo-risk: sets a 5.7mmol/180min safety temp target across 7 condition blocks (A–G) ranging from an emergency-floor fallback to graded IOB/delta-based fall detection.

• **TT8.0off** — "T8.0Off3ok": cancels the manually-set 8.0mmol "hyp" temp target once BGL is confidently rising (with an activity/steps gate) and low-activity confirmed.

• **TToff-<which>** — TT57Reversal (replaces TToff2-5/HypoTTOff1/TT5.8New2-3/mildConfirmed): cancels the 5.7mmol hypo-protection TT once one of several recovery patterns is confirmed; suffix (off1–off5, N2, N3, or Mld) identifies which branch matched — off1 is the early-catch "HypoTTOff1" pattern, off2–off5 are the graded TToff2-5 recovery levels, N2/N3 are the TT5.8New2/3 high-G/low-IOB catches, and Mld hands off cleanly to BolusGivenMild's own boost.

• **TWi** — "TwilightTH15Acce0.50": drops iobTH to 15% and acce to 0.50 during a quiet early morning (06:00–08:00, low steps, flat/falling BGL) while iobTH sits in a mid-range (17–39%) band.

• **UamBst** — "UAM Boost" Tier 3 notification: reports that DetermineBasalAutoISF's own SMB-boost logic fired this cycle. DetermineBasalAutoISF.kt is a pure calculation class with no I/O of its own — it only sets a flag (`uamBoostFiredThisCycle`); OpenAPSAutoISFPlugin.kt reads that flag back after calling `determine_basal()` and is the one place that actually writes the SMS/CarePortal note. Added 2026-08-22: also raises a graph-only "B" annotation via addGraphAnnouncement() alongside this note -- no additional SMS or alert, purely a graph marker.

• **UKF1Block** — LibreUkf1ToggleTT blocked path: on a real pump (not Virtual Pump, or on AAPSCLIENT), the UKFset1 live-comparison toggle is forced off and the manual toggle attempt is blocked rather than honoured.

• **UKF1VOn / UKF1VOff** — LibreUkf1ToggleTT: on Virtual Pump only (and not AAPSCLIENT), manually setting a 5.152mmol TT toggles the UKFset1 smoothing live-comparison feature (FslUseUkfSmoothing).

• **UsuIP-N (N=1–3)** — "Usual2forTH70": restores iobTH to 70% and acce to 0.50 once BGL has recovered (≥5.5mmol, stabilised, Steroids Off); suffix identifies which of 3 recovery blocks matched.

• **W<val>%** — WizardPctDownTT/UpTT: remote ±5 nudge (via a 5.046/5.048mmol TT) on the Wizard bolus percentage (OverviewBolusPercentage); suffix is the new percentage, zero-padded to 3 digits.

• **X50-N (N=1–3)** — "Extra50%": deepens hypo protection (acce 0.07, iobTH 50%) when BGL is falling on the 100% profile; suffix identifies which of 3 condition blocks matched (steep multi-delta fall, moderate fall, or overnight falling 01:00–05:00).

Appendix E — List 1 and List 2 Temporary-Target Command Catalogue [↑ TOC]

The mmol/L numbers below are command signals, not clinical glucose targets. On a pump or Virtual build, List 1 and the clinical List 2 actions invoke the matching local handler directly and create no temporary target. On AAPSClient, the same selection creates a five-minute relay TT because preferences/actions do not sync directly; the pump receives the code, performs the action, cancels the relay target and applies its relay repeat guard. Every command still passes through its confirmation popup. List 2 graph controls are always local display settings and never use relay TTs.

List 1 - settings and command codes [↑ TOC]

List 1 availability note. Code 5.152 is offered only on a non-Client Virtual-Pump comparison device. Code 5.154 is intentionally unused: UKF2 graph history is maintained without switching the live dosing engine, so checking a graph option no longer changes the glucose source.

List 2 - actions, relay codes and local graph controls [↑ TOC]

Appendix F — Automation States In Use [↑ TOC]

A reference for every named state this fork's coded automations (and, for a few, native Automation-tab actions) read or write via the AutomationStateService — what it means, every automation that writes each value, and every automation that reads it as a gate. Generated by a full trace of every checkAutomationState()/setAutomationState()/automationStateService.* call site in OpenAPSAutoISFPlugin.kt, current as of the 2026-08-22 cleanup that removed three dead states (see §15g below).

a. MJ [↑ TOC]

Values: NOMJremains (default), MJ active, MJ2, MJ3, MJ4, MJ5, MJ6

Written by:

• handleDirectMjUserAction (Overview MJ button handler, not a readyToRun automation) — Action.START → "MJ active" (line 348); Action.RESTORE → "NOMJremains" (line 357)

• "MJ4" — MJ == MJ4 → "NOMJremains" (line 1922; port of the old "Test" automation)

• "MJ5" — MJ == MJ5 → "NOMJremains", also switches to Standard profile for 30 min (line 1930; port of "Test2")

• "Test3" — MJ == MJ6 → "NOMJremains" (line 1946; diagnostic port, also fires a URGENT notification + targeted SMS)

• "MJ2old" — MJ == MJ active AND time 02:10–03:10 → "MJ2" (line 1953)

• "MJ3old" — MJ == MJ2 AND time 01:05–02:05 → "MJ3" (line 1961)

• "MJoff" — MJ == MJ3 AND (12:00–21:04 with BG ≥10.5mmol, OR 00:00–01:00) → "NOMJremains" (line 1975)

• "MjStateNoMjTT" — manual remote-control TT code (≈5.144mmol) → "NOMJremains", unconditional (line 2799)

• "MjStateMj3TT" — manual remote-control TT code (≈5.146mmol) → "MJ3", unconditional (line 2807)

• "MoreMJ" — complex daytime condition tree (low acce ≤0.11, low activity, BG falling <5.5mmol on 50% profile, no COB, IOB≥0.2, LowBG=50recent, etc.) OR an independent "no recent genuine high" path → "MJ3" (line 4734)

Read by:

• handleDirectMjUserAction / handleDirectSteroidUserAction — re-validate the MJ/Steroid button press is still valid before acting (lines 326, 380, 390, 396, 402, 408)

• "MJ4"/"MJ5"/"Test3"/"MJ2old"/"MJ3old"/"MJoff" — each keys off the current MJ value to know when to advance/reset the cycle (lines 1920, 1927, 1940, 1951, 1959, 1968)

• TOD-offset auto-clear block — nomjRemains gates both the negative- and positive-offset clearing logic (line 2690)

• "Extra50" blocks 1/2 — require an MJ cycle to be *active* (!NOMJremains) as part of the fall-protection trigger (line 3423)

• "HighOldPod" — requires NOMJremains (line 3588)

• "HighNight00AM" branch 2 — requires NOMJremains (line 3659; block is permanently disabled via highNight00AmEnabled = false)

• "BolusGiven" bg2 branch — requires MJ NOT NOMJremains (an MJ cycle active) (line 3839)

• mjActive() helper (checks MJ == "MJ active" specifically) — blocks "BolusGiven" bg3 and "BolusGivenMild"'s mildConfirmed while MJ is freshly active (lines 3902, 4223)

• "MJrecentCurrProfAcce" — requires MJ NOT NOMJremains as one of several gates (line 4641)

• "BasalUp" — NOMJremains is one OR-branch of its cannulaOrStateOk gate (line 4670)

• "MoreMJ" — requires NOMJremains in both its main condition tree and its independent noRecentHighTrigger path (lines 4710, 4731)

• "AcceUp0.5" — requires NOMJremains OR SteroidsON (line 4854)

• "EveningIobCeiling" — requires NOMJremains (line 4974)

• "EveningTH" — evB1/evB2 require MJ NOT "MJ active"; evB3 requires "MJ active" (lines 5047, 5051, 5054)

• "Bolus2" branch b2 — requires "MJ active" (line 5213)

• autoIsfSettingsSnapshot() — logs the current MJ value into the AAPSClient settings snapshot (diagnostic only, line 5838)

Summary: MJ tracks the multi-stage state of a specific manual dosing protocol ("MJ 0.35/.70"), triggered by an Overview button press. NOMJremains is the neutral/default value; pressing the button advances it to "MJ active", and a chain of time-gated automations (MJ2old→MJ2, MJ3old→MJ3, MJoff→NOMJremains) walks it back down across the night, while the remaining numbered values (MJ4/MJ5/MJ6) are wired to test/diagnostic automations rather than the real cycle. The state is read constantly throughout the file as a gate: dozens of unrelated automations (evening/overnight profile, acceleration-weight and iobTH adjustments, post-bolus SMB boosts, TOD-offset clearing, basal-up recovery) branch on whether an MJ cycle is currently in progress, because active MJ dosing changes what "normal" dosing behaviour should look like.

b. Steroids [↑ TOC]

Values: Steroids Off (default), SteroidsON

Written by:

• handleDirectSteroidUserAction Action.START_110 (Overview steroid button) → "SteroidsON" (line 424)

• handleDirectSteroidUserAction Action.TURN_OFF (Overview steroid button) → "Steroids Off" (line 468)

• "StepsSteroidsOff" — sustained steps ≥1000 in the last hour, IOB 2.0–4.0U, BG ≤8.0mmol rising ≥0.5mmol, no COB → "Steroids Off" (line 5360). Per the source comment this is the *only* coded automation that ever asserts this state; turning steroids ON is always a manual user action with no automated counterpart.

Read by (all as a "steroids not currently active" precondition, unless noted):

• handleDirectSteroidUserAction's own self-checks for each escalation step 110→130→150→190→250→off (lines 381, 384, 389, 395, 401, 407)

• "SkittlesHypoRisk" block F (line 3350)

• "iobTHDaytimeFloor" (line 3402)

• "OffHighProf" (line 3513)

• "Shower12" (line 3614)

• "HighNight00AM" (disabled block) (line 3649)

• "Usual2forTH" (line 4469)

• "CarbsTHoff" (line 4517)

• "MJrecentCurrProfAcce" (line 4639)

• "AcceUp0.5" — OR'd with MJ NOMJremains (line 4854)

• "TwilightTH15Acce" (line 5104)

• "NightAcce" (line 5126)

• "SemiTwilightAcce" — all 3 branches (lines 5159, 5161, 5165)

Summary: Steroids records whether the user is currently on a steroid course, which materially raises insulin resistance and warrants the much higher iobTH/profile-percentage escalation ladder reachable via the manual 110%→130%→150%→190%→250% steroid buttons. Nearly every "back off"/hypo-protective automation elsewhere in the file (the Twilight/SemiTwilight/NightAcce resets, CarbsTHoff, Usual2forTH, OffHighProf, Shower12, etc.) gates on Steroids Off so it can't accidentally undo steroid-driven dosing; only one coded automation (StepsSteroidsOff) can clear it automatically, based on sustained-activity evidence that resistance has resolved.

c. LowBG [↑ TOC]

Values: 50recent, NO50rec (default)

Written by:

• "PrepareSet50" → 50recent (line 2015; deep-hypo entry: acce→0.07, iobTH→50, forces the Low profile)

• "PP50Off" → NO50rec (line 3306; 6-block recovery exit from the 50% prepare state)

• "SkittlesHypoRisk" → 50recent (line 3362; 7-block hypo-risk TT-5.7 set)

• "50SetRecent" → 50recent (line 3377; fires when profile actually dropped to 50% while state was still NO50rec)

• "Not50Recently" → NO50rec (line 3386; profile back at 100% and BG delta ≥0.1mmol -- as of 2026-08-23, also requires BG ≥5.5mmol; see the summary below)
• "Extra50" → 50recent (line 3435; deepens hypo protection on a confirmed fall)
• "TT57Reversal" off1–off5 branches → NO50rec (line 4233; the 5.7mmol TT genuinely ended/recovered)
• "CarbsStopTT57" → NO50rec (line 4450)
• "Usual2forTH" → NO50rec (line 4504)
• "CarbsTHoff" → NO50rec (line 4540)
• "HighPP130Off" → NO50rec (line 4829; also resets Profile to C100)
• "AlarmHypo1" → 50recent (line 5296; any of 4 hypo-alarm branches)
• "AlarmHypo2" → 50recent (line 5341; low-glucose + recent-steps alarm)

Read by:

• "PP50Off" gate — must be 50recent (line 3265)
• "50SetRecent" gate — must be NO50rec (plus profile==50) (line 3376)
• "Not50Recently" gate — must be 50recent (plus profile==100) (line 3384)
• "iobTHDaytimeFloor" gate — must be NO50rec (line 3401)
• "Extra50" gate — must be NO50rec (line 3418)
• "OvernightDuraRescue" — noRecentLow requires NOT 50recent, excluding rebound-highs from the rescue (line 3486)
• DetermineBasalAutoISF.kt's recent-low rebound guard — fed via the recentLowActive = checkAutomationState("LowBG","50recent") constructor param (line 5732)
• BolusWizard (external, per source comment) also reads this same flag to halve carb-bolus insulin after a recent low
• rt.reason export line, appended every cycle regardless of outcome: "LowBGrecent: Y/N" for the AIV history/CSV export (line 5807)

Summary: LowBG is the "was there a recent low" flag driving the file's 50%-profile hypo-protection state machine, and it also feeds two safety guards outside this state machine: the recent-low rebound guard in determine_basal() and BolusWizard's carb-bolus halving. It is set to 50recent whenever any of several hypo-risk/alarm automations fire (PrepareSet50, Skittles, Extra50, AlarmHypo1/2, 50SetRecent) and cleared back to NO50rec only once BG has genuinely recovered (PP50Off, Not50Recently, TT57Reversal, CarbsStopTT57, Usual2forTH, CarbsTHoff, HighPP130Off) — deliberately maintained as an actively-cleared two-way flag (unlike the removed BGLstate one-way latch), so downstream consumers can trust a "not recent" reading as much as a "recent" one.

The BG floor (≥5.5mmol) was chosen from real evidence: checked against all LowBG Y-to-N clear events across a real 4-day sample, clears below 5.5mmol were consistently followed by a large rebound correction, clears at or above 6.0mmol generally weren't. Since this flag gates the recent-low rebound guard (SMB halved while armed), keeping it armed longer means a correction landing during that window delivers at half strength rather than being skipped -- see the changelog for the full evidence writeup.

BG floor added 2026-08-23. Checked against all 20+ LowBG Y-to-N clear events across a real 4-day sample (20-23 Aug): the delta-only bar cleared the flag on the very first uptick off a low, often while BG was still clearly in low/low-normal range (4.2-5.0mmol seen repeatedly) -- not once the low had actually resolved. Clears below 5.5mmol were consistently followed by a large (≥0.8U) correction into a continuing rebound; clears at or above 6.0mmol generally weren't.

Worst confirmed case: 23 Aug, cleared at 4.9mmol/11:07am, an autonomous carbless rebound (COB=0.0 throughout) climbed unprotected to 7.6mmol by 12:41pm, ~1.2U was delivered, and that IOB decayed BG back down to 5.6mmol -- still falling -- three-plus hours later. Since this same flag gates DetermineBasalAutoISF.kt's recent-low rebound guard (halves any SMB while COB>0 or an unannounced-carb-impact pattern is detected), keeping it armed longer means a real correction landing during that window now delivers at roughly half strength instead of full, rather than skipping correction altogether.

d. Steps [↑ TOC]

Values: StepsLow, StepsHigh (no default declared)

Written by: none in this file. Read-only from this file — owned externally (native Automation-tab action or other integration), per the source's own bootstrap comment.

Read by:

• TOD-offset auto-clear block — stepsHigh alone can force clearNegativeTodOffsetsNow true (protective during activity-driven fall risk); stepsLow is one of several ANDed conditions (with NOMJremains and HP2 >6.0) required before clearPositiveTodOffsetsNow fires (lines 2714–2719). This is Steps' only use site in the file.

Confirmed misconfigured on the live phone (2026-08-22): the States tab there has no Steps state at all — instead there is a separate, self-named StepsLow state (state name StepsLow, holding value StepsLow), almost certainly from whatever native Automation-tab action writes step data being configured to set state StepsLow directly rather than state Steps with value StepsLow/StepsHigh. Since the code only ever checks state name Steps, checkAutomationState("Steps", "StepsLow") and ("Steps", "StepsHigh") both read false unconditionally on that device. Practical effect: clearPositiveTodOffsetsNow (which requires stepsLow with no alternative path) can never fire at all, so positive TOD offsets never auto-clear; clearNegativeTodOffsetsNow still works via its other two OR-branches (MJ state, HP2<4.0) but permanently loses the stepsHigh trigger path. Fix is phone-side: retarget the writing action to state=Steps, value=StepsLow, and add a StepsHigh-writing action if one doesn't already exist elsewhere.

Summary: Steps is a read-only external activity classification (StepsLow/StepsHigh) consumed solely by the time-of-day ISF-offset auto-clear logic: a high-activity reading alone is enough to force-clear negative TOD offsets, while a low-activity reading is one of several conditions required before positive TOD offsets are cleared.

e. Profile [↑ TOC]

Values: PP130, C100, AllOK (default), Batt1%, Bolus

Written by:

• "BatteryOver1pc" → AllOK (line 3579; battery recovered above 1%)

• "Pod2" (POD 78 hours) → PP130 (line 4598)

• "ExportSettingsPodActivation" → PP130 (line 4946; fires when a new pod has been activated since the last settings export)

• "PodChangeHighPP130" → C100 (line 4802; starts a 130% boost around a pod change)

• "HighPP130Off" → C100 (line 4826; exits the 130%/110% boost)

• "BolusGiven" (bg1/bg2/bg3 branches) → Bolus (line 3934; marks a post-bolus 6.8mmol TT so it isn't misread as the manual Activity TT)

• "Bolus2" → Bolus (line 5219; same marker, for its own post-bolus TT)

• Standalone unconditional block (no readyToRun key of its own) → C100 once >15min have passed since the last bolus, clearing the Bolus marker (line 5233)

• "HighNight00AM" (permanently disabled block, highNight00AmEnabled = false) → C100 (line 3665) — dead in practice

Read by:

• "Battery1pc" precondition — must currently be PP130, C100, or AllOK (a "normal running" state) before the battery-critical profile drop is allowed (line 3554)

• "ActivityProf50" — must NOT be Bolus, so a real Activity TT isn't confused with a brief post-bolus TT sharing the same 6.8mmol value (line 3681)

• "PodChangeHighPP130" — must be PP130 to start its own 130% boost (line 4795)

• The Bolus-clear block itself reads Bolus to know whether there's anything to clear (line 5230)

Note: Batt1% is declared as a valid value but is never actually written by any automation — Battery1pc's own source comment (line 3570) confirms this was a stale guard requirement that the writer never satisfied, since removed from the read side.

Summary: Profile is a catch-all marker state, mainly used to disambiguate *why* the effective profile-percentage is currently elevated (PP130/C100 from pod-change- or export-triggered boosts) and to flag a brief post-bolus temp target (Bolus) so it can't be confused with the manually-set Activity TT that happens to share its 6.8mmol value. AllOK is the neutral/default value.

f. Sleeping [↑ TOC]

Values: True (only value declared — there is no explicit "not sleeping" value; absence of a current value / a false inState read represents "not sleeping")

Written by: none in this file. Read-only from this file — owned externally (native Automation-tab trigger/action or other integration).

Read by:

• activityMonitor() — existSleepState = automationStateService.hasStateValues("Sleeping") and useSleepState = automationStateService.inState("Sleeping", "True") (lines 5984–5985), logged every cycle via aapsLogger.debug. useSleepState combined with recent steps ≤200 disables the inactivity-boost detection outright ("sleeping state", line 5999). existSleepState (whether the state has been declared at all, independent of its value) also gates the idle-window inactivity-detection branch via !existSleepState (line 6003).

Summary: Sleeping is a read-only external flag that the Activity Monitor's inactivity-sensitivity logic reads to suppress its own inactivity-boost detection while the user is asleep, since a stationary phone / low step count overnight should not be treated the same as daytime inactivity.

g. Removed 2026-08-22 [↑ TOC]

Three states were removed from REQUIRED_AUTOMATION_STATES (and their remaining code references cleaned up) after an audit found no live use:

• MJstate (values MJon/MJoff) — every setState call for it was already commented out in the source; MJ (NOMJremains/MJ active/MJ2/MJ3/...) is the real, actively-used state. This looks like an earlier, abandoned naming attempt that was never fully cleaned up.

• BGLstate (value BGLlastLOW) — a write-only one-way latch, set in GentleHypoRisk/AlarmHypo1/AlarmHypo2 and cleared nowhere, so once any hypo alert had ever fired it stayed BGLlastLOW permanently and carried no information. Its one remaining reader (in MoreMJ) had already been swapped to the properly-maintained LowBG=50recent flag before this cleanup; the writes themselves were removed 2026-08-22 to make it genuinely inert rather than just functionally dead.

• query_got_up (value query_it) — a one-shot signal written in activityMonitor() whenever Sleeping=True and no steps were recorded between the 15-minute and 30-minute windows, but never read back by anything, here or elsewhere in the codebase. Sleeping itself remains live and is documented above.

Appendix G — Current Live Settings Snapshot [↑ TOC]

Snapshot source: AutoISF_settings_Live_20260824_155639.txt
Captured: 24 August 2026, 3:56 PM — MAIN_PHONE_LOCAL (Live), full configuration, version 3.4.2.6+aisf321UK_654.
Important: These are point-in-time values captured from the Live device, not recommended defaults. Later app activity, profile changes, imports, or manual edits may change them.

Setting	Live value
main_phone_snapshot_time	8/24/26 03:56 PM
source	MAIN_PHONE_LOCAL
configuration_flavor	full
configuration_version	3.4.2.6+aisf321UK_654
configuration_application_id	info.nightscout.androidaps
configuration_profile	Current ProfileReal
Enable alternative activation of SMB always	true
autoISF_max	2.60
autoISF_max_low	1.00
autoISF_min	0.30
autoisf_bgaccel_isf_weight_high	0.95
autoisf_bgaccel_isf_weight_normal	0.70
autoisf_boost_automations_enabled	true
autoisf_clean_graph_requested	false
autoisf_custom_automations_enabled	true
autoisf_dura_isf_weight_normal	2.50
autoisf_fslcal_offset_normal	1.40
autoisf_fslcal_slope_normal	0.72
autoisf_libre_offset_orig	1.40
autoisf_libre_slope_orig	0.72
autoisf_low_profile_name	Current Profile
autoisf_low_raw_24_override_active	false
autoisf_low_storage_notified	false
autoisf_mild_boost_ratio	0.21
autoisf_mild_offset_zero_active	false
autoisf_mj_kotlin_buttons_enabled	true
autoisf_old_pod_notified	false
autoisf_old_sensor_adj_active	false
autoisf_old_sensor_adj_enabled	true
autoisf_pp_isf_weight_high	0.15
autoisf_pp_isf_weight_normal	0.10
autoisf_sensor_age_code_enabled	true
autoisf_smb_delivery_baseline	0.13
autoisf_smb_offset_override	0.80
autoisf_smb_offset_override_enabled	true
autoisf_standard_profile_name	Current ProfileReal
autoisf_steroid_kotlin_button_enabled	true
autoisf_tdd_factor	true
autoisf_tdd_factor_fallback	1.00
autoisf_tdd_sensitivity	true
autoisf_tod_offset_0002	0.00
autoisf_tod_offset_0204	0.00
autoisf_tod_offset_0406	0.00
autoisf_tod_offset_0609	0.00
autoisf_tod_offset_0912	0.00
autoisf_tod_offset_1218	0.00
autoisf_tod_offset_1822	0.00
autoisf_tod_offset_2200	0.00
autoisf_uam_boost_enabled	false
bgAccel_ISF_weight	0.70
bgBrake_ISF_weight	0.25
boost_bolus_cap	2.50
boost_max_iob	1.00
boost_scale_value	1.00

Setting	Live value
dura_ISF_weight	2.50
fslCal_Offset	1.40
fslCal_Slope	0.72
graph5_bgl_only	false
half_basal_exercise_target	6.90
high_temptarget_raises_sensitivity	true
higher_ISFrange_weight	0.80
iob_threshold_percent	70
low_temptarget_lowers_sensitivity	true
lower_ISFrange_weight	0.70
openapsama_enable_autoISF	true
openapsama_smb_delivery_ratio	0.13
openapsama_smb_delivery_ratio_bg_range	0.00
openapsama_smb_delivery_ratio_max	0.50
openapsama_smb_delivery_ratio_min	0.95
openapsama_smb_max_range_extension	4.00
pp_ISF_weight	0.10
show_carb_model_curve	true
show_graph5	true
split_bolus_enabled	true
split_bolus_interval	7

End [↑ TOC]